



US012404307B2

(12) **United States Patent**
Rauscher et al.

(10) **Patent No.: US 12,404,307 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **METHODS AND COMPOSITIONS FOR USE OF RECOMBINANT BACTERIAL EFFECTOR PROTEINS AS ANTI-INFLAMMATORY AGENTS**

(71) Applicant: **The Wistar Institute of Anatomy and Biology**, Philadelphia, PA (US)

(72) Inventors: **Frank J. Rauscher**, Wayne, PA (US);
Peter Mondics, Newtown Square, PA (US)

(73) Assignees: **The Wistar Institute of Anatomy and Biology**, Philadelphia, PA (US); **Innate Biologics LLC**, Harrisburg, PA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/686,110**

(22) Filed: **Mar. 3, 2022**

(65) **Prior Publication Data**

US 2022/0306702 A1 Sep. 29, 2022

Related U.S. Application Data

(63) Continuation of application No. 16/964,762, filed as application No. PCT/US2019/015224 on Jan. 25, 2019, now abandoned.

(60) Provisional application No. 62/622,068, filed on Jan. 25, 2018.

(51) **Int. Cl.**
C07K 14/255 (2006.01)
A61K 38/00 (2006.01)
C07K 14/25 (2006.01)

(52) **U.S. Cl.**
CPC **C07K 14/255** (2013.01); **C07K 14/25** (2013.01); **A61K 38/00** (2013.01); **C07K 2319/03** (2013.01); **C07K 2319/30** (2013.01); **C07K 2319/70** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,635,497 A	6/1997	Molenaar
8,840,901 B2	9/2014	Ruter et al.
2004/0086976 A1	5/2004	Fleer et al.
2004/0147719 A1	7/2004	Cornelis
2004/0208889 A1	10/2004	Sutton et al.
2011/0183908 A1	7/2011	Ruter et al.
2012/0064572 A1	3/2012	Finlay

FOREIGN PATENT DOCUMENTS

WO	WO 2001/19393	4/2001
WO	WO 2014/076642	5/2014

OTHER PUBLICATIONS

Bowie et al. (Science, 1990, 247:1306-1310).*

Burgess et al. (J. Cell Biol. 111:2129-2138, 1990).*

Lazar et al. (Mol. Cell. Biol., 8:1247-1252, 1988).*

Jung et al., Identification of tissue-specific targeting peptide, Journal of Computer-Aided Molecular Design, vol. 26:1267-1275, Oct. 2012.

Zakrewsky et al., Choline and Geranate Deep Eutectic Solvent as a Broad-Spectrum Antiseptic Agent for Preventative and Therapeutic Applications, Advanced Healthcare Materials, vol. 5(11):1282-1289, Jun. 2016.

Banerjee et al., Transdermal Protein Delivery Using Choline and Geranate (CAGE) Deep Eutectic Solvent, Advance Healthcare Matter, vol. 6(15):1601411, Aug. 2017.

Badenhorst et al., Pharmaceutical Strategies for the Topical Dermal Delivery of Peptides/Proteins for Cosmetic and Therapeutic Applications, Austin Journal of Pharmacology and Therapeutics, vol. 2(6): 1-100, Aug. 2014.

Cornelis, G., The Yersinia Ysc-Yop virulence apparatus, International Journal of Medical Microbiology, vol. 291(6-7):455-62, Feb. 2002.

Cataliotti et al., Oral Brain Natriuretic Peptide: A Novel Strategy for Chronic Protein Therapy for Cardiovascular Disease, Trends in Cardiovascular Medicine, vol. 17(1): 10-14, Jan. 2007.

Veronese, F., & Mero, A., The impact of PEGylation on biological therapies, BioDrugs, vol. 22(5):315-29, Aug. 2012.

Miller et al., Amphiphilic conjugates of human brain natriuretic peptide designed for oral delivery: in vitro activity screening, Bioconjugate Chemistry, vol. 17(2):267-74, Mar.-Apr. 2006.

Veronese, F., & Pasut, G., PEGylation, Successful Approach to Drug Delivery, Drug Discovery Today, vol. 10(21): 1451-8, Dec. 2005.

Wang et al., AlbuBNP, a recombinant B-type natriuretic peptide and human serum albumin fusion hormone, as a long-term therapy of congestive heart failure, Pharmaceutical Research, vol. 21(11):2105-11, Nov. 2004.

Guidotti et al., Cell-Penetrating Peptides: From Basic Research to Clinics, Trends in Pharmacological Sciences, vol. Trends in Pharmacological Sciences, vol. 38(4):406-424, Apr. 2017.

Norkowski i et al., Bacterial LPX motif-harboring virulence factors constitute a species-spanning family of cell-penetrating effectors, Cellular and Molecular Life Sciences, vol. 75(66-73):1-17, Dec. 2017.

Dixon et al., Correction for Dixon et al., Highly efficient delivery of functional cargoes by the synergistic effect of GAG binding motifs and cell-penetrating peptides, PNAS, vol. 113(3):E291-299, Jan. 2016.

Tezgel et al., Synthetic Protein Mimics for Functional Protein Delivery, Biomacromolecules, vol. 18(3):819-825, Mar. 2017.

Bolhassani et al., In vitro and in vivo delivery of therapeutic proteins using cell penetrating peptides, Peptides, vol. 87:50-63, Nov. 2016.

(Continued)

Primary Examiner — Brian Gangle
(74) *Attorney, Agent, or Firm* — Howson & Howson LLP; Colleen M. Schaller; Richard F. Kane

(57) **ABSTRACT**

Provided herein are methods and compositions comprising a set of paired peptides comprising a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. The paired peptides can be linked to a protein transduction domain. The compositions can be formulated as pharmaceuticals. The compositions are useful for the treatment of inflammatory disorders.

12 Claims, 29 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

OTHER PUBLICATIONS

Yao et al., Structure and specificity of the bacterial cysteine methyltransferase effector NleE suggests a novel substrate in human DNA repair pathway, PLOS Pathogens, vol. 10(11):e1004522, Nov. 2014.

Ruter, P., 'Drugs from Bugs': bacterial effector proteins as promising biological (immune-) therapeutics, FEMS Microbiology Letters, vol. 351(2):126-132, Feb. 2014.

International Search Report and Written Opinion issued in International Patent Application No. PCT/US2019/015224, dated May 14, 2019.

* cited by examiner

Bacterial Effector Constructs

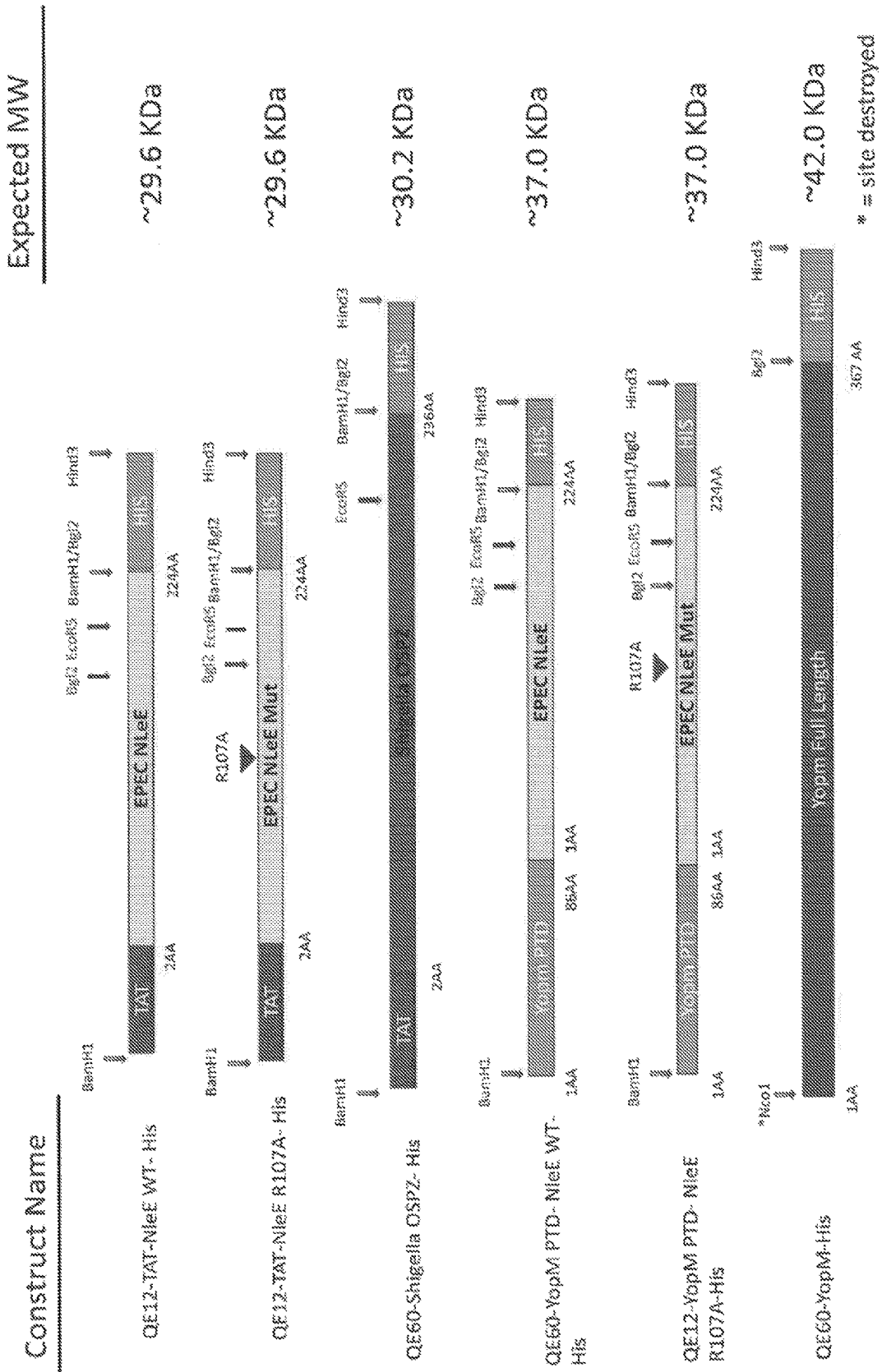
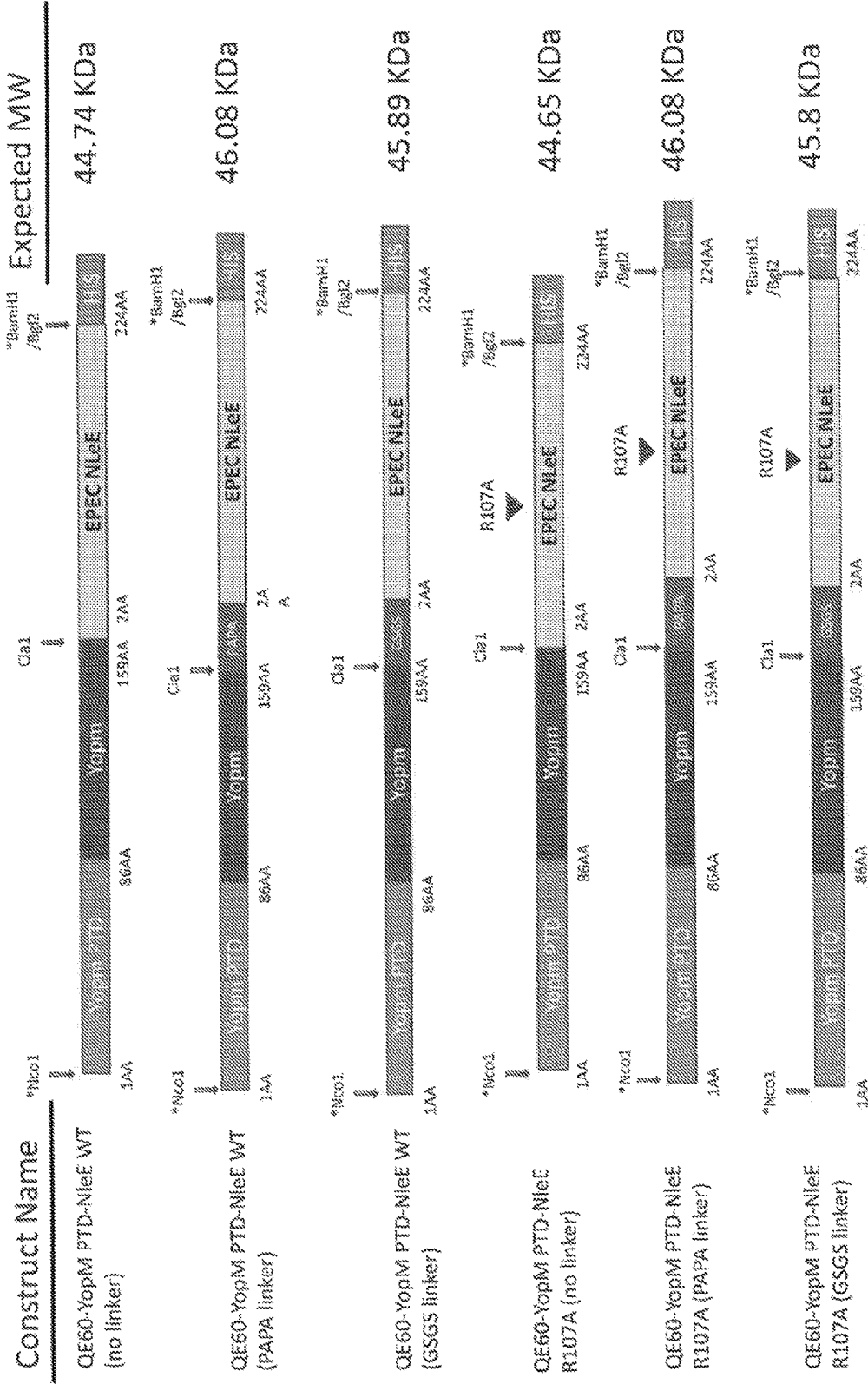


FIG. 1

Bacterial Effector Constructs



* = site destroyed

FIG. 2

Cloning Strategy for TAT-NleE WT (& Mutant R107A)

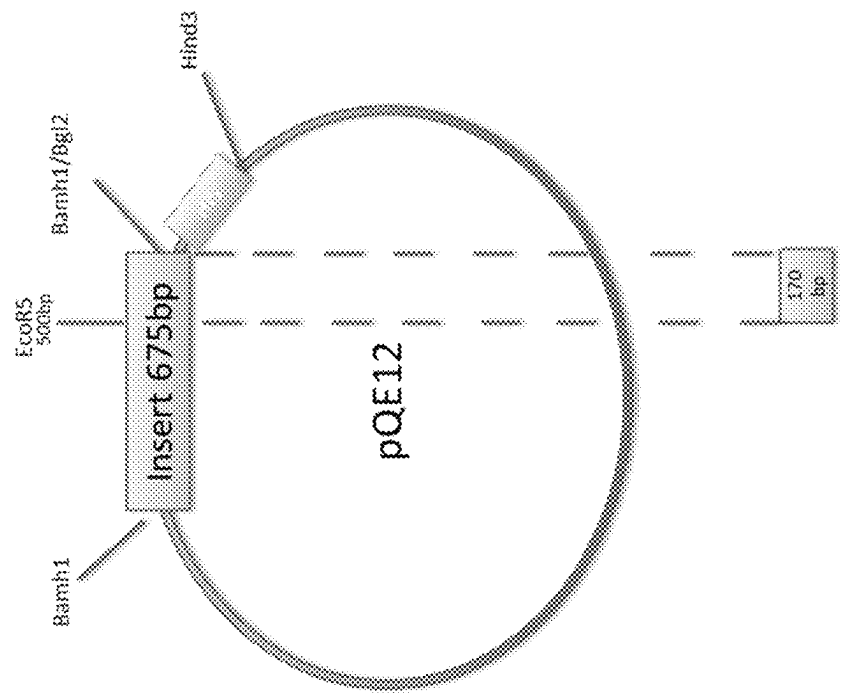


FIG. 3

Cloning Strategy for TAT-Shigella OSPZ

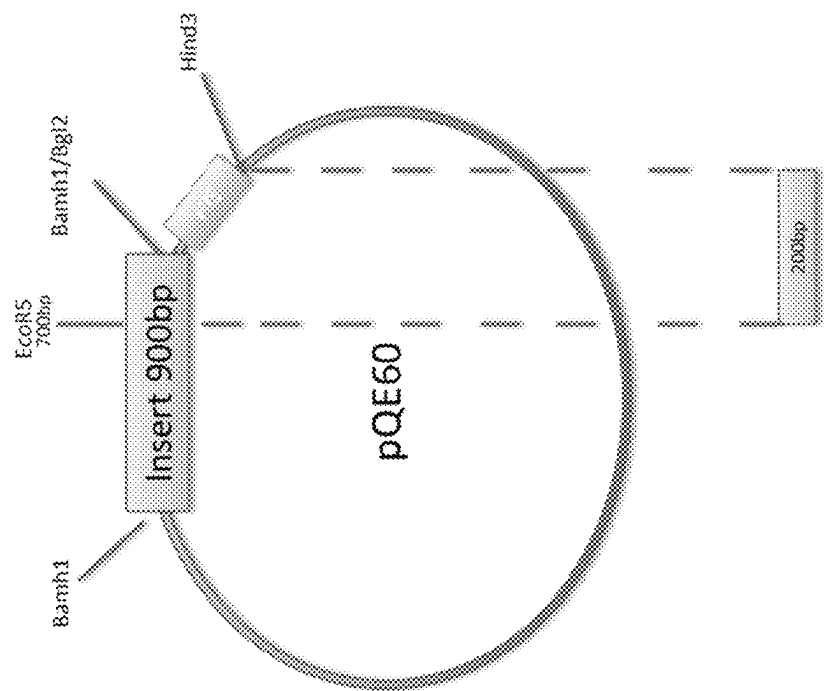


FIG. 4

Cloning Strategy for YopM PTD-NleE WT (& Mutant R107A)

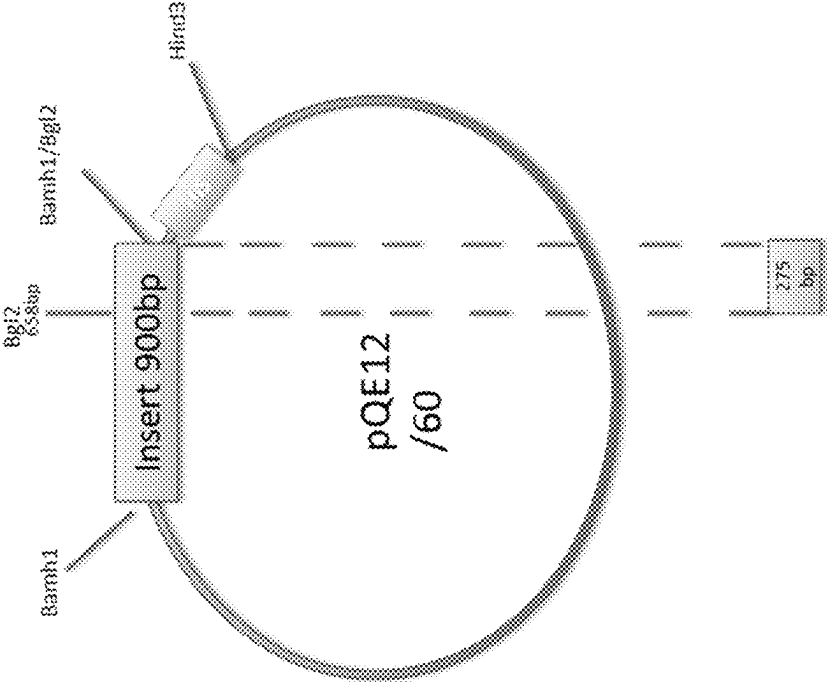


FIG. 5

Cloning Strategy for YopM PTD-NleE WT No Linker, PAPA Linker, GSGS Linker (Also for Mutant R107A)

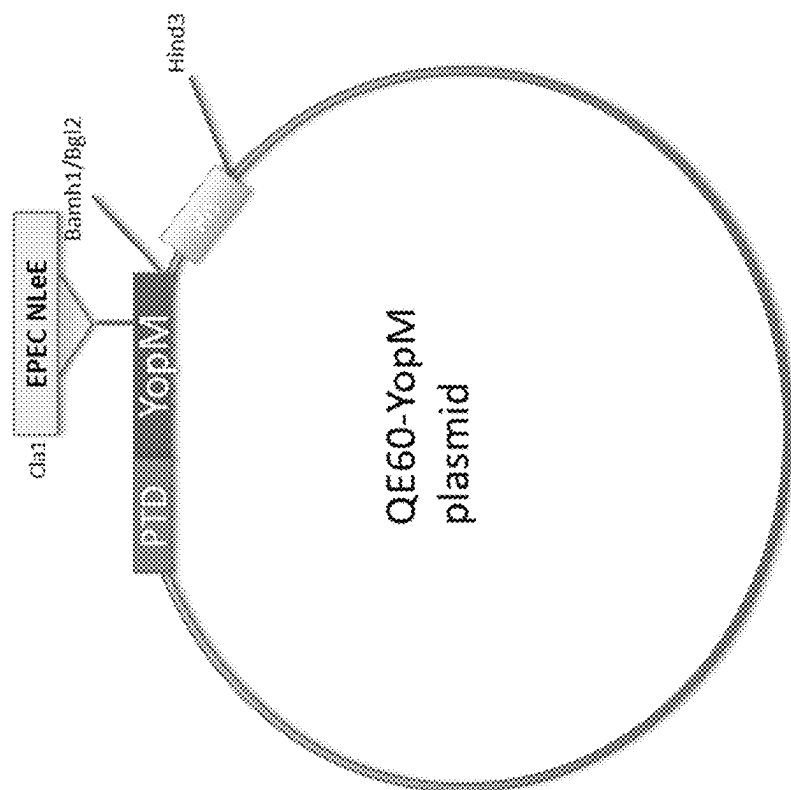


FIG. 6

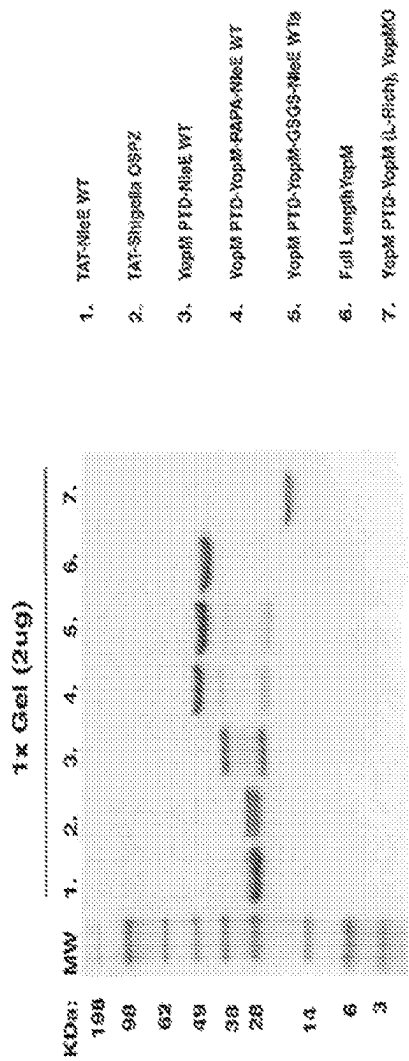


FIG. 7

Effect of paired fusion proteins on TNF-Alpha

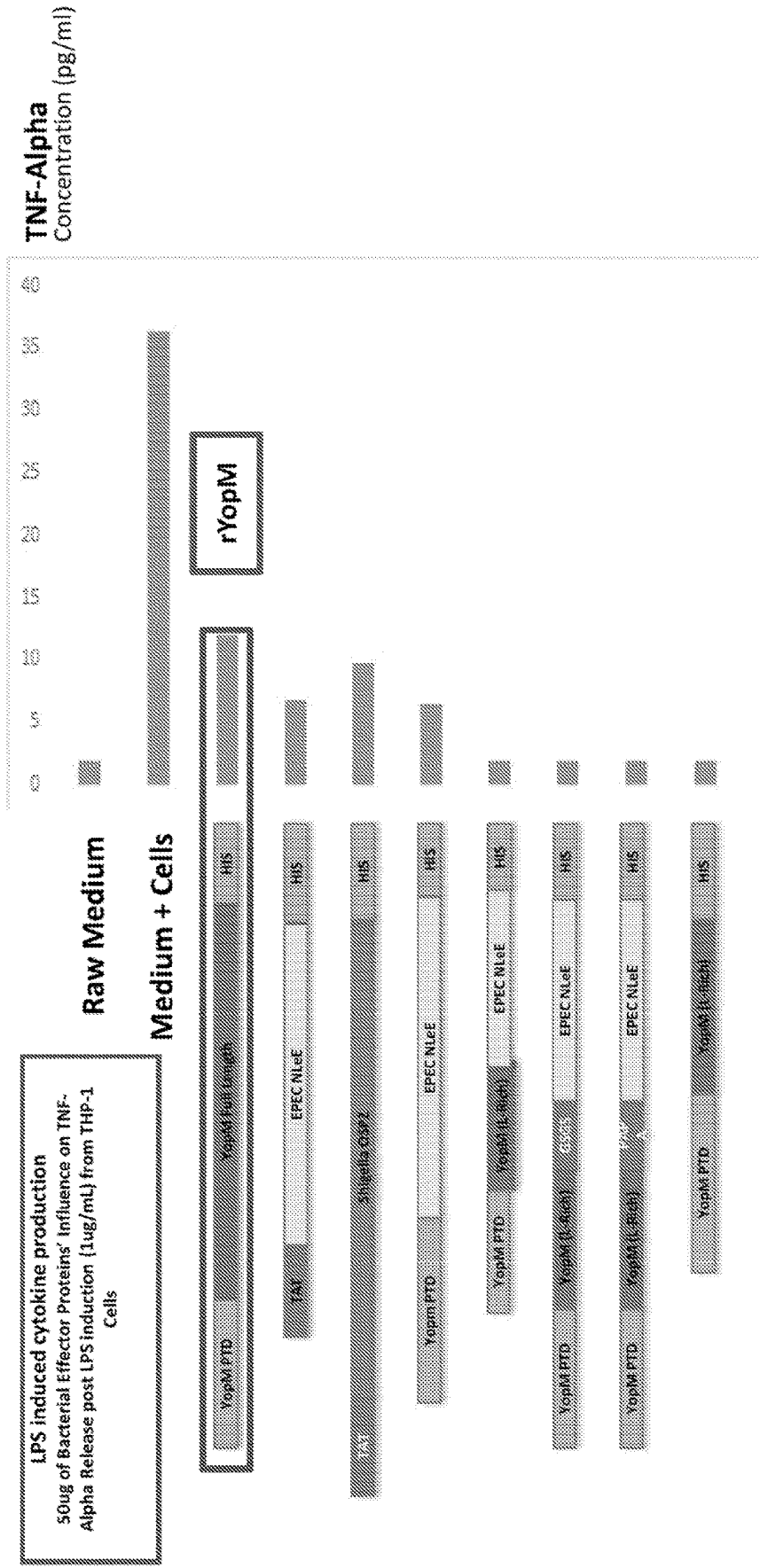


FIG. 9

Effect of paired fusion proteins on IL-6

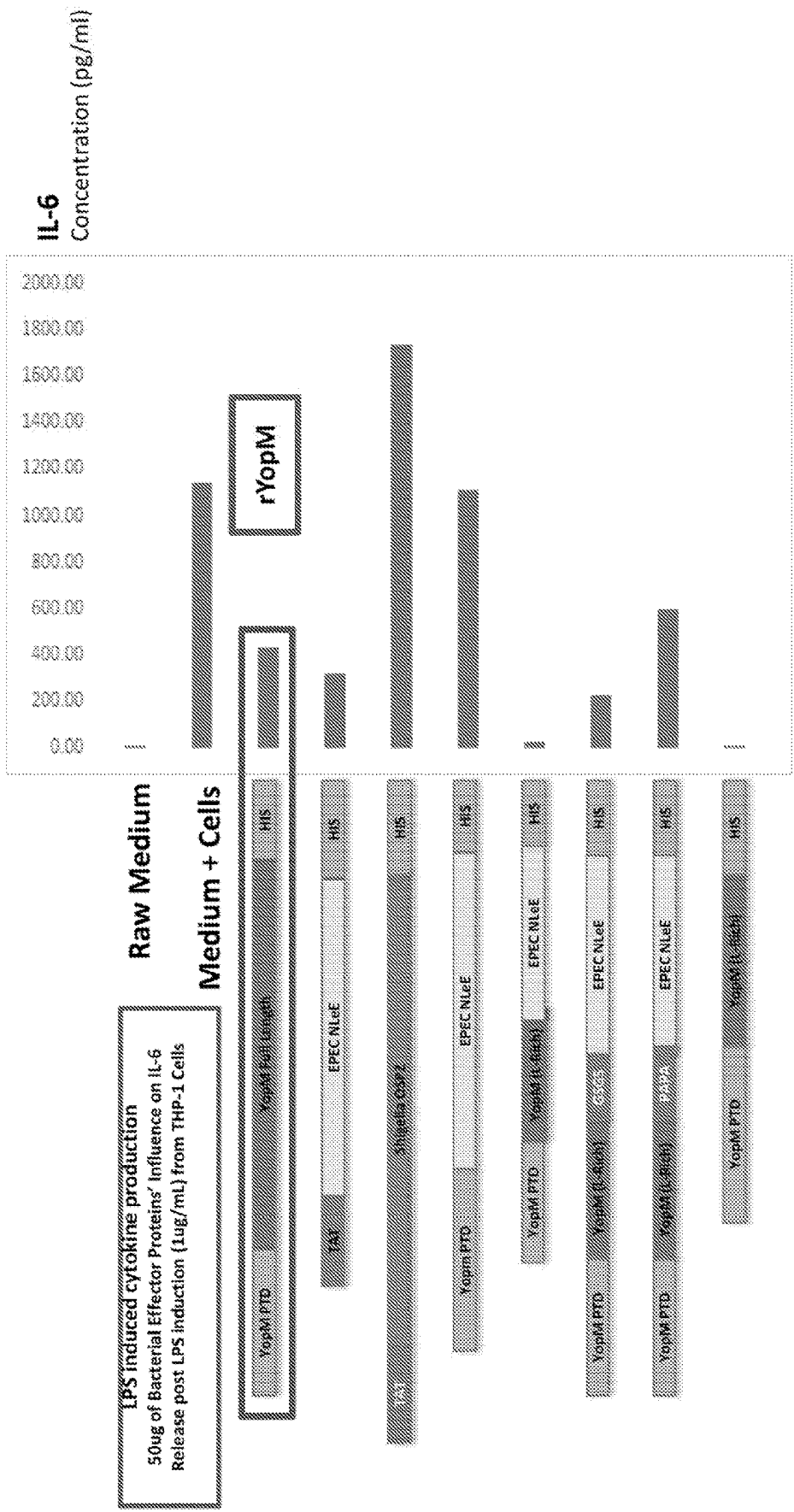


FIG. 10

Effect of paired fusion proteins on MCP-1

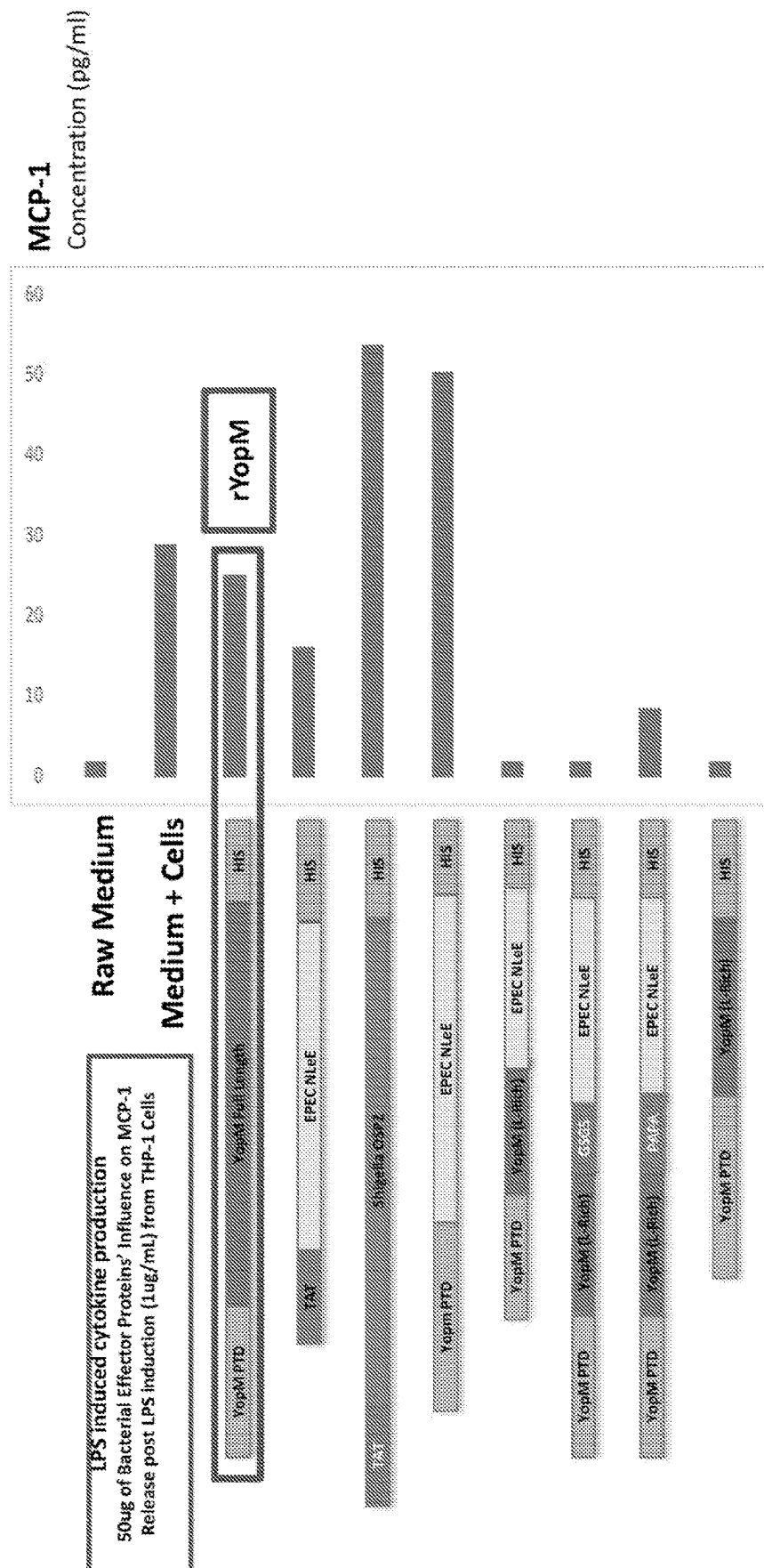


FIG. 17

Effect of paired fusion proteins on IL-23

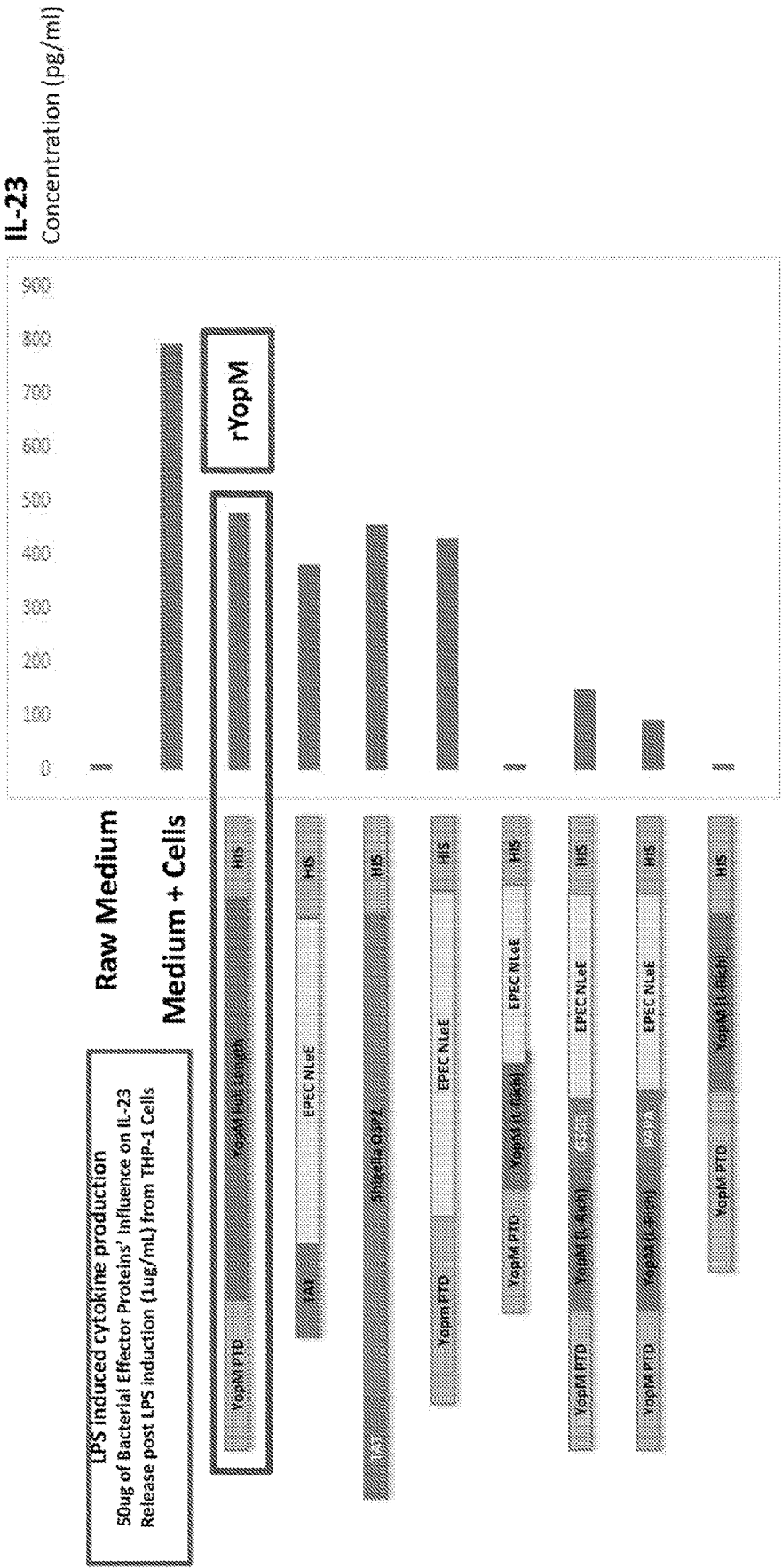


FIG. 12

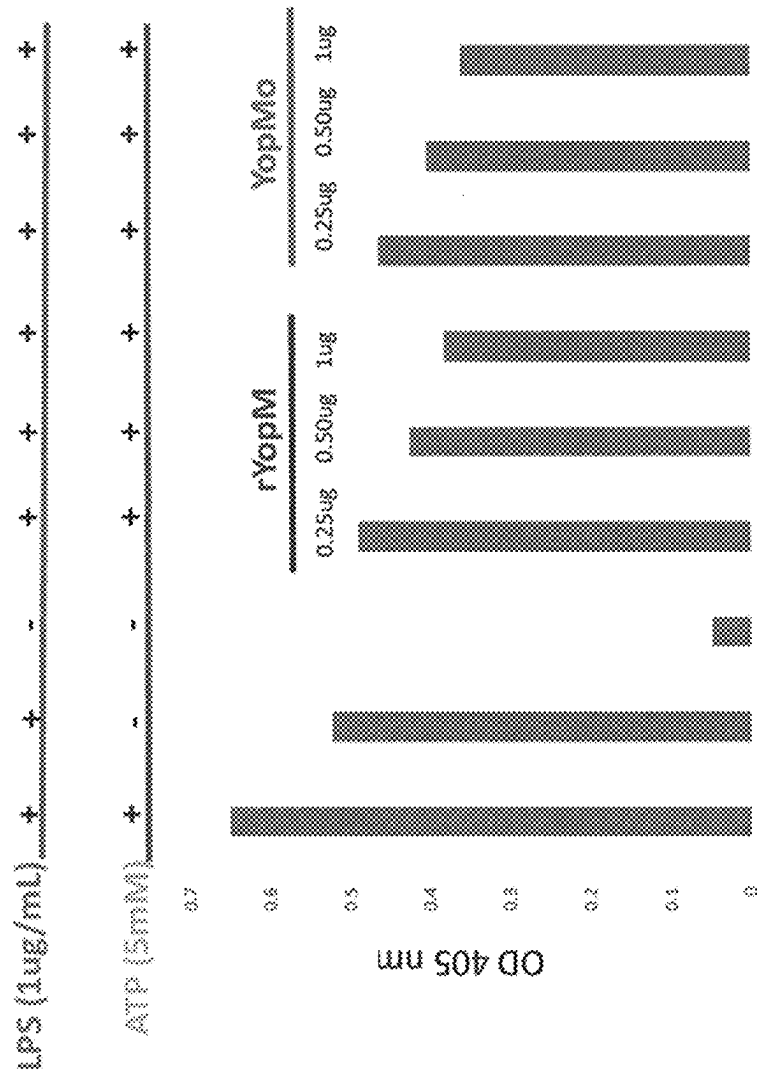


FIG. 13

Uptake of FITC-TAT-NleE-WT-His protein by HacaT cells

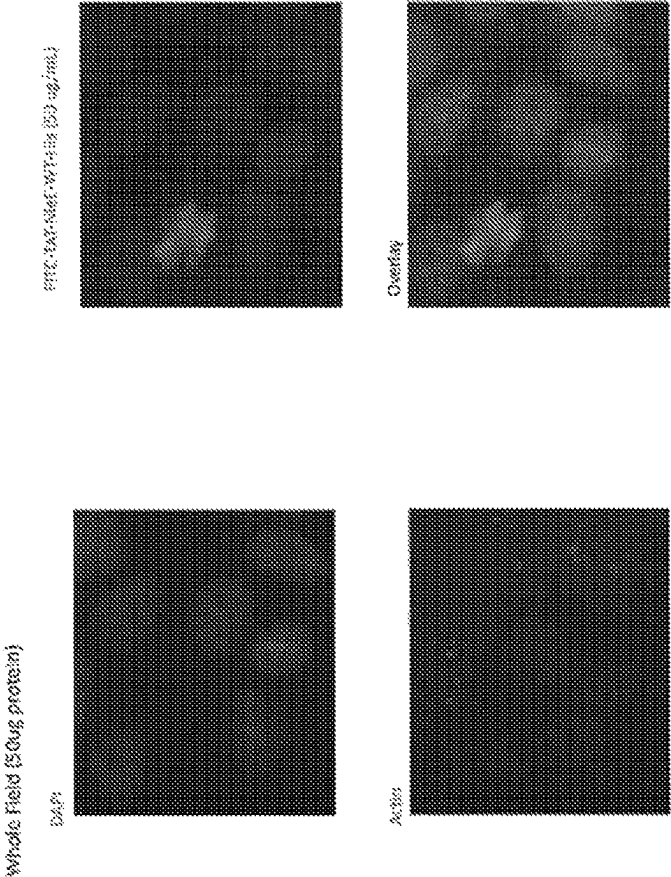


FIG. 14

Uptake of FITC-YopM PTD-YopM (L-Rich)-GSGS Linker NleE-WT-His protein by Haca
cells

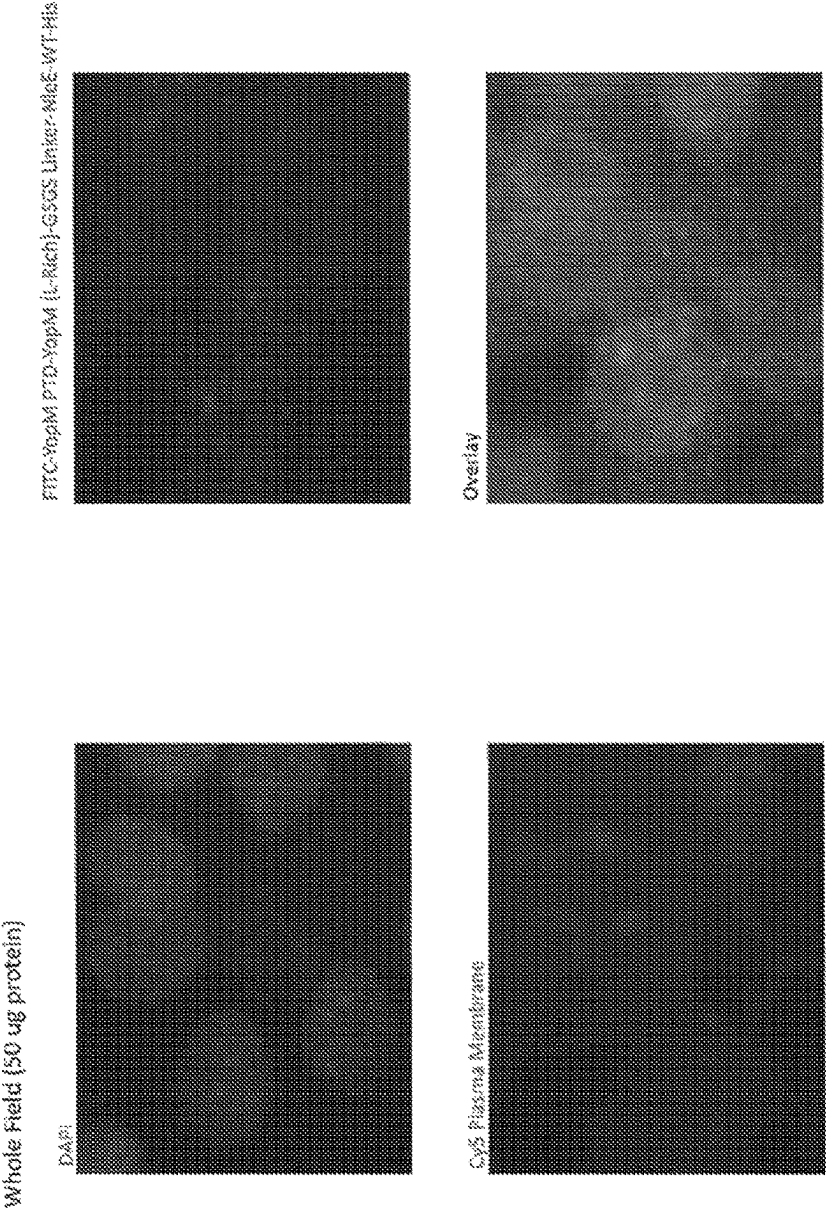


FIG. 15

Uptake of FITC-YopM PTD-YopM (L-Rich)-PAPA Linker-NleE-WT-His protein by Hacat cells

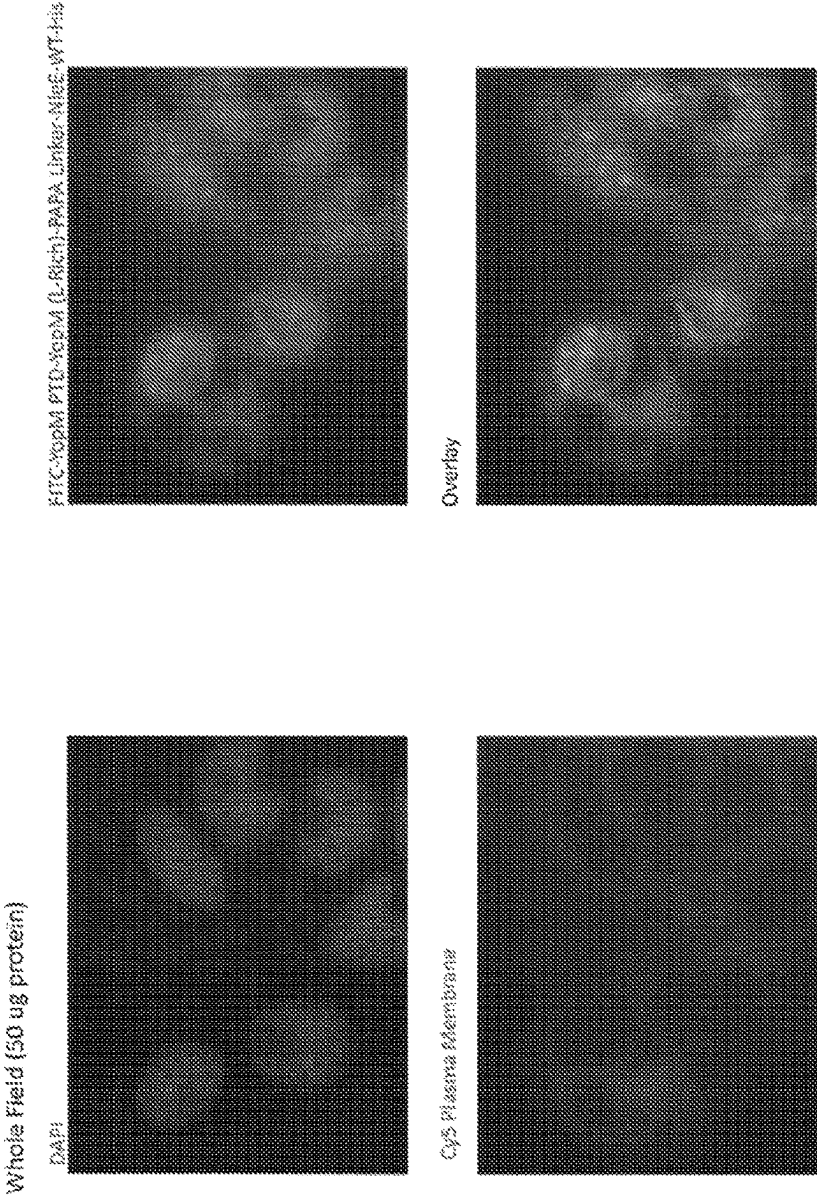


FIG. 16

Uptake of FITC-YopM PTD-YopM (L-Rich)-GSGS Linker-NleE-WT-His protein in H9c2 cells

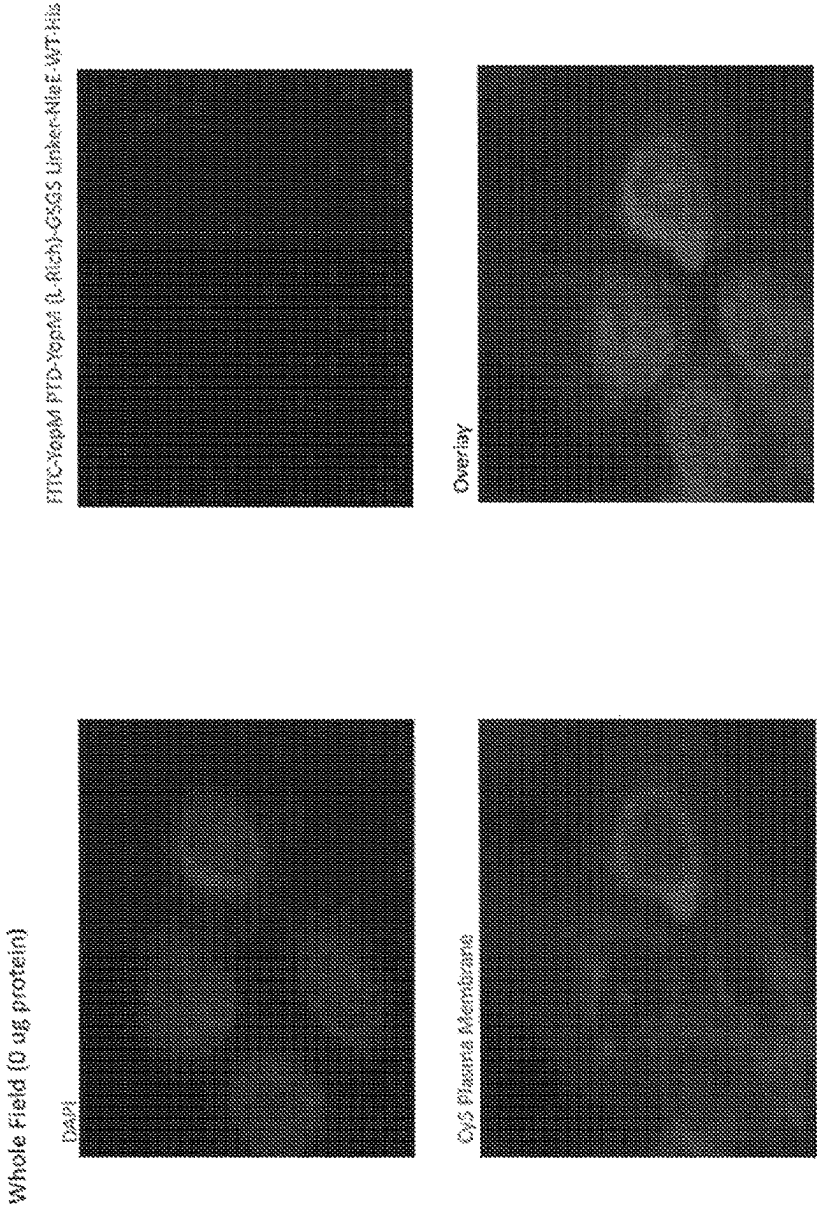


FIG. 17

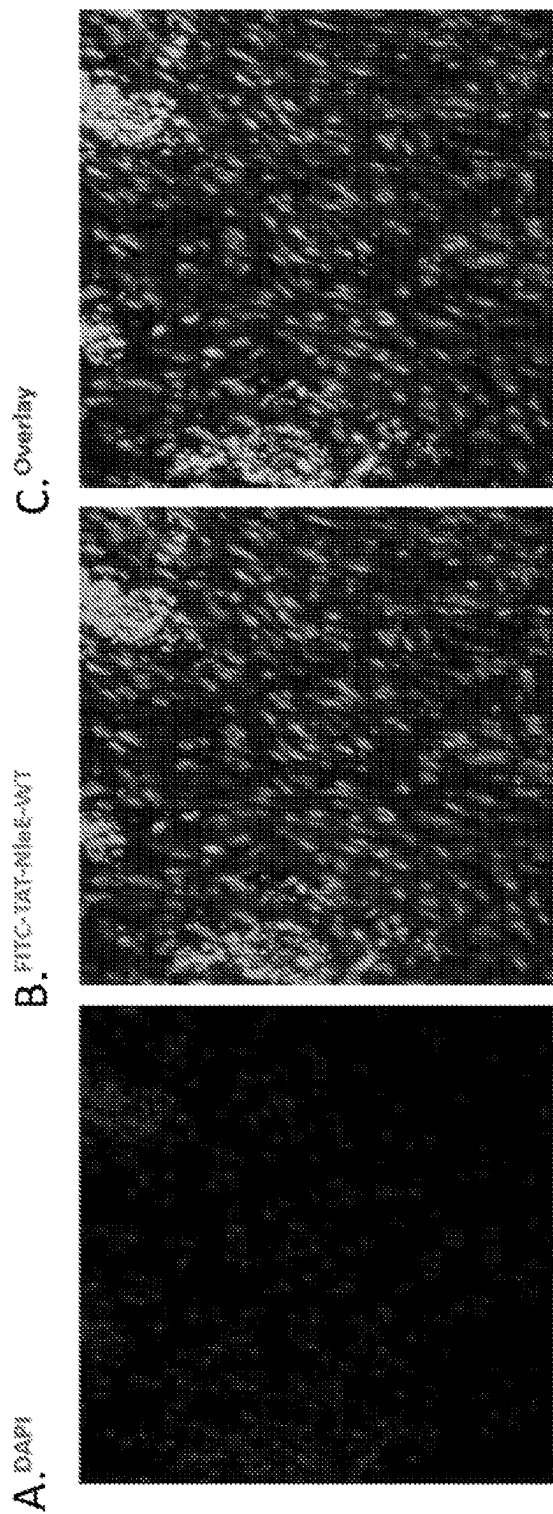


FIG. 18

2-photon Microscopy of 1cm x 1cm segment of mouse skin

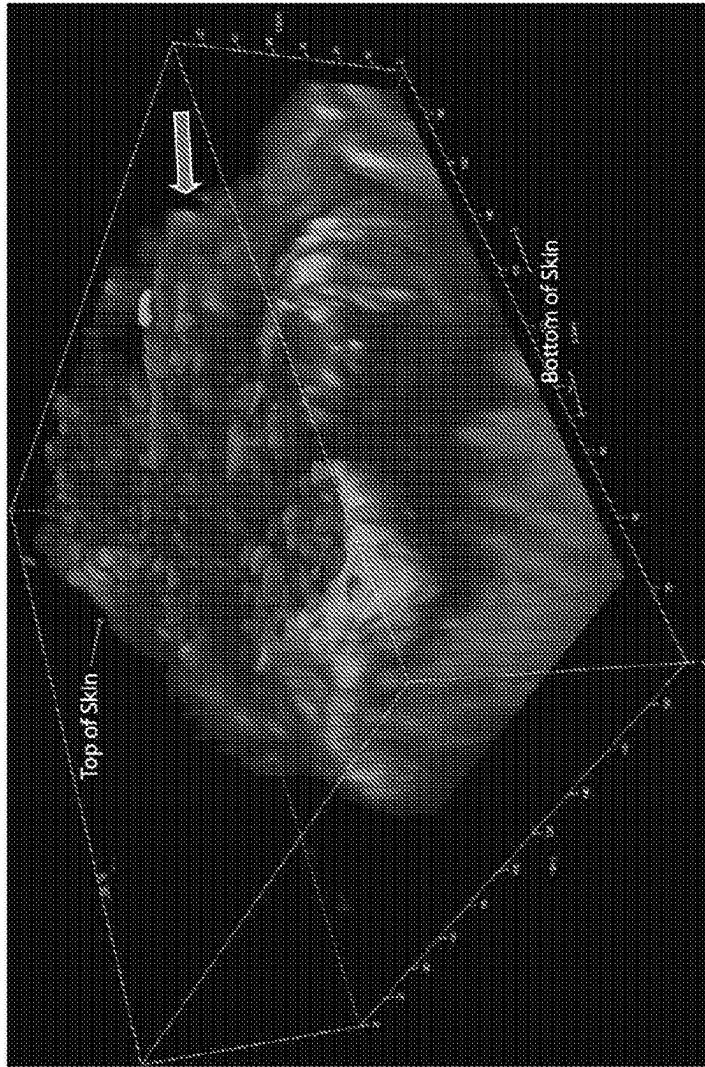


FIG. 19

2-Photon Microscopy of 10 micron slice from Mouse Skin

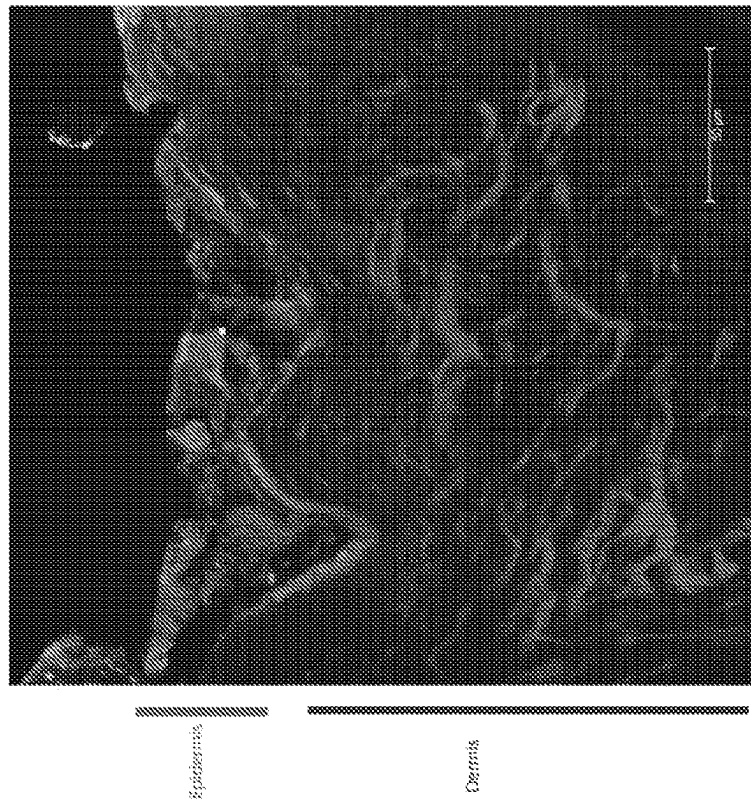


FIG. 20

YopM PTD-YopM (L-Rich)-PAPA/GSGS-NleE-WT-His Enzymatic Activity Assay

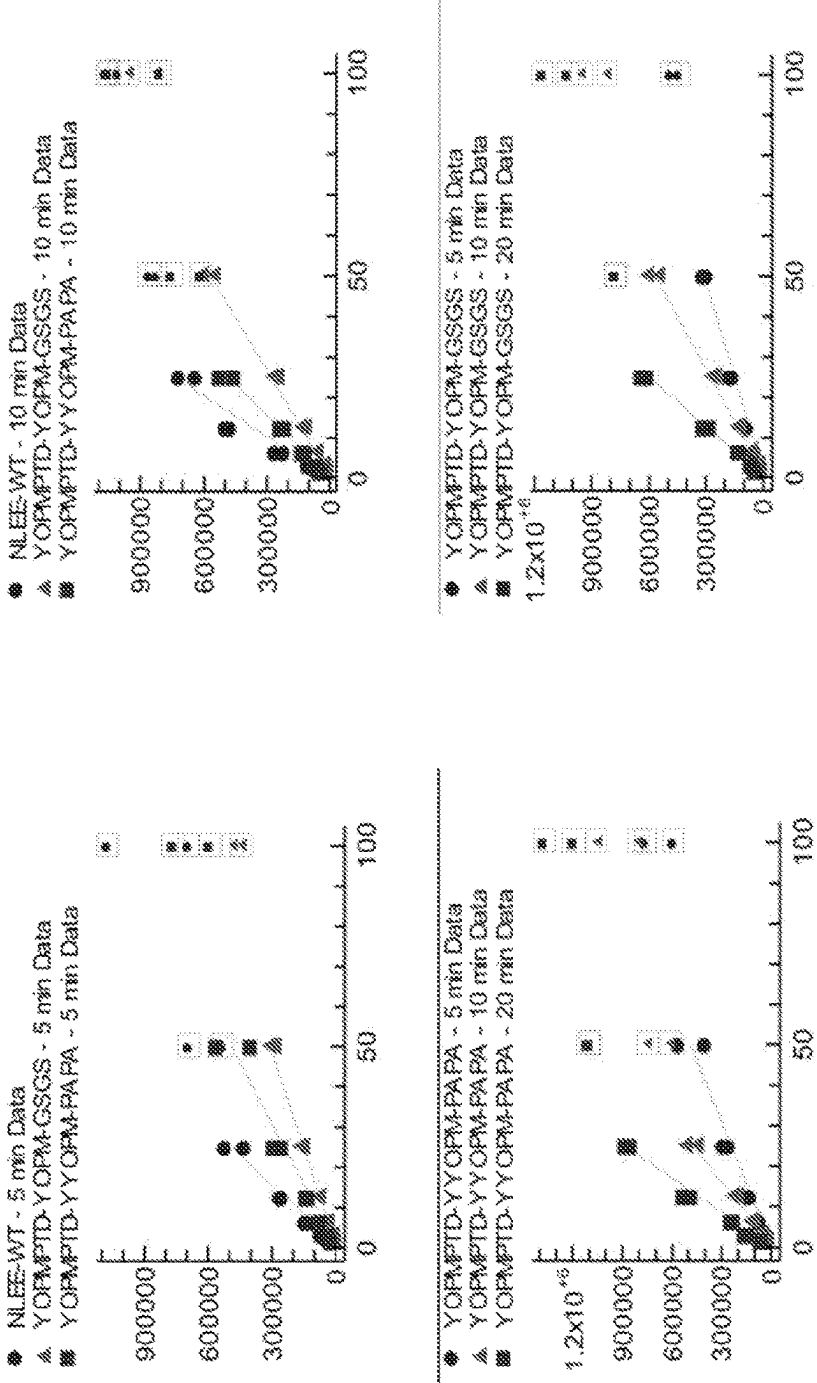


FIG. 21

NleE-R107A mutant and YopM Methyltransferase Activity Assay

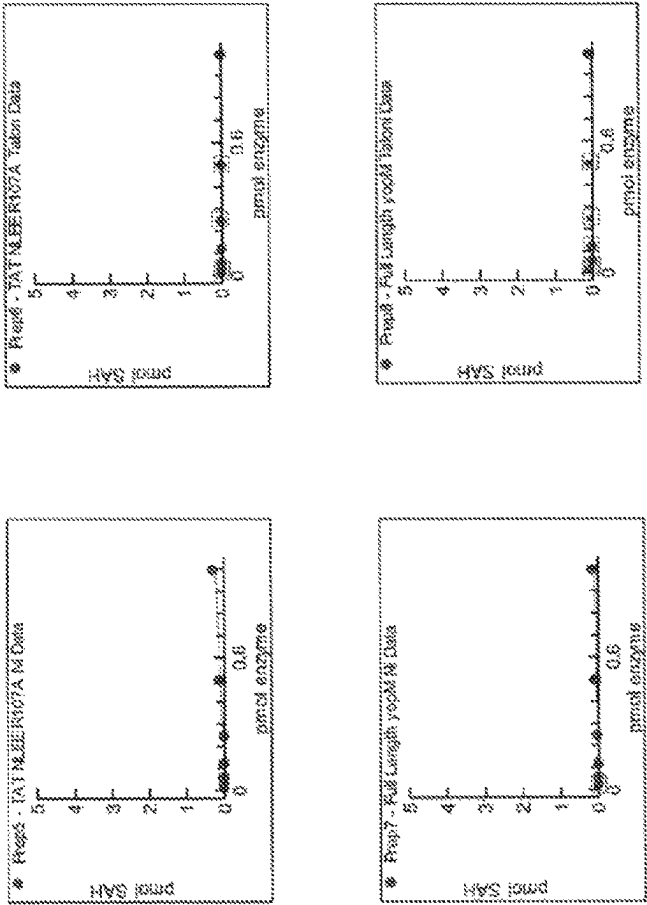


FIG. 22

NleE and Shigella Methyltransferase Activity Assay

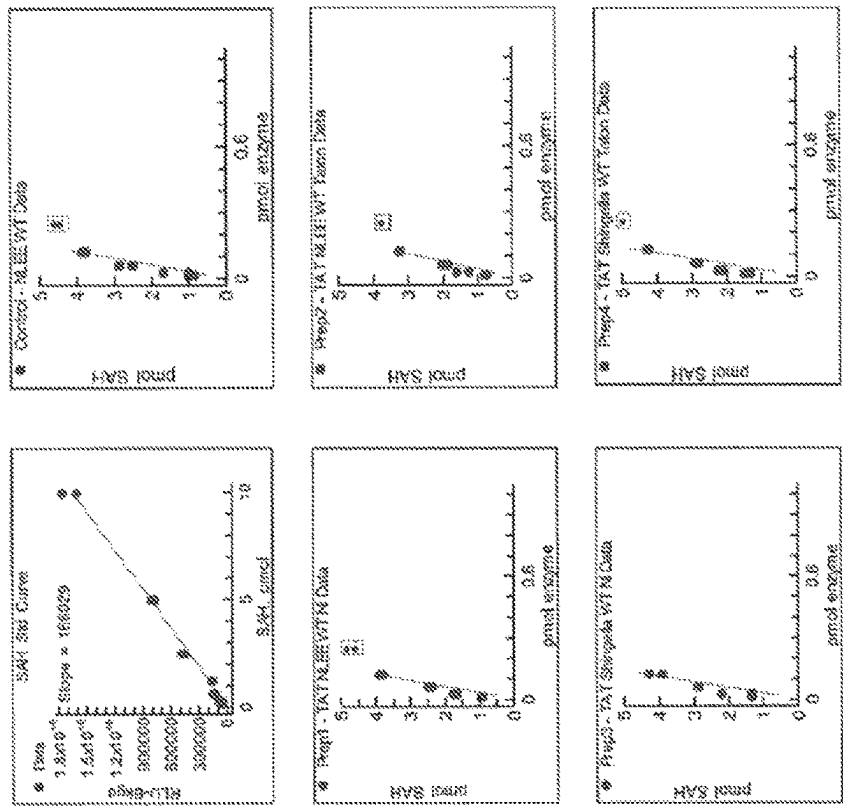


FIG. 23

Comparison across vehicles

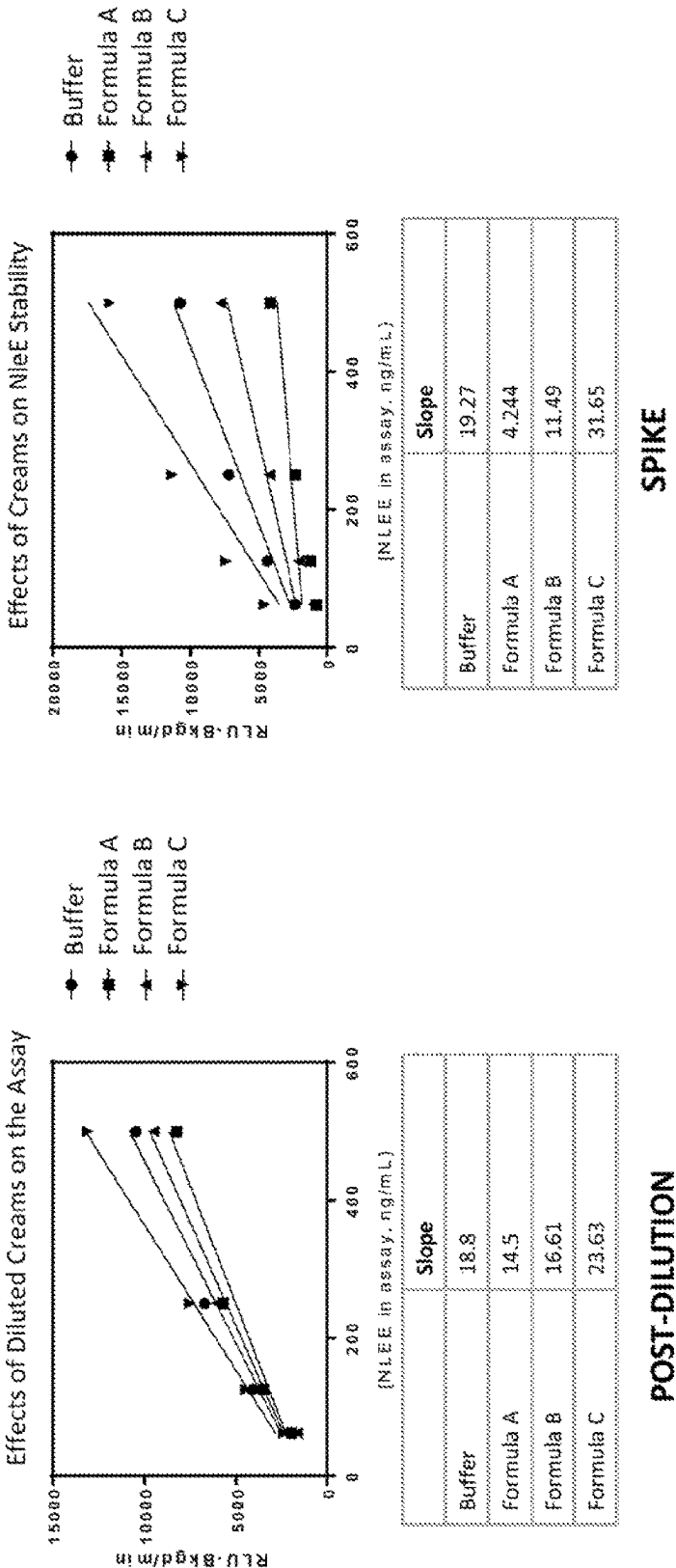


FIG. 24

Transdermal penetration of CAGE-NleE protein

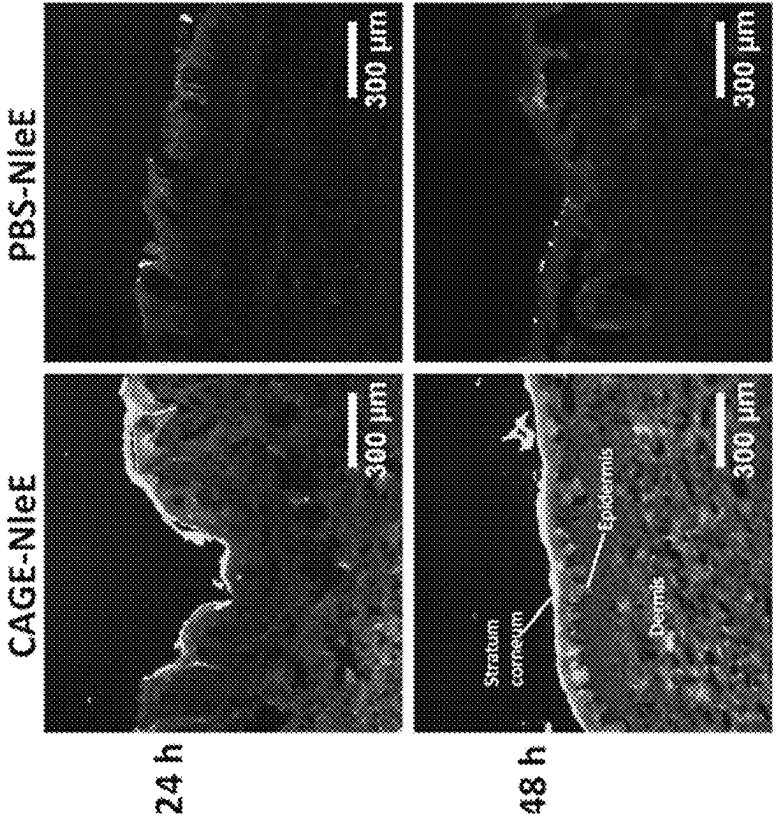


FIG. 25

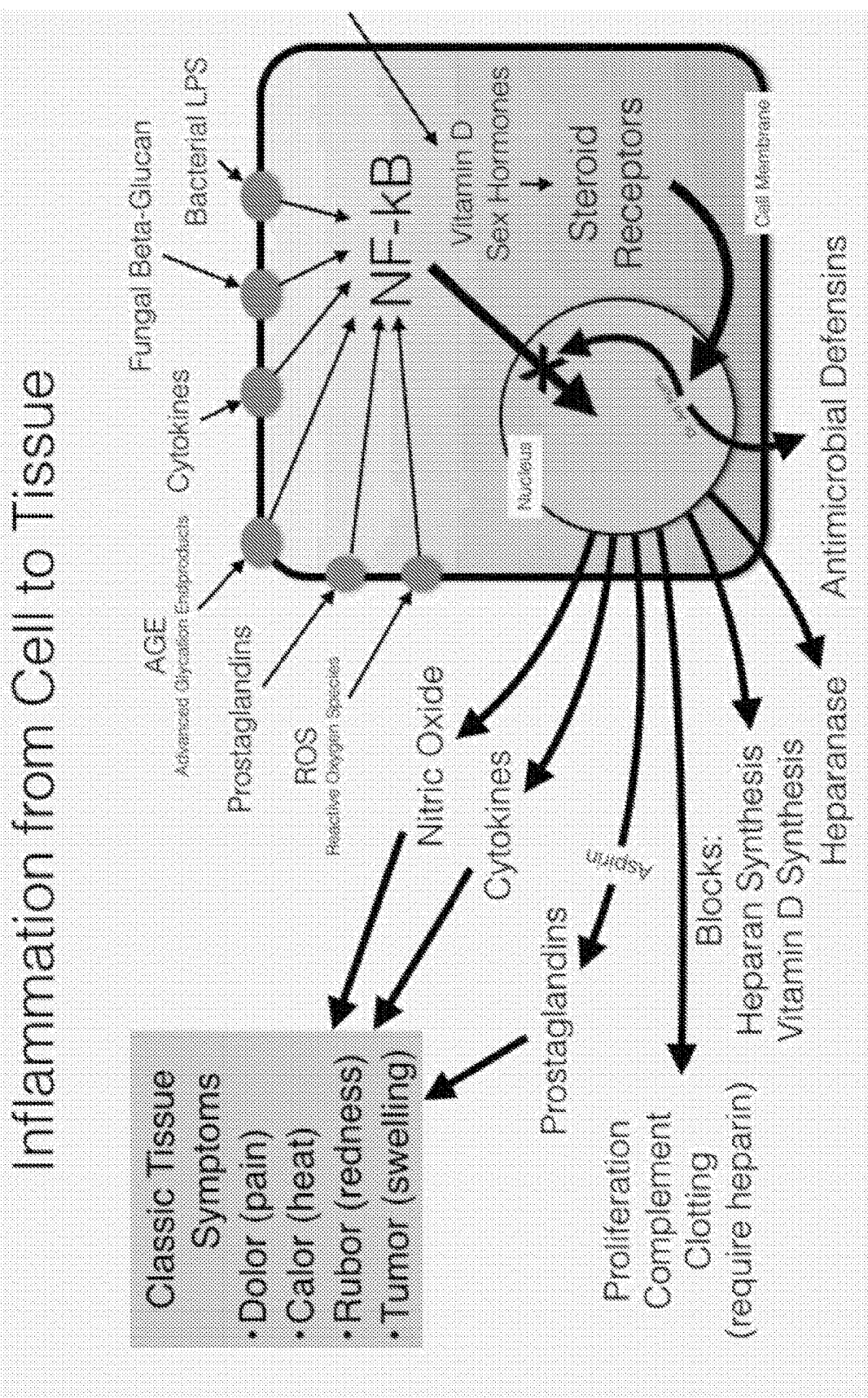


FIG. 26

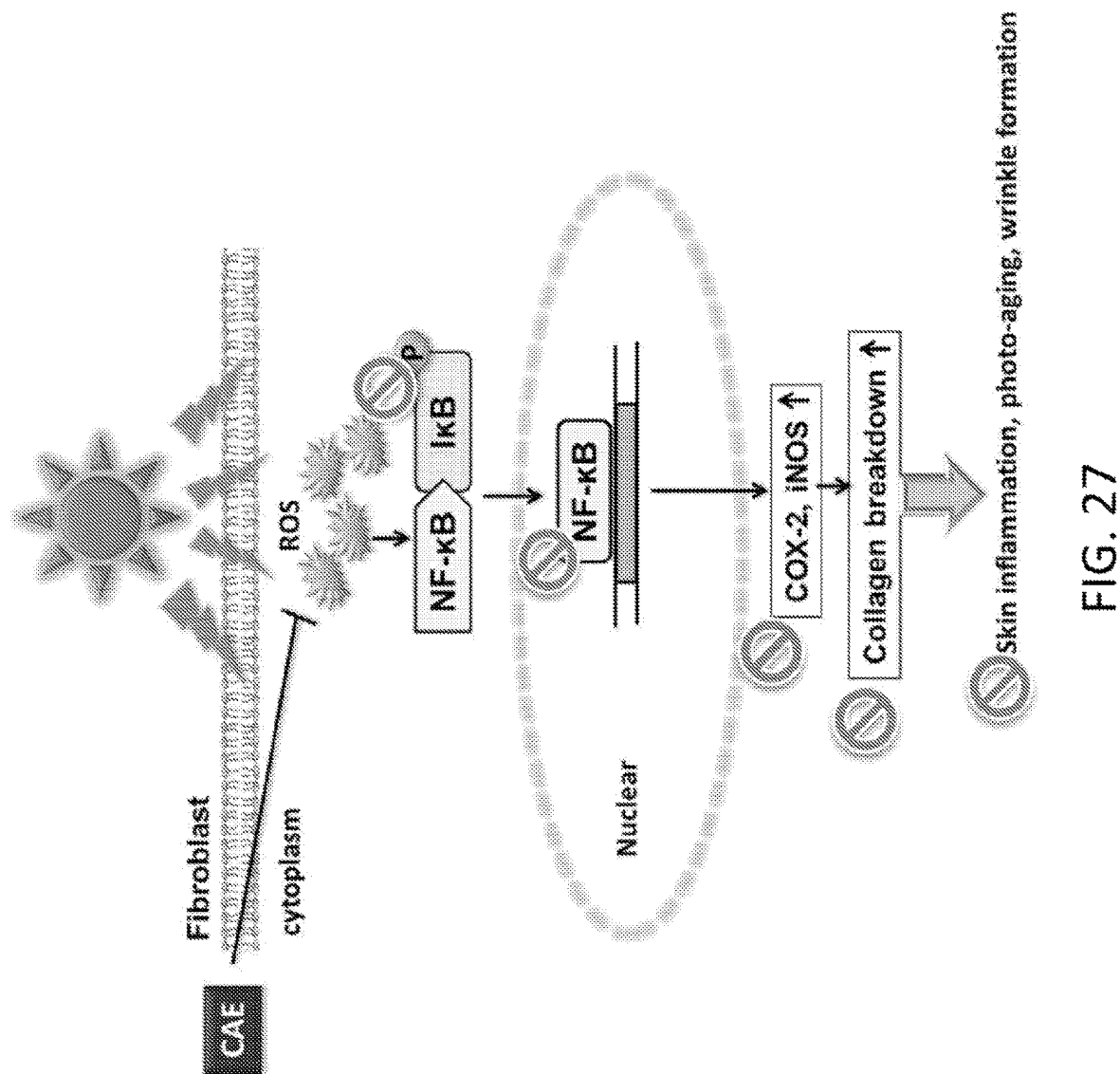


FIG. 27

E I Y
M I N P V T N T I Q G V S P I N T K Y A E H V V K N I Y P E I K H D Y F N E S P N I Y D K
G G F V
K Y I S G I T R G V A E L K Q E E F V N E K A R R F S Y M K T M Y S V C P E A F E P I S
R N E A S T P E G S W L T V I S G K R P M G Q F S V D S L Y N P D L H A L C E L P D I C
C K I F P K E N N D F L Y I V V V Y R N D S P L G E Q R A N R F I E L Y N I K R D I M Q
E L N Y E L P E L K A V K S E M I I A R E M G E I F S Y M P G E I D S Y M K Y I N N K
L
S K I E *

FIG. 28

YopM Amino Acid Sequence (SEQ ID NO: 2)

MFINPRNVSNTEFLQEPLRHSSNLTEMPVEAENVKSKTEYYNAWSEWERN
PPGNGEQREMAVSRLRDCIDRQAHELELNNLGLSSLPELPPHLESLVASCN
SLTELPELPQSLKSLLVDNNNNLKALSDDLPLLLEYLGVSNQLEKLPQLQNSS
FLKIIDVDNNSLKKLPDLPPSLEFFAAGNNQLEELPELQNLPELTAIYADNNS
LKKLPDLPLSLESIVAGNNILEELPELQNLPELTTIYADNNLLKTLPLDLPPSL
EALNVRDNYLTDLPELPQSLTFLDVSENI FSGLSELPNNLYYLNASSNEIRSL
CDLPPSLEELNVSNKKLIELPALPPRLERLIASFNHLAEVPELPQNLKQLHVE
YNPLREFPDIPESVEDLRMNSERVVDPPYEF AHETTDKLEDDDVFE

FIG. 29

1

METHODS AND COMPOSITIONS FOR USE OF RECOMBINANT BACTERIAL EFFECTOR PROTEINS AS ANTI-INFLAMMATORY AGENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of pending U.S. patent application Ser. No. 16/964,762, filed Jul. 24, 2020, which is a national stage of International Patent Application No. PCT/US2019/015224, filed Jan. 25, 2019, which claims the benefit of the priority of U.S. Provisional Patent Application No. 62/622,068, filed Jan. 25, 2018, which applications are incorporated herein by reference.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED IN ELECTRONIC FORM

Applicant hereby incorporates by reference the Sequence Listing material filed in electronic form herewith. This file is labeled "WST171USC1_SeqList.txt", created Jul. 24, 2020, and having 223 KB.

FIELD OF THE INVENTION

The present invention relates to methods and compositions for treating inflammatory disorders.

BACKGROUND OF THE INVENTION

Most treatments for acute inflammation, such as skin inflammatory conditions, only treat the symptoms (swelling, redness, pain, heat) by using non-specific drugs like corticosteroids and emollients/skin softeners. None of these non-specific drugs affect the underlying mechanism of activation of the inflammatory pathway, e.g., NFkB/JNK/p38. Moreover, normal aging processes chronically activate ectopically these three signal transduction pathways, resulting in inflammation. The inflammation makes these pathways a target for inhibition as part of an anti-aging/wellness program.

To date, the cosmeceutical industry's approach to reducing skin inflammation has many flaws. The "Active Compounds" contained in most cremes and topicals do not have defined, specific targets in the inflamed cell/tissue the cell and are extremely complex mixtures/extracts/serums containing millions of ingredients with no defined targets, or ingredients which are not absorbed by skin and have no effect at all. Because these compositions do not target the actual pathways which are causing inflammation, they have the potential to alter many processes non-specifically leading to toxic side effects.

A continuing need in the art exists for new and effective tools and methods for treating the causes of inflammation.

SUMMARY OF THE INVENTION

Disclosed herein are compositions of paired peptides comprising a first bacterial effector polypeptide linked to a second bacterial effector polypeptide that are useful for treating an inflammatory disorder. Accordingly, disclosed are compositions that include a set of paired peptides, wherein the set of paired peptides is linked to a protein transduction domain, and wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment

2

thereof linked to a second bacterial effector polypeptide or fragment thereof. The first and second bacterial effector polypeptides can be different, that is they can recognize a different molecular targets or modulate different inflammatory pathways. In an embodiment, the protein transduction domain and the set of paired peptides can be a fusion protein. The fusion protein can include one or more linkers. The protein transduction domain can be a YopM protein transduction domain, an SspH1 protein transduction domain, or an IpaH protein transduction domain. The first bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP, and the second bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79 and the second bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a YopM polypeptide or a fragment thereof and the second bacterial effector polypeptide or fragment thereof can be an NleE polypeptide or a fragment thereof. In some embodiments, the fusion protein can have an amino acid sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24.

Also provided are fusion proteins comprising a set of paired peptides wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. The first and second bacterial effector polypeptides can be different, that is they can recognize a different molecular targets or modulate different inflammatory pathways. The fusion protein can include one or more linkers. The first bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP, and the second bacterial effector polypeptide or fragment thereof can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79 and the second bacterial effector polypeptide or fragment thereof can be a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79. In some embodiments, the first bacterial effector polypeptide or fragment thereof can be a YopM polypeptide or a fragment thereof and the second bacterial effector polypeptide or fragment thereof can be an NleE polypeptide or a fragment thereof. In some embodiments, the fusion protein comprises

ing a set of paired peptides can have an amino acid sequence as set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88.

Also provided are nucleic acids encoding a set of paired peptides, wherein the set of paired peptides is linked to a protein transduction domain, and wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. Also provided are nucleic acids encoding fusion proteins comprising a set of paired peptides wherein the set of paired peptides comprises a first bacterial effector polypeptide or fragment thereof linked to a second bacterial effector polypeptide or fragment thereof. The nucleic acids can be contained within a vector, which can be expressed in a host cell. In one aspect, the compositions comprising a set of paired peptides can be formulated as pharmaceutical compositions.

Also provided are methods of treating a subject having or at risk for an inflammatory disorder, by administering to the subject a therapeutically effective amount of a pharmaceutical composition comprising the set of paired peptides. The inflammatory disorder can be a gastrointestinal disorder including inflammatory bowel disease, Crohn's disease and the ileocolitis, ileocecal, jeunoileitis, and gastroduodenal subtypes of Crohn's disease, and ulcerative colitis. The inflammatory disorder can also be a skin disorder.

Also provided are articles of manufacture, e.g., a kit. The kit can include measured amount of one or more of the compositions of the paired peptides and one or more items selected from the group consisting of packaging material, a package insert comprising instructions for use, a sterile fluid, and a sterile container.

In one aspect, a composition comprises in a pharmaceutically acceptable carrier or excipient or formulation a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain, optionally linked covalently or non-covalently to a selected protein transduction domain (PTD) or penetrating peptide (CPP). In one embodiment, the composition further comprises an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP or to an additional PTD or CPP. In one embodiment, a composition comprises multiple first and additional constructs. In another embodiment, the constructs are further associated with targeting moieties directing delivery of the constructs to a selected cell or tissue.

In another aspect, a recombinant polypeptide comprises a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain, optionally linked covalently or non-covalently to a selected protein transduction domain (PTD) or penetrating peptide (CPP) and an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP or to an additional PTD or CPP. The first construct is linked covalently or non-covalently to one or more of the additional constructs in a single polypeptide. In another embodiment, the polypeptide comprises an optional linker amino acid sequence interposed between each first and additional construct. In another embodiment, the polypeptide is further associated with targeting moieties directing delivery of the polypeptide to a selected cell or tissue.

In another aspect, a recombinant nucleic acid molecule is provided which encodes one of the constructs or polypep-

tides described herein. These nucleic acid molecules can be further associated with regulatory sequences for expressing the constructs in vivo or in vitro.

In a further aspect, a pharmaceutical or cosmeceutical composition comprises as an active agent a polypeptide as described above, or a mixture of constructs as described above in a formulation suitable for delivery of the active agent into and through the layers of the skin. In one embodiment, the formulation contains a CAGE solvent (defined below) or other components suitable for topical administration.

In yet another aspect, a pharmaceutical or cosmeceutical composition comprises as an active agent a polypeptide as described above, or a mixture of constructs as described above in a formulation suitable for delivery to a selected cell or tissue.

In still other aspects, methods for making the compositions, constructs, polypeptides and nucleic acid molecules are provided.

In yet a further aspect, a method for treating or ameliorating or suppressing an inflammatory response comprises administering to a subject in need thereof a composition, construct, polypeptide or nucleic acid molecule described herein.

Still other aspects and advantages of these compositions and methods are described further in the following detailed description of the preferred embodiments thereof.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features and advantages of the present invention will be more fully disclosed in, or rendered obvious by, the following detailed description of the preferred embodiment of the invention, which is to be considered together with the accompanying drawings wherein like numbers refer to like parts and further wherein:

FIG. 1 is a schematic of bacterial effector constructs.

FIG. 2 is a schematic of bacterial effector constructs.

FIG. 3 is a schematic of the cloning strategy for TAT-NleE WT and Mutant R107A.

FIG. 4 is a schematic of the cloning strategy for TAT-Shigella OSPZ.

FIG. 5 is a schematic of the cloning strategy for YopM PTD-NleE WT and Mutant R107A.

FIG. 6 is a schematic of the cloning strategy for YopM PTD-NleE VVT No Linker, PAPA Linker, GSGS Linker and for Mutant R107A.

FIG. 7 is an exemplary SDS gel showing purified fusion proteins.

FIG. 8 is a graph showing the results of a dose-response analysis of IL-6 production in cells treated with recombinant purified effector proteins.

FIG. 9 is a graph showing the results of an analysis of the effect of paired fusion proteins on TNF-alpha.

FIG. 10 is a graph showing the results of an analysis of the effect of paired fusion proteins on IL-6.

FIG. 11 is a graph showing the results of an analysis of the effect of paired fusion proteins on MCP-1.

FIG. 12 is a graph showing the results of an analysis of the effect of paired fusion proteins on IL-23.

FIG. 13 is a graph showing the results of a dose response analysis of rYopM and YopMo on caspase 1 activity.

FIG. 14 shows the uptake of FITC-TAT-NleE-WT-His protein by Hacat cells.

FIG. 15 shows the uptake of FITC-YopM PTD-YopM (L-Rich)-GSGS Linker NleE-WT-His protein by Hacat cells.

FIG. 16 shows the uptake of FITC-YopM PTD-YopM (L-Rich)-PAPA Linker-NleE-WT-His protein by Hacat cells

FIG. 17 shows the uptake of FITC-YopM PTD-YopM (L-Rich)-GSGS Linker-NleE-WT-His protein by Hacat cells.

FIG. 18 shows the uptake of TAT-NleE fusion polypeptide into intact mouse skin.

FIG. 19 shows is a two-photon microscopy image of uptake of TAT-NleE fusion polypeptide into intact mouse skin.

FIG. 20 shows a two-photon microscopy image of a 10 micron slice of mouse skin showing uptake of TAT-NleE fusion polypeptide into intact mouse skin.

FIG. 21 shows graphs illustrating an analysis of methylase activity of YopM PTD-YopM (L-Rich)-PAPA/GSGS-NleE-WT-His fusion proteins.

FIG. 22 shows graphs illustrating an analysis of methyltransferase activity NleE-R107A mutant and YopM.

FIG. 23 shows graphs illustrating a NleE and *Shigella* Methyltransferase activity assay.

FIG. 24 depicts the results of an experiment comparing the effect of formulations on NleE methylation activity.

FIG. 25 shows four micrograph panels showing transdermal penetration of CAGE-NleE protein.

FIG. 26 is a schematic showing how inflammation develops from cell to tissue in the skin.

FIG. 27 is a schematic showing the impact of a skin irritant on the NFkB pathways.

FIG. 28 shows the amino acid sequence of NleE SEQ ID NO: 1.

FIG. 29 shows the publicly available UniProt P17778 amino acid sequence of YopM SEQ ID NO: 2.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

This description of preferred embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description of this invention. The drawing figures are not necessarily to scale and certain features of the invention may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness. In the description, relative terms such as "horizontal," "vertical," "up," "down," "top" and "bottom" as well as derivatives thereof (e.g., "horizontally," "downwardly," "upwardly," etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation. Terms including "inwardly" versus "outwardly," "longitudinal" versus "lateral" and the like are to be interpreted relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as "connected" and "interconnected," refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. The term "operatively connected" is such an attachment, coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship. When only a single machine is illustrated, the term "machine" shall also be

taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. In the claims, means-plus-function clauses, if used, are intended to cover the structures described, suggested, or rendered obvious by the written description or drawings for performing the recited function, including not only structural equivalents but also equivalent structures.

Technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs and by reference to published texts, which provide one skilled in the art with a general guide to many of the terms used in the present application. The definitions contained in this specification are provided for clarity in describing the components and compositions herein and are not intended to limit the claimed invention.

The present invention is based in part on the inventors' finding that combinations of bacterial effector polypeptides can have synergistic immunomodulatory activity. Many bacterial pathogens, including enteropathogenic *Escherichia coli* (EPEC), enterohemorrhagic *E. coli*, and *Shigella*, utilize a type III secretion system (T3SS) to deliver multiple virulence proteins directly into host cells. These virulence proteins, also referred to as effector proteins, are produced by bacteria during the infection of a eukaryotic host. The effector proteins down-regulate the host's immune system, typically at the site of the infection. Thus, effector proteins that target specific inflammatory pathways can function as immunomodulators, for example, in the treatment of inflammatory disorders. The inventors have found that combinations of effector proteins that target multiple inflammatory pathways produced augmented immunomodulatory effects.

Accordingly, the invention features compositions and methods that are useful for the treatment of inflammatory disorders. The compositions can include fusion proteins comprising bacterial effector polypeptides or a fragment of a bacterial effector polypeptide. More specifically, the compositions can include a set of paired peptides configured as a first bacterial effector polypeptide or a fragment thereof and a second bacterial effector polypeptides or a fragment thereof. For ease of reading, we will not repeat the phrase "or a fragment thereof" on every occasion. It is to be understood that where we refer to a first bacterial effector polypeptide, we refer to the first bacterial like and effector polypeptide or a fragment thereof. Similarly, it is to be understood that where we refer to a second bacterial effector polypeptide, we refer to the second bacterial effector polypeptide or a fragment thereof. Also featured are compositions comprising a set of paired peptides configured as a first bacterial effector polypeptide or a fragment thereof and a second bacterial effector polypeptides or a fragment thereof, and linked to a protein transduction domain. The methods can include a method of treating a subject at risk for or having an inflammatory disorder.

The compositions and methods described herein deliver a combinatorial construct of small recombinant protein effectors that directly target cellular pathways involved in inflammation, e.g., the NFkB pathway activated in skin inflammation. As described in detail below, the delivery of a recombinant construct comprising one or multiple effector proteins linked or fused to a PTD or CPP is useful to treat inflammation. Still other embodiments include the association of an optional targeting moiety, directing the construct to a specific cell or tissue type, with the effector protein and/or formulation in a topical carrier. While these compositions and methods of treatment have a number of advan-

tages, a significant advantage is that the protein effectors used are not able to enter into the circulation. In one embodiment, as discussed in detail below, the compositions and methods involve the use of NleE incorporated into a cream or oil formulation that reduces skin inflammation and may be widely used for many applications, both as a cosmetic beauty crème to reduce redness and irritation and as a treatment of a disease or cause of skin irritation.

The present compositions and methods using multiple e.g., NFkB/MAPK inhibitors, for treatment of inflammatory responses (e.g., pain, redness, swelling, heat) is based upon the fact that inflammation caused by infection, injury, autoimmunity, sunburn, aging, etc is detected at the cell membrane. Signals that are received by the skin cell are funneled thru the NFkB (sometimes MAPK) pathway. This is a relay mechanism in the cell that must be highly regulated. Shutting off the NFkB pathway is the key to controlling skin inflammation. Naturally occurring bacterial effector proteins (e.g., NleE, YopM, SSPH1) are very potent inhibitors of the NFkB system. Their sole target and purpose is to dampen the inflammatory response.

As described herein, when these bacterial proteins are recombinantly engineered into a selected polypeptide and delivered to inflamed skin via fusion with a PTD/CPP and optional targeting moiety, they abolish the inflammatory response. The combination of two or more effector proteins, each with a different substrate in the cell, may be combinatorially fused together in a single polypeptide. In some embodiments, such combination achieves a synergistic effect, which is a significant improvement in activity beyond that accomplished by delivery of a single effector.

The bacterial effector, NleE, is characterized by specificity, potency and efficiency in shutting down NFkB and hence inflammatory reactions. The presence of NleE and/or a combination with other bacterial effectors in an anti-inflammatory formulation has a number of advantages. Among the advantages are extremely high substrate specificity, which results in only a very low, if any, chance of off-target effects and toxicity for therapeutic use. In one embodiment, fusing PTDs to NleE and its related effectors is useful to deliver the effector to sites of inflammation. We and others have shown that a number of recombinant effector proteins, when fused to PTDs can indeed cross the cell and tissue boundary and be taken up by cells resulting in NFkB/JNK/p38 pathway inhibition.

The methods and compositions described below provide combinations of multiple effectors, or single or multiple effector(s) fused with a PTD, or single or multiple effector(s) fused with a targeting moiety, or single or multiple effector(s) fused with a PTD and a targeting moiety, in a chimeric recombinant protein, along with an emollient compound (such as CAGE) for use as a topical anti-inflammatory crème for many different ailments.

Compositions

Provided herein are compositions comprising engineered bacterial effector polypeptides for use in the treatment of inflammation. The engineered bacterial effector polypeptides can be configured as a set of paired peptides. More specifically, a set of paired peptides can be a construct comprising a first bacterial effector polypeptide or a fragment thereof and a second bacterial effector polypeptides or a fragment thereof. The set of paired peptides can be linked to one or more polypeptide sequences that facilitate intracellular delivery of the paired peptides, for example, a protein transduction domain (PTD) or a cell penetrating peptide (CPP).

Bacterial effector polypeptides. The first bacterial effector polypeptide and the second bacterial effector polypeptide can be a bacterial effector polypeptide selected from the exemplary bacterial effector polypeptides from a variety of bacteria as shown in Tables 1 and 2. Their enzymatic activity and host targets are also shown in Tables 1 and 2. Representative Uniprot or Genbank references for the polypeptides are shown in Tables 3 and 4. Additional amino acid sequences for, and nucleic acid sequences encoding, these bacterial effector polypeptides can be identified from databases such as UniProt, NCBI, GenBank and publications extant in the art.

TABLE 1

BACTERIAL T3SS EFFECTORS			
Effector	Bacteria	Intracellular Activity	Host Target
OspF	<i>Shigella flexneri</i>	Phosphothreonine lyase	ERK, p38 MAPKs
OspG	<i>Shigella flexneri</i>	Serine/threonine kinase	E2 ubiquitin ligases
NleH1	EPEC-EHEC	Serine/threonine kinase	RPS3
NleE/OspZ	EPEC-EHEC/ <i>Shigella</i>	Cysteine methylase	TAB2/NFkB
NleB	EPEC-EHEC	O-GlcNAc transferase	FADD, GAPDH, RIPK1, TRADD
NleC	EPEC-EHEC	Zinc metalloprotease	NFkB
YopH	<i>Yersinia</i>	Phosphotyrosine phosphatase	Akt/FAK
YopE	<i>Yersinia</i>	Rho GAP	Rho GTPases/caspases
YopP/YopJ	<i>Yersinia</i>	Acetyltransferase	MAPKs
YopM	<i>Yersinia</i>	LRR motif	PKN/RSK

TABLE 2

BACTERIAL T3SS E3 UBIQUITIN LIGASE EFFECTORS				
E3 Ligase Factor		Bacteria	Intracellular Activity	Host Target
Type/Family	Factor			
HECT	SopA	<i>Salmonella typhimurium</i>	Regulation of host inflammation	TRIM65/56
	NleL	EPEC/EHEC	Formation of actin pedestal	Unknown
RING U-Box	NleG	EPEC/EHEC	Unknown	Unknown
	LubX	<i>Legionella pneumophila</i>	Regulation of another effector function	Cdh1, SidH
NEL	Gob X	<i>L. pneumophila</i>	Unknown	Unknown
	IpaH1A	<i>Shigella flexneri</i>	Inhibition of NF-Kb activation	HOIP
	IpaH2.5	<i>S. flexneri</i>	Inhibition of NF-kB activation	HOIP
	IaH4.5	<i>S. flexneri</i>	Inhibition of NF-kB and I-IFN activation	p65, TBK1
	IpaH7.8	<i>S. flexneri</i>	Induction of pyroptosis	GLMN
	IpaH9.8	<i>S. flexneri</i>	Inhibition of NF-kB activation	NEMO
	IpaH0722	<i>S. flexneri</i>	Inhibition of NF-kB activation	TRAF2
	SspH1	<i>S. typhimurium</i>	Inhibition of androgen receptor	PKN1
	SspH2	<i>S. typhimurium</i>	Promotion of IL-8 secretion	Nod1, SGT1
	SlrP	<i>S. typhimurium</i>	Induction of host cell death	Trx
	SidE family	SidC (SdcA)	<i>L. pneumophila</i>	Unknown

TABLE 3

Representative Amino Acid Sequences of Bacterial T3SS Effectors			
Effector	Bacteria	Host Target	UniProt or Genbank Reference ID
OspF	<i>Shigella flexneri</i>	ERK, p38 MAPKs	Q8VSP9 (OSPF_SHIFL)
OspG	<i>Shigella flexneri</i>	E2 ubiquitin ligases	Q99PZ6 (OSPG_SHIFL)
NleH1	EPEC-EHEC	RPS3	Q8X831 (Q8X831_ECO57)
NleE/OspZ	EPEC-EHEC/ <i>Shigella</i>	TAB2/NfKB	Q7DBA6 (Q7DBA6_ECO57)
NleB	EPEC-EHEC	FADD, GAPDH, RIPK1, TRADD	VEC94465.1 (Genbank)
NleC	EPEC-EHEC	NfKB	CBG88408.1 (Genbank)
YopH	<i>Yersinia</i>	Akt/FAK	P15273 (YOPH_YEREN)
YopE	<i>Yersinia</i>	Rho	P31492 (YOPE_YEREN)
YopP/YopJ	<i>Yersinia</i>	GTPases/caspases MAPKs	Q34336 (YOPP_BACSU)
YopM	<i>Yersinia</i>	PKN/RSK	P17778 (YOPM_YERPE)

TABLE 4

Representative Amino Acid Sequences of Bacterial T3ss E3 Ubiquitin Ligase Effectors				
E3 Ligase Factor				
Type/Family	Factor	Bacteria	Host Target	UniProt or Genbank Reference ID
HECT	SopA	<i>Salmonella typhimurium</i>	TRIM65/56	Q8ZNR3 (SOPA_SALTY)
	NleL	EPEC/EHEC	Unknown	A0A0D6ZN92 (A0A0D6ZN92_ECOLX)
RING	NleG	EPEC/EHEC	Unknown	A0A023YUN6 (A0A023YUN6_ECOLX)
U-Box	LubX	<i>Legionella pneumophila</i>	Cdh1, SidH	Q5ZRQ0 (LUBX_LEGPH)
	Gob X	<i>pneumophila</i>	Unknown	
NEL	IpaH1.4	<i>Shigella flexneri</i>	HOIP	A0A380D0I4 (A0A380D0I4_SHIFL)
	IpaH2.5	<i>S. flexneri</i>	HOIP	Q99Q42 (Q99Q42_SHIFL)
	IaH4.5	<i>S. flexneri</i>	p65, TBK1	P18009 (IPA4_SHIFL)
	IpaH7.8	<i>S. flexneri</i>	GLMN	P18014 (IPA7_SHIFL)
	IpaH9.8	<i>S. flexneri</i>	NEMO	Q8VSC3 (IPA9_SHIFL)
	IpaH0722	<i>S. flexneri</i>	TRAF2	D0ZVG2
	SspH1	<i>S. typhimurium</i>	PKN1	(SSPH1_SALT1)
	SspH2	<i>S. typhimurium</i>	Nod1, SGT1	P0CE12 (SSPH2_SALTY)
	SlrP	<i>S. typhimurium</i>	Trx	Q8ZQQ2 (SLRP_SALTY)
SidE family	SidC (SdcA)	<i>L. pneumophila</i>	Unknown	Q6RCR3 (Q6RCR3_LEGPN)

Thus a bacterial effector peptide can be an SspH1; SspH2; SlrP, IpaH1.4, IpaH2.5, IpaH3, IpaH4.5, IpaH7.8, IpaH9.8, NleE, NleC; NleD; NleB; NleH, NleH1, YopM; YopE; YopH; YopJ; YopP; OspG; OspF; OspZ; OspL, SopE; SopB; SopE2; SipA; AvrA; SseL; EspT; or a TiR polypeptide.

In some embodiments, the bacterial effector polypeptide can have an amino acid sequence at least 90% identical to an amino acid sequence set forth in any of SEQ ID NOs. 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79. In some embodiments, the bacterial

effector polypeptide have an amino acid sequence as set forth in any of SEQ ID NOs. 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79.

The term "type III secretion system or T3SS" refers to a highly specialized molecular needle construct containing a *Yersinia* injectisome spanning the bacterial membranes, *Yersinia* outer protein (Yop) effectors and Yop translocators needed to deliver the effectors across the membrane (Camelis G R., Int J Med Microbial. 2002 February; 291(6-7): 455-62). Pathogenic *Yersinia* require this T3SS to survive and replicate extracellularly within lymphoid tissues of their animal or human hosts. See, also, U.S. Pat. No. 8,840,901.

The term "immunomodulatory effector protein" refers to small proteins, generally bacterial in origin, that suppress the human innate immune system during infection. These effector proteins activate the NFkB, JNK, and p38 signaling pathways during infection. These effector proteins are often secreted into the cells targeted for infection by T3SS. Once inside the cell, each effector protein targets a single host protein required for innate immunity, which it inactivates using a myriad of mechanisms including acetylation, methylation, action of phosphatases on P04 proteins, induced protein degradation etc.

All T3SS bacterial effector proteins are very small, globular, highly stable, highly catalytic, have high substrate specificity, bind co-factors very tightly, and can be injected into the cell in denatured form. They modify every substrate molecule in the cell. For instance, the EPEC effector NleE is a cysteine methyltransferase which has a single target in the cell, i.e., the TAB2 scaffold protein in the NFkB pathway. EPEC are mildly infective gut bacteria that attach to colon cells and directly inject virulence proteins thru a T3SS to control innate immune pathways as a survival strategy. During EPEC infection, the injected NleE protein methylates every molecule of cellular TAB2; thereby completely shutting off NFkB signaling, as described in Yao, Q. et al., Structure and Specificity of the Bacterial Cysteine Methyltransferase Effector NleE Suggests a Novel Substrate in Human DNA Repair Pathway., PLoS Pathogens (November 2014) 10(11):e1004522; doi:10.1371/journal.ppat.1004522.

Still another effector protein is YopM, discussed in U.S. Pat. No. 8,840,901 and Rüter, C & Hardwidge, P R, 'Drugs from Bugs': bacterial effector proteins as promising biological (immune-) therapeutics. FEMS Microbiol Lett 351 (December 2013/January 2014) 126-132.

In some embodiments, the first and second bacterial effector polypeptides are different. For example, they can have different amino acid sequences, different structures, different functions, different molecular targets, or have non-overlapping redundant roles in inhibiting an inflammatory pathway, for example, the NFkB, JNK, p38, and STING pathways.

Also included as effectors or effector proteins of the compositions and methods described herein are functional equivalents of the proteins described above. By the term "functional equivalent" is meant any amino acid sequence or modification thereof that has the same targeting and immune suppressing function of the naturally occurring effector protein. In one embodiment, such functional equivalents can have modifications of one or more amino acids from the known sequences. In one embodiment, such functional equivalents can be a smaller fragment of the known sequences. In one embodiment, such functional equivalents can be a derivative of the naturally occurring sequences or be derived from other than human sources. In one embodiment, such functional equivalents can be altered by chemical modification or be altered by recombinant production to be

associated with sequences with which the effector proteins are not associated in nature. Similarly, chemical or structural changes or fragments of the nucleic acid sequences that encode the effector proteins are also considered functional equivalents herein.

The paired peptides can be joined by a linker. A linker can be any reagent, molecule or macromolecule that connects the first and second bacterial effector polypeptides such that the linker does not substantially alter the physiological activity of the effector polypeptides. A linker can be a peptide bond. That is, the first and second bacterial effector polypeptides or fragments thereof can be a fusion polypeptide comprising one or more amino acid segments from the first bacterial effector polypeptide and one or more amino acid segments from second bacterial effector polypeptide. The term "amino acid segment" as used herein refers to a contiguous stretch of amino acids within a polypeptide. For example, the amino acid residues 30 to 40 within a 100 amino acid polypeptide would be considered an amino acid segment. An amino acid segment can be a length greater than eight amino acid residues (e.g., greater than about nine, ten, 15, 20, 25, 30, 40, 50, 75, 100, 150, 200, 500, 1000, or more amino acid residues). In some embodiments, an amino acid segment can have a length less than 1000 amino acid residues (e.g., less than 500, less than 400, less than 350, less than 300, less than 200, or less than 100 amino acid residues). In other embodiments, an amino acid segment can have a length from about 20 to about 200 amino acid residues (e.g., about 30 to about 180 amino acid residues, or about 40 to about 150 amino acid residues).

The amino acid segments of the first bacterial effector polypeptide can be contiguous with the amino acid segments of the second or they can be separated by amino acids inserted as a structural spacer. A spacer segment can be one or more amino acids. The one or more amino acids can include amino acids that are the same or that are different. For example, a spacer can be a repeating series of a neutral amino acid (e.g., glycine, alanine, valine, isoleucine or leucine) ranging in number from 1 to 10 or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more). Another example of a spacer configuration can be a series of interspersed amino acids that may be neutral (e.g., glycine-alanine-glycine-alanine-glycine-alanine, or glycine-glycine-glycine-valine-valine-valine) or charged amino acids (e.g., glutamate-glutamate-arginine-arginine-arginine, or aspartate-lysine-aspartate-lysine-aspartate-lysine) or amino acids with other functional groups (e.g., proline-proline-proline-serine-serine-serine or tyrosine-glutamine-cysteine-methionine-tryptophan) ranging in number from 1 to 10 or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more). In another embodiment, a spacer configuration can be a sequence of amino acids derived from a naturally occurring protein such as the hinge region joining the heavy chain CH1 and CH2 domains of immunoglobulin G. In some embodiments, the linker can be a GSGS linker (SEQ ID NO. 92) or a PAPA linker (SEQ ID NO. 91). In some embodiments, the fusion protein comprising a set of paired peptides can exclude a linker.

A fusion protein can be produced in vitro by continuous peptide synthesis according to standard chemical methods known to those in the art. Synthetic polypeptides can also be purchased from commercial sources. A fusion protein can also be produced by recombinant DNA techniques. Nucleic acid segments encoding the first bacterial effector polypeptide can be operably linked in the same open reading frame to nucleic acid sequences encoding the second bacterial effector polypeptide in a vector that includes the requisite regulatory elements, e.g., promoter sequences, transcription

initiation sequences, and enhancer sequences, for expression in prokaryotic or eukaryotic cells.

The paired peptide constructs can include a combination of any of an SspH1; SspH2; SlrP; IpaH1.4; IpaH2.5; IpaH3; IpaH4.5; IpaH7.8; IpaH9.8, NleE, NleC; NleD; NleB; NleH, NleH1; YopM; YopE; YopH; YopJ; YopP; OspG; OspF; OspZ; OspI; SopE; SopB; SopE2; SipA; AvrA; SseL; EspT; or a Tir polypeptide. Thus, the first bacterial effector polypeptide can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP, and the second bacterial effector polypeptide can be a polypeptide selected from the group consisting of NleE, NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP.

In some embodiments, the paired peptide construct can include a first bacterial effector polypeptide or fragment thereof is a YopM polypeptide or a fragment thereof or an NleE polypeptide or a fragment thereof and a second bacterial effector polypeptide or fragment thereof is a YopM polypeptide or a fragment thereof or an NleE polypeptide or a fragment thereof. In some embodiments, the first bacterial effector polypeptide or fragment thereof is a YopM polypeptide or a fragment thereof and the second bacterial effector polypeptide or fragment thereof is an NleE polypeptide or a fragment thereof.

In some embodiments, the paired peptide constructs can be configured as summarized in Table 5 below.

TABLE 5

Paired Peptide Fusion Constructs				
Construct name	Effector 1	Linker	Effector 2	SEQ ID NO
YopM PTD-NleE (No linker)	YopM (L-rich)	—	EPEC NleE	80
YopM PTD-NleE (GSGS linker)	YopM (L-rich)	GSGS	EPEC NleE	82
YopM PTD-NleE (PAPA linker)	YopM (L-rich)	PAPA	EPEC NleE	81

In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 85% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 90% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 95% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 99% identical to the sequence set forth in SEQ ID NO. 80, 81, 82, 83, 84, 85, 86, 87, or 88. In some embodiments, the paired peptide fusion protein can have an amino acid sequence as set forth in SEQ ID NOs. 80, 81, 82, 83, 84, 85, 86, 87, or 88.

Protein transduction domains. The set of paired peptides can be linked to one or more polypeptide sequences that facilitate intracellular delivery of the paired peptide. The terms "protein transduction domain (PTDs)" and/or "cell-penetrating peptide (CPPs)" refers to powerful sequences that allow intracellular delivery of conjugated cargoes to

modify cell behavior. These small peptides can transport a wide variety of biologically active conjugates into the cell. Heterologous CPP coding sequences are added to effectors or effector-fusions to facilitate cellular uptake of the proteins into cells and tissues, including use of endogenous CPPs encoded in native effector proteins. This includes addition of CPP sequences or modules to the effector via chemical crosslinking, attachment to a nano-particle or other scaffold chemically or via PPIs for the purpose of transporting the effector across tissue and cell membranes. Among useful PTD or CPPs for the present methods and compositions are those known and identified in the art, including, without limitation, HIV Tat protein basic domain, (HIV Tat amino acids 48-60 or 49-57), poly-Arg or polyLys, penetratin, MPG, Pep-1, MAP, and transportan. See, e.g., Table 1 of Guidotti, G. et al, Trends in Pharmacological Sciences (April 2017), 38(4):406-424, which includes additional examples of CPPs and sequences origins and properties. Other CPPs are described in Norkowski, S. et al, Bacterial LPX motif-harboring virulence factors constitute a species-spanning family of cell-penetrating effectors, Cellular and Molecular Life Sciences (December 2017) doi.org/10.1007/s00018-017-2733-4, which described prototypes of such bacteria-derived cell-penetrating effectors (CPEs) including the *Yersinia enterocolitica*-derived YopM, the *Salmonella typhimurium* effector SspH1, and the *Shigella* IpaH proteins. Still other protein transporter molecules include those previously described in Dixon, J E et al, Proc. Natl Acad Sci, (January 2016), E291-299; as well as synthetic protein mimics described by Tezgel, A O et al BioMacromolecules (2017) 16:819-825. See, also, Bolhassani, A. et al, In vitro and in vivo delivery of therapeutic proteins using cell penetrating peptides. Peptides (November 2016), 87:50-63, which discusses useful CPPs for the present compositions and methods, including without limitation, covalent bonded CPPs, such as Poly-Arg peptides, Tat and VP22, df Tat, Cyclic CPPs, IMT-P8 (particularly useful for transdermal delivery), seven arginine (R7) and Streptolysin O (SLO)-mediated systems and elastin like polypeptide, CPP-adaptor system, 1,2-Benzisothiazolin-3-one (BIT) and Tat, activatable cell-penetrating peptides, LDP12, M918, BR2, peptide for ocular delivery (POD), native protein independent of R11-CPP, Poly-arginine/Tat and Tat-PTD among others. Also identified are non-covalent bonded CPPs such as Pep-1, CADY-2, R8 and azo-R8, Penetratin, HR9 and IR9 peptides and pVEC. All of these documents are incorporated by reference herein for detailed descriptions of known CPPs and PTDs. It is also anticipated that novel PTD/CPPs will prove useful with the compositions described herein.

Exemplary protein transduction domains include a YopM protein transduction domain, an SspH1 protein transduction domain, or an IpaH protein transduction domain. A useful YopM protein transduction domain can have an amino acid sequence as set forth in SEQ ID NO 5.

In some embodiments, a fusion protein comprising a set of paired peptides linked to a protein transduction domain can have an amino acid sequence is at least 85% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 90% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 95% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24. In some embodiments, paired peptide fusion protein can have an amino acid sequence is at least 99% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24.

In some embodiments, the paired peptide fusion protein can have an amino acid sequence as set forth in SEQ ID NOS. 10, 13, 16, 19, 22, or 24.

Polypeptides. We tend to use the term “protein” to refer to longer or larger amino acid polymers, and we tend to use the term “polypeptide” to refer to shorter sequences or to a chain of amino acid residues within a larger molecule (e.g., within a fusion protein) or complex. Both terms, however, are meant to describe an entity of two or more subunit amino acids, amino acid analogs, or other peptidomimetics, regardless of post-translational modification (e.g., amidation, phosphorylation or glycosylation). The subunits can be linked by peptide bonds or other bonds such as, for example, dicysteine, ester or ether bonds. The terms “amino acid” and “amino acid residue” refer to natural and/or unnatural or synthetic amino acids, which may be D- or L-form optical isomers. Full-length proteins, analogs, mutants, and fragments thereof are encompassed by this definition.

The amino acid sequence of the bacterial effector polypeptides disclosed herein can be identical to the wild-type sequences of appropriate components. Alternatively, any of the components can contain mutations such as deletions, additions, or substitutions. All that is required is that the variant bacterial effector polypeptide have at least 5% (e.g., 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 99%, 100%, or even more) of the ability of the bacterial effector polypeptide containing only wild-type sequences to specifically bind the target. Substitutions will preferably be conservative substitutions. Conservative substitutions typically include substitutions within the following groups: glycine and alanine; valine, isoleucine, and leucine; aspartic acid and glutamic acid; asparagine, glutamine, serine and threonine; lysine, histidine and arginine; and phenylalanine and tyrosine.

Variant bacterial effector polypeptides, e.g., those having one or more amino acid substitutions relative to a native bacterial effector polypeptide amino acid sequence, can be prepared and modified as described herein. Amino acid substitutions can be made, in some cases, by selecting substitutions that do not differ significantly in their effect on maintaining (a) the structure of the peptide backbone in the area of the substitution, (b) the charge or hydrophobicity of the molecule at the target site, or (c) the bulk of the side chain. For example, naturally occurring residues can be divided into groups based on side-chain properties: (1) hydrophobic amino acids (norleucine, methionine, alanine, valine, leucine, and isoleucine); (2) neutral hydrophilic amino acids (cysteine, serine, and threonine); (3) acidic amino acids (aspartic acid and glutamic acid); (4) basic amino acids (asparagine, glutamine, histidine, lysine, and arginine); (5) amino acids that influence chain orientation (glycine and proline); and (6) aromatic amino acids (tryptophan, tyrosine, and phenylalanine). Substitutions made within these groups can be considered conservative substitutions. Non-limiting examples of useful substitutions include, without limitation, substitution of valine for alanine, lysine for arginine, glutamine for asparagine, glutamic acid for aspartic acid, serine for cysteine, asparagine for glutamine, aspartic acid for glutamic acid, proline for glycine, arginine for histidine, leucine for isoleucine, isoleucine for leucine, arginine for lysine, leucine for methionine, leucine for phenylalanine, glycine for proline, threonine for serine, serine for threonine, tyrosine for tryptophan, phenylalanine for tyrosine, and/or leucine for valine. Variant bacterial effector polypeptides having conservative and/or non-conservative substitutions (e.g., with respect to any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64,

66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24), as well as fragments of any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, fragments of variants of any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88 and polypeptides comprising any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, variants or fragments of any of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, can be screened for biological activity using suitable assays, including those described herein. For example, the activity of a bacterial effector polypeptide, for example, NLeE or a mutant or fragment thereof, can be evaluated *in vitro* by assaying for methylase activity or in cell based systems to characterize its effect on cytokine release.

In some embodiments, a bacterial effector polypeptide can comprise an amino acid sequence as set forth in SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, but with a particular number of amino acid substitutions. For example, a bacterial effector polypeptides can have the amino acid sequence of any one of SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88, but with one, two, three, four, or five amino acid substitutions.

In some embodiments, a bacterial effector polypeptide as provided herein can include an amino acid sequence with at least 85% (e.g., 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 97.5%, 98%, 98.5%, 99.0%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9%, or 100%) sequence identity to a region of a reference bacterial effector polypeptide sequence (e.g., SEQ ID NOS: 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 79, 10, 13, 16, 19, 22, or 24, 80, 81, 82, 83, 84, 85, 86, 87, or 88). Methods of determining percent sequence identity are discussed below.

In some embodiments, a polypeptide provided herein can be a substantially pure polypeptide. As used herein, the term "substantially pure" with reference to a polypeptide means that the polypeptide is substantially free of other polypeptides, lipids, carbohydrates, and nucleic acid with which it is naturally associated. Thus, a substantially pure polypeptide is any polypeptide that is removed from its natural environment and is at least 60 percent pure or is any chemically synthesized polypeptide. A substantially pure polypeptide can be at least about 60, 65, 70, 75, 80, 85, 90, 95, or 99 percent pure. Typically, a substantially pure polypeptide will yield a single major band on a non-reducing polyacrylamide gel.

A variety of methods can be used to make a polypeptide including, for example, expression by prokaryotic systems, expression by eukaryotic systems, and chemical synthesis techniques. Exemplary methods for polypeptide purification purificinclud, without limitation, fractionation, centrifugation, and chromatography, e.g., gel filtration, ion exchange chromatography, reverse-phase HPLC and immunoaffinity purification.

A polypeptide can be modified by linkage to a polymer such as polyethylene glycol (PEG), or by fusion to another polypeptide such as albumin, for example. For example, one or more PEG moieties can be conjugated to a bacterial effector polypeptide or fusion protein via lysine residues. Linkage to PEG or another suitable polymer, or fusion to albumin or another suitable polypeptide can result in a modified bacterial effector polypeptide or fusion protein having an increased half life as compared to an unmodified bacterial effector polypeptide or fusion protein. Without being bound by a particular mechanism, an increased serum half life can result from reduced proteolytic degradation, immune recognition, or cell scavenging of the modified bacterial effector polypeptide or fusion protein. Methods for modifying a polypeptide by linkage to PEG (also referred to as "PEGylation") or other polymers include those set forth in U.S. Pat. No. 6,884,780; Cataliotti et al. ((2007) Trends Cardiovasc. Med. 17:10-14; Veronese and Mero (2008) BioDrugs 22:315-329; Miller et al. (2006) Bioconjugate Chem. 17:267-274; and Veronese and Pasut (2005) Drug Discov. Today 10:1451-1458, all of which are incorporated herein by reference in their entirety. Methods for modifying a polypeptide by fusion to albumin include those set forth in U.S. Patent Publication No. 20040086976, and Wang et al. (2004) Pharm. Res. 21:2105-2111, both of which are incorporated herein by reference in their entirety.

Nucleic acids. We may use the terms "nucleic acid" and "polynucleotide" interchangeably to refer to both RNA and DNA, including cDNA, genomic DNA, synthetic DNA, and DNA (or RNA) containing nucleic acid analogs, any of which may encode a polypeptide of the invention and all of which are encompassed by the invention. Polynucleotides can have essentially any three-dimensional structure. A nucleic acid can be double-stranded or single-stranded (i.e., a sense strand or an antisense strand). Non-limiting examples of polynucleotides include genes, gene fragments, exons, introns, messenger RNA (mRNA) and portions thereof, transfer RNA, ribosomal RNA, siRNA, micro-RNA, ribozymes, cDNA, recombinant polynucleotides, branched polynucleotides, plasmids, vectors, isolated DNA of any sequence, isolated RNA of any sequence, nucleic acid probes, and primers, as well as nucleic acid analogs. In the context of the present invention, nucleic acids can encode a bacterial effector polypeptide, paired peptide fusion protein, or construct comprising a paired peptide fusion protein linked to a protein transduction domain.

An "isolated" nucleic acid can be, for example, a naturally-occurring DNA molecule or a fragment thereof, provided that at least one of the nucleic acid sequences normally found immediately flanking that DNA molecule in a naturally-occurring genome is removed or absent. Thus, an isolated nucleic acid includes, without limitation, a DNA molecule that exists as a separate molecule, independent of other sequences (e.g., a chemically synthesized nucleic acid, or a cDNA or genomic DNA fragment produced by the polymerase chain reaction (PCR) or restriction endonuclease treatment). An isolated nucleic acid also refers to a DNA molecule that is incorporated into a vector, an autonomously replicating plasmid, a virus, or into the genomic DNA of a prokaryote or eukaryote. In addition, an isolated nucleic acid can include an engineered nucleic acid such as a DNA molecule that is part of a hybrid or fusion nucleic acid. A nucleic acid existing among many (e.g., dozens, or hundreds to millions) of other nucleic acids within, for example, cDNA libraries or genomic libraries, or gel slices containing a genomic DNA restriction digest, is not an isolated nucleic acid.

Isolated nucleic acid molecules can be produced by in several ways. For example, polymerase chain reaction (PCR) techniques can be used to obtain an isolated nucleic acid containing a nucleotide sequence described herein, including nucleotide sequences encoding a polypeptide described herein. PCR can be used to amplify specific sequences from DNA as well as RNA, including sequences from total genomic DNA or total cellular RNA. Generally, sequence information from the ends of the region of interest or beyond is employed to design oligonucleotide primers that are identical or similar in sequence to opposite strands of the template to be amplified. Various PCR strategies also are available by which site-specific nucleotide sequence modifications can be introduced into a template nucleic acid.

Isolated nucleic acids also can be chemically synthesized, either as a single nucleic acid molecule (e.g., using automated DNA synthesis in the 3' to 5' direction using phosphoramidite technology) or as a series of oligonucleotides. For example, one or more pairs of long oligonucleotides (e.g., >50-100 nucleotides) can be synthesized that contain the desired sequence, with each pair containing a short segment of complementarity (e.g., about 15 nucleotides) such that a duplex is formed when the oligonucleotide pair is annealed. DNA polymerase is used to extend the oligonucleotides, resulting in a single, double-stranded nucleic acid molecule per oligonucleotide pair, which then can be ligated into a vector.

Two nucleic acids or the polypeptides they encode may be described as having a certain degree of identity to one another. For example, a bacterial effector polypeptide and a biologically active variant thereof may be described as exhibiting a certain degree of identity. Alignments may be assembled by locating short sequences in the Protein Information Research (PIR) site (<http://pir.georgetown.edu>), followed by analysis with the "short nearly identical sequences" Basic Local Alignment Search Tool (BLAST) algorithm on the NCBI website (<http://www.ncbi.nlm.nih.gov/blast>).

As used herein, the term "percent sequence identity" refers to the degree of identity between any given query sequence and a subject sequence. For example, a bacterial effector polypeptide disclosed herein can be the query sequence and a fragment of a bacterial effector polypeptide can be the subject sequence. Similarly, a fragment of bacterial effector polypeptide can be the query sequence and a biologically active variant thereof can be the subject sequence.

To determine sequence identity, a query nucleic acid or amino acid sequence can be aligned to one or more subject nucleic acid or amino acid sequences, respectively, using the computer program ClustalW (version 1.83, default parameters), which allows alignments of nucleic acid or protein sequences to be carried out across their entire length (global alignment).

ClustalW calculates the best match between a query and one or more subject sequences and aligns them so that identities, similarities and differences can be determined. Gaps of one or more residues can be inserted into a query sequence, a subject sequence, or both, to maximize sequence alignments. For fast pair wise alignment of nucleic acid sequences, the following default parameters are used: word size: 2; window size: 4; scoring method: percentage; number of top diagonals: 4; and gap penalty: 5. For multiple alignments of nucleic acid sequences, the following parameters are used: gap opening penalty: 10.0; gap extension penalty: 5.0; and weight transitions: yes. For fast pair wise alignment of protein sequences, the following parameters are used:

word size: 1; window size: 5; scoring method: percentage; number of top diagonals: 5; gap penalty: 3. For multiple alignment of protein sequences, the following parameters are used: weight matrix: blosum; gap opening penalty: 10.0; gap extension penalty: 0.05; hydrophilic gaps: on; hydrophilic residues: Gly, Pro, Ser, Asn, Asp, Gln, Glu, Arg, and Lys; residue-specific gap penalties: on. The output is a sequence alignment that reflects the relationship between sequences. ClustalW can be run, for example, at the Baylor College of Medicine Search Launcher site (searchlauncher.bcm.tmc.edu/multi-align/multi-align.html) and at the European Bioinformatics Institute site on the World Wide Web (ebi.ac.uk/clustalw).

To determine a percent identity between a query sequence and a subject sequence, ClustalW divides the number of identities in the best alignment by the number of residues compared (gap positions are excluded), and multiplies the result by 100. The output is the percent identity of the subject sequence with respect to the query sequence. It is noted that the percent identity value can be rounded to the nearest tenth. For example, 78.11, 78.12, 78.13, and 78.14 are rounded down to 78.1, while 78.15, 78.16, 78.17, 78.18, and 78.19 are rounded up to 78.2.

The nucleic acids and polypeptides described herein may be referred to as "exogenous". The term "exogenous" indicates that the nucleic acid or polypeptide is part of, or encoded by, a recombinant nucleic acid construct, or is not in its natural environment. For example, an exogenous nucleic acid can be a sequence from one species introduced into another species, i.e., a heterologous nucleic acid. Typically, such an exogenous nucleic acid is introduced into the other species via a recombinant nucleic acid construct. An exogenous nucleic acid can also be a sequence that is native to an organism and that has been reintroduced into cells of that organism. An exogenous nucleic acid that includes a native sequence can often be distinguished from the naturally occurring sequence by the presence of non-natural sequences linked to the exogenous nucleic acid, e.g., non-native regulatory sequences flanking a native sequence in a recombinant nucleic acid construct. In addition, stably transformed exogenous nucleic acids typically are integrated at positions other than the position where the native sequence is found.

Nucleic acids of the invention, that is, nucleic acids having a nucleotide sequence of any paired peptides fusion proteins and constructs disclosed herein, can include nucleic acids sequences that are at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 99% identical to the reference sequences disclosed herein.

A nucleic acid, i.e., an oligonucleotide (e.g., a probe or a primer) that is specific for a target nucleic acid will hybridize to the target nucleic acid under suitable conditions. We may refer to hybridization or hybridizing as the process by which an oligonucleotide single strand anneals with a complementary strand through base pairing under defined hybridization conditions. It is a specific, i.e., non-random, interaction between two complementary polynucleotides. Hybridization and the strength of hybridization (i.e., the strength of the association between the nucleic acids) is influenced by such factors as the degree of complementarity between the nucleic acids, stringency of the conditions involved, and the melting temperature (T_m) of the formed hybrid. The hybridization products can be duplexes or triplexes formed with targets in solution or on solid supports.

Vectors. Vectors containing nucleic acids such as those described herein also are provided. A "vector" is a replicon,

such as a plasmid, phage, or cosmid, into which another DNA segment may be inserted so as to bring about the replication of the inserted segment. Generally, a vector is capable of replication when associated with the proper control elements. Suitable vector backbones include, for example, those routinely used in the art such as plasmids, viruses, artificial chromosomes, BACs, YACs, or PACs. The term "vector" includes cloning and expression vectors, as well as viral vectors and integrating vectors. An "expression vector" is a vector that includes a regulatory region. A wide variety of host/expression vector combinations may be used to express the nucleic acid sequences described herein. Suitable expression vectors include, without limitation, plasmids and viral vectors derived from, for example, bacteriophage, baculoviruses, and retroviruses. Numerous vectors and expression systems are commercially available from such corporations as Novagen (Madison, Wis.), Clontech (Palo Alto, Calif.), Stratagene (La Jolla, Calif.), and Invitrogen/Life Technologies (Carlsbad, Calif.).

The vectors provided herein also can include, for example, origins of replication, scaffold attachment regions (SARs), and/or markers. A marker gene can confer a selectable phenotype on a host cell. For example, a marker can confer biocide resistance, such as resistance to an antibiotic (e.g., kanamycin, G418, bleomycin, or hygromycin). As noted above, an expression vector can include a tag sequence designed to facilitate manipulation or detection (e.g., purification or localization) of the expressed polypeptide. Tag sequences, such as green fluorescent protein (GFP), glutathione S-transferase (GST), polyhistidine, c-myc, hemagglutinin, or Flag™ tag sequences typically are expressed as a fusion with the encoded polypeptide. Such tags can be inserted anywhere within the polypeptide, including at either the carboxyl or amino terminus.

Additional expression vectors also can include, for example, segments of chromosomal, non-chromosomal and synthetic DNA sequences. Suitable vectors include derivatives of SV40 and known bacterial plasmids, e.g., *E. coli* plasmids col E1, pCR1, pBR322, pMal-C2, pET, pGEX, pMB9 and their derivatives, plasmids such as RP4; phage DNAs, e.g., the numerous derivatives of phage 1, e.g., NM989, and other phage DNA, e.g., M13 and filamentous single stranded phage DNA; yeast plasmids such as the 2p plasmid or derivatives thereof, vectors useful in eukaryotic cells, such as vectors useful in insect or mammalian cells; vectors derived from combinations of plasmids and phage DNAs, such as plasmids that have been modified to employ phage DNA or other expression control sequences.

Yeast expression systems can also be used. For example, the non-fusion pYES2 vector (XbaI, SphI, ShoI, NotI, GstXI, EcoRI, BstXI, BamHI, SacI, KpnI, and HindIII cloning sites; Invitrogen) or the fusion pYESHisA, B, C (XbaI, SphI, ShoI, NotI, BstXI, EcoRI, BamHI, SacI, KpnI, and HindIII cloning sites).

The vector can also include a regulatory region. The term "regulatory region" refers to nucleotide sequences that influence transcription or translation initiation and rate, and stability and/or mobility of a transcription or translation product. Regulatory regions include, without limitation, promoter sequences, enhancer sequences, response elements, protein recognition sites, inducible elements, protein binding sequences, 5' and 3' untranslated regions (UTRs), transcriptional start sites, termination sequences, polyadenylation sequences, nuclear localization signals, and introns.

As used herein, the term "operably linked" refers to positioning of a regulatory region and a sequence to be transcribed in a nucleic acid so as to influence transcription

or translation of such a sequence. For example, to bring a coding sequence under the control of a promoter, the translation initiation site of the translational reading frame of the polypeptide is typically positioned between one and about fifty nucleotides downstream of the promoter. A promoter can, however, be positioned as much as about 5,000 nucleotides upstream of the translation initiation site or about 2,000 nucleotides upstream of the transcription start site. A promoter typically comprises at least a core (basal) promoter. A promoter also may include at least one control element, such as an enhancer sequence, an upstream element or an upstream activation region (UAR). The choice of promoters to be included depends upon several factors, including, but not limited to, efficiency, selectability, inducibility, desired expression level, and cell- or tissue-preferential expression.

Vectors include, for example, viral vectors (such as adenoviruses ("Ad"), adeno-associated viruses (AAV), and vesicular stomatitis virus (VSV) and retroviruses), liposomes and other lipid-containing complexes, and other macromolecular complexes capable of mediating delivery of a polynucleotide to a host cell. Vectors can also comprise other components or functionalities that further modulate gene delivery and/or gene expression, or that otherwise provide beneficial properties to the targeted cells.

A "recombinant viral vector" refers to a viral vector comprising one or more heterologous gene products or sequences. Since many viral vectors exhibit size-constraints associated with packaging, the heterologous gene products or sequences are typically introduced by replacing one or more portions of the viral genome. Such viruses may become replication-defective, requiring the deleted function(s) to be provided in trans during viral replication and encapsidation (by using, e.g., a helper virus or a packaging cell line carrying gene products necessary for replication and/or encapsidation).

Suitable nucleic acid delivery systems include recombinant viral vector, typically sequence from at least one of an adenovirus, adenovirus-associated virus (AAV), helper-dependent adenovirus, retrovirus, or hemagglutinating virus of Japan-liposome (HVJ) complex. In such cases, the viral vector comprises a strong eukaryotic promoter operably linked to the polynucleotide e.g., a cytomegalovirus (CMV) promoter. The recombinant viral vector can include one or more of the polynucleotides therein, preferably about one polynucleotide. In some embodiments, the viral vector used in the invention methods has a pfu (plaque forming units) of from about 10^8 to about 5×10^{10} pfu. In embodiments in which the polynucleotide is to be administered with a non-viral vector, use of between from about 0.1 nanograms to about 4000 micrograms will often be useful e.g., about 1 nanogram to about 100 micrograms.

Additional vectors include retroviral vectors such as Moloney murine leukemia viruses and HIV-based viruses. One HIV-based viral vector comprises at least two vectors wherein the gag and pol genes are from an HIV genome and the env gene is from another virus. DNA viral vectors include pox vectors such as orthopox or avipox vectors, herpesvirus vectors such as a herpes simplex I virus (HSV) vector.

Pox viral vectors introduce the gene into the cells cytoplasm. Avipox virus vectors result in only a short term expression of the nucleic acid. Adenovirus vectors, adeno-associated virus vectors and herpes simplex virus (HSV) vectors may be an indication for some invention embodiments. The adenovirus vector results in a shorter term expression (e.g., less than about a month) than adeno-associated virus, in some embodiments, may exhibit much

longer expression. The particular vector chosen will depend upon the target cell and the condition being treated. The selection of appropriate promoters can readily be accomplished. An example of a suitable promoter is the 763-base-pair cytomegalovirus (CMV) promoter. Other suitable promoters which may be used for gene expression include, but are not limited to, the Rous sarcoma virus (RSV), the SV40 early promoter region, the herpes thymidine kinase promoter, the regulatory sequences of the metallothionein (MTT) gene, prokaryotic expression vectors such as the β -lactamase promoter, the tac promoter, promoter elements from yeast or other fungi such as the Gal 4 promoter, the ADC (alcohol dehydrogenase) promoter, PGK (phosphoglycerol kinase) promoter, alkaline phosphatase promoter; and the animal transcriptional control regions, which exhibit tissue specificity and have been utilized in transgenic animals: elastase I gene control region which is active in pancreatic acinar cells, insulin gene control region which is active in pancreatic beta cells, immunoglobulin gene control region which is active in lymphoid cells, mouse mammary tumor virus control region which is active in testicular, breast, lymphoid and mast cells, albumin gene control region which is active in liver, alpha-fetoprotein gene control region which is active in liver, alpha 1-antitrypsin gene control region which is active in the liver, beta-globin gene control region which is active in myeloid cells, myelin basic protein gene control region which is active in oligodendrocyte cells in the brain, myosin light chain-2 gene control region which is active in skeletal muscle, and gonadotropic releasing hormone gene control region which is active in the hypothalamus. Certain proteins can be expressed using their native promoter. Other elements that can enhance expression can also be included such as an enhancer or a system that results in high levels of expression such as a tat gene and tat element. This cassette can then be inserted into a vector, e.g., a plasmid vector such as, pUC19, pUC118, pBR322, or other known plasmid vectors, that includes, for example, an *E. coli* origin of replication. The plasmid vector may also include a selectable marker such as the β -lactamase gene for ampicillin resistance, provided that the marker polypeptide does not adversely affect the metabolism of the organism being treated.

Pharmaceutical carriers. The compositions also include a pharmaceutically acceptable carrier. We use the terms "pharmaceutically acceptable" (or "pharmacologically acceptable") to refer to molecular entities and compositions that do not produce an adverse, allergic or other untoward reaction when administered to an animal or a human, as appropriate. The term "pharmaceutically acceptable carrier," as used herein, includes any and all solvents, dispersion media, coatings, antibacterial, isotonic and absorption delaying agents, buffers, excipients, binders, lubricants, gels, surfactants and the like, that may be used as media for a pharmaceutically acceptable substance.

Thus, the invention also includes pharmaceutical compositions which contain, as the active ingredient, a fusion protein comprising a set of paired peptides or a fusion protein comprising a set of paired peptides linked to a protein transduction domain, in combination with one or more pharmaceutically acceptable carriers. An active ingredient can be a composition comprising a set of paired peptides linked to a protein transduction domain and wherein prepared peptides comprise a first bacterial effector polypeptide linked to a second bacterial effector polypeptide. In some embodiments, the polypeptide compositions can be sterilized using conventional sterilization techniques before or after it is combined with the pharmaceutically acceptable

carrier. In making the compositions of the invention, the polypeptide compositions are typically mixed with an excipient, diluted by an excipient or enclosed within such a carrier in the form of, for example, a capsule, tablet, sachet, paper, or other container. When the excipient serves as a diluent, it can be a solid, semisolid, or liquid material (e.g., normal saline), which acts as a vehicle, carrier or medium for the active ingredient. Thus, the compositions can be in the form of tablets, pills, powders, lozenges, sachets, cachets, elixirs, suspensions, emulsions, solutions, syrups, aerosols (as a solid or in a liquid medium), ointments, soft and hard gelatin capsules, suppositories, sterile injectable solutions, and sterile packaged powders. As is known in the art, the type of diluent can vary depending upon the intended route of administration. The resulting compositions can include additional agents, such as preservatives. The excipient or carrier is selected on the basis of the mode and route of administration. Suitable pharmaceutical carriers, as well as pharmaceutical necessities for use in pharmaceutical formulations, are described in Remington's Pharmaceutical Sciences (E. W. Martin), a well-known reference text in this field, and in the USP/NF (United States Pharmacopeia and the National Formulary). Some examples of suitable excipients include lactose, dextrose, sucrose, sorbitol, mannitol, starches, gum acacia, calcium phosphate, alginates, tragacanth, gelatin, calcium silicate, microcrystalline cellulose, polyvinylpyrrolidone, cellulose, water, syrup, and methyl cellulose. The formulations can additionally include: lubricating agents such as talc, magnesium stearate, and mineral oil; wetting agents; emulsifying and suspending agents; preserving agents such as methyl- and propylhydroxy-benzoates; sweetening agents; and flavoring agents. The pharmaceutical compositions can also be formulated so as to provide quick, sustained or delayed release of the active ingredient after administration to the patient.

Pharmaceutically acceptable compositions for use in the present methods, including those in which the polypeptides are entrapped in a colloid for oral delivery, can be prepared according to standard techniques. The polypeptides can be dried and compacted by grinding or pulverizing and inserted into a capsule for oral administration. In some embodiments, the polypeptides can be combined one or more excipients, for example, a disintegrant, a filler, a glidant, or a preservative. Suitable capsules include both hard shell capsules or soft-shelled capsules. Any lipid-based or polymer-based colloid may be used to form the capsule. Exemplary polymers useful for colloid preparations include gelatin, plant polysaccharides or their derivatives such as carrageenans and modified forms of starch and cellulose, e.g., hypromellose. Optionally, other ingredients may be added to the gelling agent solution, for example plasticizers such as glycerin and/or sorbitol to decrease the capsule's hardness, coloring agents, preservatives, disintegrants, lubricants and surface treatment. In some embodiments, the capsule does not include gelatin. In other embodiments, the capsule does not include plant polysaccharides or their derivatives.

Regardless of their original source or the manner in which they are obtained, the polypeptides of the invention can be formulated in accordance with their use. These compositions can be prepared in a manner well known in the pharmaceutical art, and can be administered by a variety of routes, depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration may be oral or topical (including ophthalmic and to mucous membranes including intranasal, vaginal and rectal delivery). In some embodiments, administration can be pulmonary (e.g., by inhalation or insufflation of powders or aero-

sols, including by nebulizer; intratracheal, intranasal, epidermal and transdermal) or ocular. Methods for ocular delivery can include topical administration (eye drops), subconjunctival, periocular or intravitreal injection or introduction by balloon catheter or ophthalmic inserts surgically placed in the conjunctival sac. Parenteral administration includes intravenous, intraarterial, subcutaneous, intraperitoneal or intramuscular injection or infusion; or intracranial, e.g., intrathecal or intraventricular administration. Parenteral administration can be in the form of a single bolus dose, or may be, for example, by a continuous perfusion pump. Pharmaceutical compositions and formulations for topical administration may include transdermal patches, ointments, lotions, creams, gels, drops, suppositories, sprays, liquids, powders, and the like. Conventional pharmaceutical carriers, aqueous, powder or oily bases, thickeners and the like may be necessary or desirable.

The compositions can be formulated in a unit dosage form, each dosage containing, for example, from about 0.005 mg to about 2000 mg of polypeptides per daily dose. The term "unit dosage forms" refers to physically discrete units suitable as unitary dosages for human subjects and other mammals, each unit containing a predetermined quantity of active material calculated to produce the desired therapeutic effect, in association with a suitable pharmaceutical excipient. For preparing solid compositions such as tablets, the principal active ingredient is mixed with a pharmaceutical excipient to form a solid preformulation composition containing a homogeneous mixture of a compound of the present invention. When referring to these preformulation compositions as homogeneous, the active ingredient is typically dispersed evenly throughout the composition so that the composition can be readily subdivided into equally effective unit dosage forms such as tablets, pills and capsules. This solid preformulation is then subdivided into unit dosage forms of the type described above containing from, for example, 0.005 mg to about 1000 mg of the compositions of the present invention.

The compositions can be formulated in a unit dosage form, each dosage containing, for example, from about 0.1 mg to about 50 mg, from about 0.1 mg to about 40 mg, from about 0.1 mg to about 20 mg, from about 0.1 mg to about 10 mg, from about 0.2 mg to about 20 mg, from about 0.3 mg to about 15 mg, from about 0.4 mg to about 10 mg, from about 0.5 mg to about 1 mg; from about 0.5 mg to about 100 mg, from about 0.5 mg to about 50 mg, from about 0.5 mg to about 30 mg, from about 0.5 mg to about 20 mg, from about 0.5 mg to about 10 mg, from about 0.5 mg to about 5 mg; from about 1 mg from to about 50 mg, from about 1 mg to about 30 mg, from about 1 mg to about 20 mg, from about 1 mg to about 10 mg, from about 1 mg to about 5 mg; from about 5 mg to about 50 mg, from about 5 mg to about 20 mg, from about 5 mg to about 10 mg; from about 10 mg to about 100 mg, from about 20 mg to about 200 mg, from about 30 mg to about 150 mg, from about 40 mg to about 100 mg, from about 50 mg to about 100 mg of the active ingredient.

In some embodiments, tablets or pills of the present invention can be coated or otherwise compounded to provide a dosage form affording the advantage of prolonged action. For example, the tablet or pill can comprise an inner dosage and an outer dosage component, the latter being in the form of an envelope over the former. The two components can be separated by an enteric layer which serves to resist disintegration in the stomach and permit the inner component to pass intact into the duodenum or to be delayed in release. A variety of materials can be used for such enteric layers or coatings, such materials including a number of

polymeric acids and mixtures of polymeric acids with such materials as shellac, cetyl alcohol, and cellulose acetate.

The liquid forms in which the compositions of the present invention can be incorporated for administration orally or by injection include aqueous solutions, suitably flavored syrups, aqueous or oil suspensions, and flavored emulsions with edible oils such as cottonseed oil, sesame oil, coconut oil, or peanut oil, as well as elixirs and similar pharmaceutical vehicles.

The proportion or concentration of the compositions of the invention in a pharmaceutical composition can vary depending upon a number of factors including dosage, chemical characteristics (e.g., hydrophobicity), and the route of administration.

Methods of Treatment

The compositions disclosed herein are generally and variously useful for treatment of inflammatory disorders and conditions. Inflammatory disorders and conditions encompass a wide range of disorders covering many different systems and organs, including the gastrointestinal tract, the skin, the lungs, and the musculoskeletal system. Exemplary inflammatory disorders include inflammatory bowel disease, rheumatoid arthritis, allergy, asthma, autoimmune diseases, coeliac disease, glomerulonephritis, hepatitis, preperfusion injury and transplant rejection. An inflammatory disorder can be a gastrointestinal disorder, for example, inflammatory bowel disease, Crohn's disease, and the ileocolitis, ileocecal, jejunoileitis, and gastroduodenal subtypes of Crohn's disease, and ulcerative colitis and subtypes of ulcerative colitis.

A subject is effectively treated whenever a clinically beneficial result ensues. This may mean, for example, a complete resolution of the symptoms associated with an inflammatory disorder, a decrease in the severity of the symptoms associated with an inflammatory disorder, or a slowing of the progression of symptoms associated with an inflammatory disorder. These methods can further include the steps of a) identifying a subject (e.g., a patient and, more specifically, a human patient) who has an inflammatory disorder; and b) providing to the subject a composition comprising a paired peptide composition disclosed herein in a physiologically acceptable carrier. An amount of such a composition provided to the subject that results in a complete resolution of the symptoms associated with an inflammatory disorder, a decrease in the severity of the symptoms associated with an inflammatory disorder, or a slowing of the progression of symptoms associated with an inflammatory disorder considered a therapeutically effective amount. The present methods may also include a monitoring step to help optimize dosing and scheduling as well as predict outcome.

The methods disclosed herein can be applied to a wide range of species, e.g., humans, non-human primates (e.g., monkeys), horses, pigs, cows or other livestock, dogs, cats or other mammals kept as pets, rats, mice, or other laboratory animals. The compositions described herein are useful in therapeutic compositions and regimens or for the manufacture of a medicament for use in treatment of conditions as described herein (e.g., inflammatory disorders and conditions.)

When formulated as pharmaceuticals, the compositions can be administered to any part of the host's body for subsequent delivery to a target cell. A composition can be delivered to, without limitation, the brain, the cerebrospinal fluid, joints, nasal mucosa, blood, lungs, intestines, muscle tissues, skin, or the peritoneal cavity of a mammal. In terms of routes of delivery, a composition can be administered by intravenous, intracranial, intraperitoneal, intramuscular, subcutaneous, intramuscular, intrarectal, intravaginal,

intrathecal, intratracheal, intradermal, or transdermal injection, by oral or nasal administration, or by gradual perfusion over time. In a further example, an aerosol preparation of a composition can be given to a host by inhalation.

Regardless of how the compositions are formulated, the dosage required will depend on the route of administration, the nature of the formulation, the nature of the subject's condition, e.g., a gastrointestinal disorder or a skin disorder, the subject's size, weight, surface area, age, and sex, other drugs being administered, and the judgment of the attending clinicians. Suitable dosages are in the range of 0.01-1,000 mg/kg. Some typical dose ranges are from about 1 µg/kg to about 1 g/kg of body weight per day. In some embodiments, the dose range is from about 0.01 mg/kg to about 100 mg/kg of body weight per day. In some embodiments, the dose can be, for example, 1 mg/kg, 2 mg/kg, 5 mg/kg, 10 mg/kg, 20 mg/kg, 50 mg/kg or 100 mg/kg. The dosage is likely to depend on such variables as the type and extent of progression of the disease or disorder, the overall health status of the particular patient, the relative biological efficacy of the compound selected, formulation of the excipient, and its route of administration.

Effective doses can be extrapolated from dose-response curves derived from in vitro or animal model test systems. For example, in vitro analysis of cytokine production by peripheral blood mononuclear cells (PBMCs) can be a useful for assaying pro- and anti-inflammatory responses, e.g., secretion of IL-1beta, IL-12, IL-4 or IL-10, IL-6, IL-23, and TNF-alpha. Compositions can also be analyzed for effects in animal models, for example, IgA production, cytokine production by explants of Peyer's patches, and dendritic cell and T-cell responses.

Wide variations in the needed dosage are to be expected in view of the variety of cellular targets and the differing efficiencies of various routes of administration. Variations in these dosage levels can be adjusted using standard empirical routines for optimization, as is well understood in the art. Administrations can be single or multiple (e.g., 2- or 3-, 4-, 6-, 8-, 10-, 20-, 50-, 100-, 150-, or more fold). Encapsulation of the compounds in a suitable delivery vehicle (e.g., polymeric microparticles or implantable devices) may increase the efficiency of delivery.

The duration of treatment with any composition provided herein can be any length of time from as short as one day to as long as the life span of the host (e.g., many years). For example, a composition can be administered once a week (for, for example, 4 weeks to many months or years); once a month (for example, three to twelve months or for many years); or once a year for a period of 5 years, ten years, or longer. It is also noted that the frequency of treatment can be variable. For example, the present compositions can be administered once (or twice, three times, etc.) daily, weekly, monthly, or yearly.

Any method known to those in the art can be used to determine if a particular response is induced. Clinical methods that can assess the degree of a particular disease state can be used to determine if a response is induced. For example, a subject can be monitored for symptomatic relief, e.g., relief from colic, diarrhea, constipation, nausea, vomiting, abdominal pain, cramping, heartburn, abdominal distention, flatulence, or incontinence, dermatitis, redness, pain, swelling. Alternatively or in addition, serum markers, imaging techniques, e.g., ultrasound, x-rays, and endoscopic methods can be used.

The compositions may also be administered in conjunction with other therapeutic modalities. Other therapeutic modalities will vary according to the particular disorder, but

can include, for example, anti-inflammatory agents, antibiotics and other dietary treatments, anti-diarrhea medications, anti-emetics, anti-cholinergic agents, Concurrent administration of two or more therapeutic agents does not require that the agents be administered at the same time or by the same route, as long as there is an overlap in the time period during which the agents are exerting their therapeutic effect. Simultaneous or sequential administration is contemplated, as is administration on different days or weeks.

Articles of Manufacture

The invention also features kits for administering the compositions. Accordingly, packaged products (e.g., sterile containers containing one or more of the compositions described herein and packaged for storage, shipment, or sale at concentrated or ready-to-use concentrations) and kits, are also within the scope of the invention. A product can include a container (e.g., a vial, jar, bottle, bag, microplate or beads) containing one or more compositions of the invention. In addition, an article of manufacture further may include, for example, packaging materials, instructions for use, syringes, delivery devices, buffers or other control reagents.

For example, the kit can include a measured amount of a composition comprising a set of paired peptides wherein the set of paired peptides comprises a first bacterial effect or polypeptide went to a second bacterial effector polypeptide and a protein transduction domain. The compounds, agents, and/or reagents can be packaged in a suitable container. The kit can further comprise instructions for administering the compositions. For example, the kit can include: a fusion protein comprising a protein transduction domain and a set of paired peptides. The kit can also include a buffering agent, a preservative, and/or a protein stabilizing agent. Each component of the kit can be enclosed within an individual container and all of the various containers can be within a single package. The product may also include a legend (e.g., a printed label or insert or other medium describing the product's use (e.g., an audio- or videotape or computer readable medium)). The legend can be associated with the container (e.g., affixed to the container) and can describe the manner in which the reagents can be used. The reagents can be ready for use (e.g., present in appropriate units), and may include one or more additional adjuvants, carriers or other diluents.

Also included as effectors or effector proteins of the compositions and methods described herein are functional equivalents of the proteins described above. By the term "functional equivalent" is meant any amino acid sequence or modification thereof that has the same targeting and immune suppressing function of the naturally occurring effector protein. In one embodiment, such functional equivalents can have modifications of one or more amino acids from the known sequences. In one embodiment, such functional equivalents can be a smaller fragment of the known sequences. In one embodiment, such functional equivalents can be a derivative of the naturally occurring sequences or be derived from other than human sources. In one embodiment, such functional equivalents can be altered by chemical modification or be altered by recombinant production to be associated with sequences with which the effector proteins are not associated in nature. Similarly, chemical or structural changes or fragments of the nucleic acid sequences that encode the effector proteins are also considered functional equivalents herein.

As used herein, the term "construct" as described herein refers to a chemically synthesized or genetically engineered assemblage that comprises one or more PTD/PPP associated with one or more effector proteins and further optionally

associated with one or more targeting moieties. The construct can be in the form of a polypeptide or a nucleic acid molecule encoding the polypeptide.

As used herein the term(s) “cosmeceutically or pharmaceutically acceptable carrier, excipient or formulation” refer to the components of a composition that provide a vehicle for delivery. For example, where the cosmeceutical or pharmaceutical product is a topical composition, the carrier or formulation can contain typical components such as cremes, saline, vitamins, oils that are normally found in cosmetic or pharmaceutical preparations for skin. See, for example, U.S. Pat. No. 5,635,497 which discloses an oil-in-water, fatty cream composition for topical administration comprising from 60 to 80 percent by weight of fatty components, from 1.5 to 5 percent by weight of at least one non-ionic, hydrophilic surfactant having an HLB of at least 14, about 6% of fatty alcohols and esters, a therapeutically effective amount of at least one topically active therapeutic agent, and water, provided the topically active therapeutic agent is not dithranol or its derivatives. See, also International Patent Publication No. WO2014/076642. Pharmaceutically or cosmetically acceptable excipients suitable for the compositions described herein can be selected from plasticizers, disintegrants, glidants, coloring agents, lubricants, stabilizers, adsorbents, preservatives, delivery retarders and mixtures thereof. Such a composition may contain a trans-epidermal or transdermal carrier agent consisting of acidic electrolyzed water having a pH of 1.0 to 4.0 and comprise clusters of water having 5 to 10 molecules of water per cluster, and a polyacrylate. Other suitable formulations may include oils, emollients, lotions for topical and transdermal applications along with buffered/aqueous and saline solutions. See, also, texts such as *Topical Drug Delivery Formulation* (eds. Osborne and Amann), 2000, publishers Taylor & Francis, *Drugs and the Pharmaceutical Sciences Series* #42.

The term “CAGE” as used herein refers to deep eutectic synthetic solvent, a choline-based oil that has antimicrobial activity, which has been shown to penetrate deep into the dermis. CAGE is described by Zakrewsky, M. et al, *Adv. Healthcare Mater.* (March 2016), 5, 1282-1289, incorporated by reference herein. CAGE has been shown to be useful in transdermal protein delivery, wherein the protein is carried 15-20 cell layers into model skin preparations. See, e.g., Banerjee, A. et al, *Adv. Healthcare Mater.* 2017, 1601411 DOI: 10.1002/adhm.201601411, incorporated by reference herein.

Still other pharmaceutical strategies to enhance dermal delivery of peptides or proteins including carrier peptides, signaling peptides neurotransmitter-inhibiting peptides and enzyme-inhibiting peptides, include chemical and physical penetration enhancers, such as listed in Table 2 of Badenhorst, T. et al, *Pharmaceutical Strategies for the Topical Dermal Delivery of Peptides/Proteins for Cosmetic and Therapeutic Applications*. *Austin Journal of Pharmacology and Therapeutics* (2014), 2(6):10. Also as discussed in this document are coupling with lipophilic moiety, such as lauric, palmitic and other acids, using CPP conjugates, formulation with microemulsions, encapsulation in liposomal vesicles and use of lipid particles, as well as combinations of these formulations.

The term “targeting moiety” refers to constructs useful in fusion with the effector proteins and/or PTD/CPPs described herein to direct the fusion protein or a nucleic acid sequence encoding it to a specific cell or tissue type within the body. Alterations to the fusion constructs described by the addition of amino acid segments to naturally occurring effector or

fusion effector sequence, which enables the protein to bind and to specifically target cells, tissues or other target or physiologic compartment in the human body create “targeted-effectors” and “targeted effector fusions”. Such targeting moieties include amino acid segments that enhance the efficacy of the preparation by its ability to be activated due to conditions in a specific compartment of the body could be cleaved, for instance by proteolytic cleavage, addition of post-translational modifications, or forming of PPIs with cell and tissue specific host proteins. Examples of tissue-specific targeting peptides for this use include those described in Jung, E. et al., *Identification of tissue-specific targeting peptide*. *J Comput Aided Mol Des* (October 2012) 26:1267-1275, incorporated by reference herein. Targeting moieties can also be antibodies, antibody fragments, aptamers, amino acid sequences, nucleic acid sequences that are complementary to or capable of binding a complementary sequence on a cell or tissue or chemical moieties that have a three-dimensional structure that can fit into a three-dimensional pocket on the targeted cell or tissue. For example, a targeting sequence can be a hormone, or fragment thereof, that targets or binds its naturally occurring cell surface receptor, or tissue specific markers, etc.

The term “polypeptide,” when used in singular or plural form, generally refers to a polymer of amino acids joined together by peptide bonds and may include unmodified or naturally occurring amino acids or modified or unnatural amino acids. In certain embodiment, the term polypeptide refers to a construct formed by multiple shorter peptides joined directly, or indirectly via linkers, to form a single peptide. In one embodiment, as described herein, a polypeptide is formed by the fusion of a PTD or CPP and an effector protein. In another embodiment, as described herein, a polypeptide is formed by the fusion of a PTD attached to an effector protein, wherein the effector protein is further attached to another effector protein (with or without its own PTD). In still another embodiment, the polypeptide of this invention is formed by the covalent association of a first PTD fused to a first effector protein, a linker followed by an additional (e.g., second, third, fourth, etc) effector protein. The term “first” is used only to distinguish among the effectors. In yet another embodiment, each effector protein of the polypeptide is associated with its own PTD or CPP. In yet another embodiment, only one PTD/CPP is present. In another embodiment, the single PTD/CPP is located at the N terminus of the polypeptide. In yet another embodiment, multiple PTD/CPP are present in the polypeptide, each located at the N terminus of its associated effector protein. In another embodiment, each PTD/CPP is separated from its effector protein by a linker. In another embodiment, each effector protein is separated from each additional effector protein by a linker. The first and additional effector constructs in a single polypeptide may occur in any order.

In still another embodiment of a polypeptide as described herein, the PTD/CPP are fused directly to the effector protein and each effector protein is fused to each additional effector protein. In certain embodiments, the polypeptides contain two or more different effector proteins, each having its own target. In certain embodiments, the polypeptides contain two or more different effector proteins, each targeting related targets. In still further embodiments, each polypeptide is further associated with a targeting moiety to target the polypeptide to a specific tissue or cell type, e.g., skin, epidermis, dermis.

By “homologous protein” is meant a protein having a percent sequence similarity or identity of greater than 80%, greater than 85%, greater than 90%, greater than 95%,

greater than 97%, or greater than 99% and sharing the same function as the effector protein.

As used herein, the term “polynucleotide,” when used in singular or plural form, generally refers to any polyribonucleotide or polydeoxynucleotide, which may be unmodified RNA or DNA or modified RNA or DNA that encode any of the polypeptide constructs as described above. Thus, for instance, polynucleotides as defined herein include, without limitation, single- and double-stranded DNA, DNA including single- and double-stranded regions, single- and double-stranded RNA, and RNA including single- and double-stranded regions, hybrid molecules comprising DNA and RNA that may be single-stranded or, more typically, double-stranded or include single- and double-stranded regions. In addition, the term “polynucleotide” as used herein refers to triple-stranded regions comprising RNA or DNA or both RNA and DNA. The term “polynucleotide” specifically includes cDNAs. The term includes DNAs (including cDNAs) and RNAs that contain one or more modified bases. In general, the term “polynucleotide” embraces all chemically, enzymatically and/or metabolically modified forms of unmodified polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including simple and complex cells.

By “nucleic acid molecule” as used herein is meant the nucleic acid sequence that encodes a construct or polypeptide as described above. The nucleic acid molecule can include other operative components, such as regulatory sequences directing expression of the construct or polypeptide in a cell in vivo or in vitro. The nucleic acid molecule can be in a vector.

By “vector” is meant an entity that delivers the nucleic acid molecule to cells, for therapeutic or cosmetic purposes. As used herein, a vector may include any genetic element including, without limitation, naked DNA, a phage, transposon, cosmid, episome, plasmid, or a virus. Vectors are generated using the techniques and sequences provided herein, in conjunction with techniques known to those of skill in the art. Such techniques include conventional cloning techniques of cDNA such as those described in texts such as Sambrook et al, *Molecular Cloning: A Laboratory Manual*, 3rd edition, 2001 Cold Spring Harbor Press, Cold Spring Harbor, N.Y., and current editions thereof, use of overlapping oligonucleotide sequences of the adenovirus genomes, polymerase chain reaction, CRISPR, gene editing, and any suitable method which provides the desired nucleotide sequence.

By the term “attachment” or “attach” as used herein to describe the interaction between the components of the constructs is meant covalent attachments or a variety of non-covalent types of attachment. Still another useful attachment mechanism involves via “affinity interactions”, i.e., one domain fused to an antibody fragment that recognizes an epitope on the second domain to be used instead of the two domains fused together. Other attachment chemistries useful in assembling the constructs described herein include, but are not limited to, thiol-maleimide, thiol-haloacetate, amine-NHS, amine-isothiocyanate, azide-alkyne (CuAAC), tetrazole-cyclooctene (iEDDA).

The “linker” refers to any moiety used to attach or associate different elements of the polypeptide/polynucleotide sequence components of the constructs (i.e., the effector, the PTD, the targeting moiety) to each other. Thus in one embodiment, the linker is a covalent bond. In another embodiment, the linker is a non-covalent bond. In an embodiment of a polynucleotide described herein, the linker is composed of at least one to about 20 nucleic acids. Thus,

in various embodiments, the linker is formed of a sequence of at least 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 57 up to about 60 nucleic acids. In yet another embodiment of a polypeptide as described herein, the linker refers to at least one to about 20 amino acids. Thus, in various embodiments, the linker is formed of a sequence of at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or about 20 amino acids. In still other embodiments, the linker can be a larger compound or two or more compounds that associate covalently or non-covalently. In still other embodiment, the linker can be a combination of the linkers defined herein. The linkers used in the constructs of the compositions and methods are in one embodiment chemically or enzymatically cleavable, such as by redox, pH, and the like. The linkers used in the constructs of the compositions and methods are in one embodiment non-cleavable.

The term “nucleic acid molecule” refers to a recombinant assembled nucleic acid sequence encoding a construct or polypeptide described herein. The nucleic acid molecule may be naked DNA or RNA. Alternatively the nucleic acid molecule may be associated operatively with regulatory sequences permitting expression of the construct or polypeptide in vivo or in vitro. The nucleic acid molecule may be a vector, plasmid vector, or be presented in a viral vector for delivery to the subject. Generation of such nucleic acid molecules with resort to the teachings of this specification can utilize known recombinant and genetic engineering techniques. See, e.g., “Inflammatory conditions” as used herein refer, in one aspect to inflammatory skin diseases, which are the most common problem in dermatology and cause pain, redness, swelling and the sensation of heat. Such inflammatory skin conditions include, without limitation, non-specific rashes accompanied by skin itching and redness, sunburn, dermatitis, eczema, rosacea, seborrheic dermatitis, psoriasis, infection, skin injury or wounds, autoimmunity or aging effects. Other inflammatory conditions (non-skin) include, without limitation, autoimmune conditions, asthma, chronic peptic ulcer, tuberculosis, rheumatoid arthritis, periodontitis, ulcerative colitis and Crohn’s disease, sinusitis, active hepatitis, gut dysbiosis syndromes and any other disease caused by NFkB/JNK/p38 pathway activation.

“Patient” or “subject” as used herein means a mammalian animal, including a human, a veterinary or farm animal, a domestic animal or pet, and animals normally used for clinical research. In one embodiment, the subject of these methods and compositions is a human.

The terms “a” or “an” refers to one or more. For example, “an expression cassette” is understood to represent one or more such cassettes. As such, the terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein.

As used herein, the term “about” means a variability of plus or minus 10% from the reference given, unless otherwise specified.

The words “comprise”, “comprises”, and “comprising” are to be interpreted inclusively rather than exclusively, i.e., to include other unspecified components or process steps. The words “consist”, “consisting”, and its variants, are to be interpreted exclusively, rather than inclusively, i.e., to exclude components or steps not specifically recited.

With resort to the definitions of the components above, in one aspect, the inventors provide a composition comprising a first construct comprising a selected immunomodulatory effector protein or its functional equivalent thereof that targets a first functional domain optionally linked covalently or non-covalently to a selected protein transduction domain

(PTD) or penetrating peptide (CPP); an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP of (a) or to an additional PTD or CPP, or a combination of a first and one or more additional constructs.

In one embodiment, the first and additional constructs are further linked with a targeting moiety to direct activity of the composition to a specific cell or tissue, e.g., skin. In another embodiment, the constructs are admixed in a pharmaceutically or cosmeceutically acceptable carrier or excipient or formulation, such as a formulation suitable for topical administration to the skin.

One embodiment of such a composition is a single polypeptide comprising a fusion of two or more effectors. Another embodiment is a single polypeptide comprising a first effector-PTD fused construct linked covalently or non-covalently to one or more of additional constructs. The polypeptide in a further embodiment contains one or more optional linker amino acid sequences interposed between each construct of the polypeptide. In one embodiment, the polypeptide contains fused to the first construct and/or additional construct, a targeting moiety to direct the polypeptide to a specific cell or tissue. In a further embodiment, the single polypeptide is in a pharmaceutically or cosmeceutically acceptable carrier or excipient or formulation.

Whether the constructs are admixed in a composition or present in one or more polypeptides, the effector can be one or more of the effectors identified above, particularly in Tables 1 or 2. In one embodiment of the admixture or polypeptide, the first effector protein is NleE or a functional equivalent thereof. In yet other embodiment, the additional effector protein is one or more of NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In other embodiments, the admixture of constructs in the composition or present in the single polypeptide can be selected from the following embodiments. In one embodiment, the first effector is NleE and the additional effector is NleC. In another embodiment, the first effector is NleE and the additional effector is NleD. In another embodiment, the first effector is NleE and the additional effector is NleB. In another embodiment, the first effector is NleE and the additional effector is NleH. In another embodiment, the first effector is NleE and the additional effector is YopB. In another embodiment, the first effector is NleE and the additional effector is YopH. In another embodiment, the first effector is NleE and the additional effector is YopJ. In another embodiment, the first effector is NleE and the additional effector is YopP. In still another embodiment, the first effector is NleE and the additional effector is SspH1. In a further embodiment, the first effector is NleE and the additional effector is OspG. In another embodiment, the first effector is NleE and the additional effector is OspF. In another embodiment, the first effector is NleE and the additional effector is IpaH9.8. In still a further embodiment, the first effector is NleE and the additional effector is IpaH1.4. In another embodiment, the first effector is NleE and the additional effector is IpaH2.5. In another embodiment, the first effector is NleE and the additional effector is IpaH4.5. In another embodiment, the first effector is NleE and the additional effector is IpaH7.8. In yet a further embodiment, the first effector is NleE and the additional effector is SlrP. In still other embodiments of the constructs or polypeptides, three or more effectors can be delivered in a single mixture of constructs or single polypeptide. The first

and additional effectors or effector constructs may occur in any order in the fusions or in the polypeptides described herein.

In the above-listed constructs and polypeptides, a suitable PTD or CPP linked to the first construct and optionally to each additional construct or polypeptide is the naturally occurring PTD of the selected effector protein, e.g., the YopM PTD (SEQ ID NO: 2; amino acids 1-50 of YopM). See U.S. Pat. No. 8,840,901.

The PTD/CPP can also be a functional equivalent, e.g., a mutated or modified version of a naturally occurring PTD, such as the naturally occurring PTD sequence of SspH1, or of *Shigella* IpaH protein or any other of the effectors. The PTD useful in the constructs and polypeptides of the compositions can be a completely novel sequence which is developed to transport the construct or polypeptide or composition across cell membranes.

In one specific embodiment, the first construct is the YopM PTD fused to NleE. In another specific embodiment, the first construct is the SspH1 PTD fused to NleE. In still another specific embodiment, the first construct is the IpaH PTD fused to NleE. In still other embodiments, the additional construct comprises YopM, optionally associated with the YopM PTD. In still other embodiments of the single polypeptide, the polypeptide further contains at least one linker sequence as defined above interposed between the first and the one or more additional constructs. In still other embodiments of the constructs and single polypeptide, each construct or polypeptide can further contain a fused targeting moiety, such as a skin cell targeting peptide or other cell or tissue targeting peptide. Given the number of effectors, PTD/CPPs, linkers and targeting moieties identified herein and in the cited publications, any number of constructs or polypeptides may be prepared according to the teachings contained herein.

In still other embodiments of the above-listed constructs and polypeptides, the admixture of protein constructs or constructs contained on the single polypeptide is such that the targeted functional domain of each effector protein in the construct is expressed in the same cell. In another embodiment, each construct in admixture or in the single polypeptide has a non-overlapping redundant role in inhibiting NFkB, JNK and p38 pathways when present in a mammalian cell.

As described above, any of the compositions described herein, whether a mixture of constructs, a single polypeptide or nucleic acid molecules encoding them, can be prepared in a formulation comprises ingredients suitable for application to, and absorption through, the cells of the skin. One such formulation employs the above-described CAGE solvent. Other suitable formulations may include oils, emollients, lotions for topical and transdermal applications along with buffered/aqueous and saline solutions.

In yet another aspect, the compositions described herein include nucleic acid molecules comprising a nucleic acid sequence encoding a first construct, an additional construct, or the single polypeptide of any one of the above-described embodiments. Such nucleic acid molecules can also comprise operatively associated regulatory sequences, such as promoters, enhancer, etc (see, e.g., Sambrook et al) necessary to express the construct or polypeptide in a suitable cell. In certain embodiments, the nucleic acid molecule is naked DNA or RNA. In certain embodiments, the molecule is part of a plasmid or contained in a recombinant vector or virus. Methods for generating such nucleic acid molecules are within the skill of the art given the teachings herein. The nucleic acid molecules may also be delivered in a cosme-

ceutically or pharmaceutically acceptable carrier or excipient or formulation. Such formulations are described in detail in available texts as described above.

These constructs and/or polypeptides may be formulation into a wide variety of cosmeceutically or pharmaceutically acceptable carrier or excipient or formulation for many different uses and routes of administration. While topical administration is preferred for uses on skin inflammation, it is contemplated that other conventional routes of administration will be used for treating other inflammatory conditions. In some embodiments, routes of administration include transdermal (including patch formulation), intradermal injection (including infusion and subcutaneous injection). Other pharmaceutically acceptable routes of administration include, but are not limited to, systemic routes, such as intraperitoneal, intravenous, intranasal, intramuscular, intratracheal, subcutaneous, epidural, and oral routes and other parenteral routes of administration or intratumoral or intranodal administration. The composition may be administered by any convenient route, for example by infusion or bolus injection, by absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa, etc.) and may be administered together with other biologically active agents. Administration can be systemic or local. In addition, it may be desirable to introduce the pharmaceutical compounds or compositions of the invention into the central nervous system by any suitable route, including intraventricular and intrathecal injection; intraventricular injection may be facilitated by an intraventricular catheter, for example, attached to a reservoir, such as an Ommaya reservoir. Pulmonary administration can also be employed, e.g., by use of an inhaler or nebulizer, and formulation with an aerosolizing agent. Routes of administration may be combined, if desired.

Other methods of delivery of the effectors include via an attenuated bacterial strain which expresses a functional Type-Three Secretion System (TTSS)-expressing microorganism and is engineered to contain nucleic acid sequences encoding at least one of the effector immuno-modulatory proteins or fusions or polypeptide, wherein amino acid sequences also code for functional TTSS secretion signal sequences. These include attenuated bacteria engineered or induced to shed outer membrane vesicles (OMV) or other type of exosome-like, bacterial or cell-derived vesicle containing the proteins described herein.

In still other embodiments, the compositions, constructs, fusions and polypeptides and nucleic acid molecules described herein may be further manipulated by encapsulation in liposomes, micro particles, microcapsules, or in recombinant cells capable of expressing the compound, receptor-mediated endocytosis construction of a nucleic acid as part of a retroviral or other vector, etc.

Still other embodiments of compositions are provided herein. In certain embodiments, a composition of single effectors, or fusions of multiple effectors, or fusions of single effector-PTD, or fusions of multiple effectors and one or more PTD, or single/multiple effector(s)-PTD-targeting moiety fusions are prepared. The polypeptide linker regions and other non-native peptide sequences are engineered into fusion-effectors and can contain active segments which can lead to their enhanced ability to be purified in active form. In another embodiment, the effector fusions can be modified recombinantly or chemically, or by e.g., editing methodologies to provide altered subcellular localization, altered stability, or altered ability to participate in protein-protein interactions. In one embodiment, the effector fusions are modified to include amino acid segments normally used as

tags for efficient purification of proteins from complex mixtures (e.g., 6HIS, GST, and Maltose BP).

In still a further specific embodiment, each individual effector protein or multiple effector fusion has an amino acid sequence that is deliberately altered by designed site directed mutagenesis. In one embodiment, such mutagenesis alters the activity of the effector in terms of its potency as an anti-inflammatory effector. In another embodiment, the altered effector amino acid sequence alters the effector's specificity for a target, catalytic activity, antigenicity, stability, ability to bind substrate, and/or ability to work with other effectors in a preparation. Such mutagenic techniques can alter any other common property of an effector protein for the benefit of the efficacy of the resulting therapeutic preparation.

In another aspect, the compositions containing the effector fusions can lead to the inhibition of at least one or more of an NF- κ B, JNK and p38 mediated signal transduction pathway protein in selected cells and tissues. Such inhibition provides an anti-inflammatory response, an anti-apoptotic effect or a pro-apoptotic effect in a target cell.

Still other embodiments of compositions can be obtained by selection of the components as taught herein resulting in a pharmaceutical/cosmeceutical preparation, containing active and therapeutic levels of the effector fusions and constructs as described herein for use in preventing, treating, or ameliorating an immune-related disorder whose pathology stems from aberrant activation of the NF κ B, JNK or P38 pathways.

In still other aspects, methods for designing the various constructs and polypeptides described herein can be made using techniques well-known in the art of recombinant genetic engineering and manipulation of the nucleic acid techniques. Suitable techniques are known to those of skill in the art, and as provided by many of the publications incorporated by reference herein. Selection of the appropriate techniques to create the various embodiments will depend upon the effectors, PTDs, linkers, targeting moieties and other components as set out above. Molecular biology strategies to alter the proteins to make them more stable, specific, catalytic and robust are known. Further such known techniques can be used to optimize the design, construction and delivery of effector fusions based upon knowledge of the specific pathways and enzymes used by the cell in a selected inflammatory stimulus.

For example, in one embodiment the selected effectors are subject to random, high throughput mutagenesis in the catalytic active sites or the sites of substrate binding. Selection of effectors of these mutagenized populations for variants with novel substrate specificities, rates of catalysis and their combination with other known or mutagenized effectors in the fusions, constructs and polypeptides described herein can result in compositions with novel signal transduction inhibition functions.

Still other known and available methods can be employed for creating recombinant fusion effector genes/proteins and for their expression, purification, stabilization as single effectors or fusion effectors. Such methods include producing/expressing the effector protein or effector fusion proteins from cloned genes in any suitable cell based systems, including, without limitation, bacteria, yeast, insect, or mammalian cells. Methods for purification of the resulting fusion constructs resulting in highly active, stable proteins which retain the ability to target their native substrate also employ known techniques. Additional techniques for producing compositions of this invention include, without limitation, optimization of the effector preparation by incorpo-

ration into liposomes, PEGylation of the effector constructs or fusions or polypeptides. The effector constructs may be subjected to chemical derivatization to associate or conjugate the construct physically or in admixture with additional entities. Such entities include without limitation, lipids, liposomes, other drugs, and cargo molecules including nucleic acids, polypeptides, organic molecules, small organic molecules, metals, nano-particles, viruses, modified viruses, viral vectors, antibodies and/or plasmids as targets for conjugation.

Yet further aspects provide a variety of methods of using the compositions, fusions, constructs and polypeptides described above in methods for treating cosmetic conditions, such as inflammation or irritation due to normal skin agent, to the treatment of other inflammatory conditions, such as cancer, gut dysbiosis syndromes, and other conditions identified herein caused by NFkB/JNK/p38 pathway activation. For example one therapeutic use is to ameliorate the inflammation in the tumor microenvironment (TME) in any malignant or non-malignant condition that involves activation of the NFkB/JNK/p38 pathway.

The various compositions defined above may be employed therapeutically in the down regulation/normalization of the immune response elicited by the ectopic or pathologic activation of NFkB, JNK and P38 pathways. In one embodiment, such methods of treating or suppressing pathway activation can occur in subjects suffering from the consequences of normal, naturally occurring conditions such as aging associated inflammation. In yet another method, the compositions are administered for treatment of numerous conditions wherein the three pathways are aberrantly activated resulting in an inflammatory response. Such aberrant responses include without limitation, infection, wound healing, reactive dermatitis, auto-immune disease, and malignant or non-malignant proliferative disorder. These diseases, conditions and syndromes wherein the three pathways are aberrantly activated also include, but are not limited to diseases caused by autoimmunity of the patient, topical inflammation, chronic inflammation, gastroenteritis, chronic gastritis, inflammatory bowel diseases (IBD), colitis ulcerosa, psoriasis, allergic reactions, Crohns disease, dysbiosis syndromes, cancer (including glioblastoma) rheumatoid arthritis related bone diseases characterized by changes in bone resorption, reduction and relief of the signs and symptoms associated with treating inflammation, and/or suppression of the immune system.

These methods involve administering via a suitable route of administration (as described above) an amount of the composition (i.e., the effector fusions, effector-PTD fusions, effector-PTD-Targeting fusions, and the other embodiments described above) in an amount sufficient to reduce activation of the desired pathway.

Desirably, the methods further involve, in one aspect, administering a suitable dose or doses of the fusion construct(s) in a therapeutic regimen to the subject in need thereof. In one embodiment, such administration can occur once or more prior to, simultaneously with, or after any conventional additional treatment for the conditions. In one embodiment, where the condition is an infection, the additional component is an antibiotic. In another embodiment, wherein the condition is a cancer, the additional component is radiation or chemotherapy. Still other known "additional components" may be selected by one of skill in the art or the attending physician depending on the condition being treated and the physical status of the subject.

Methods for determining the timing of frequency of administration will include an assessment of subject's

response to the first administration of the composition. The dose is generally the lowest dose of the composition that is effective to suppress activation of the NFkB, JNK/p38 pathway. In still other embodiments, a suboptimal dose is delivered in a continuous infusion or a slow release formulation. The dosage required will depend primarily on factors such as the condition being treated, the age, weight and health of the patient, and may thus vary among patients. In one embodiment, where the composition comprises an admixture of peptide constructs or the single polypeptide, one such dose is about 1 to 25 μ M protein/polypeptide. In another embodiment, the dose is less than 10 μ M protein/polypeptide. In still another embodiment, the dose is between 1 μ M and 5 μ M protein/polypeptide. In another embodiment, the suboptimal dose is less than 1 μ M protein/polypeptide.

When the composition is in the form of a nucleic acid or vector or nucleic acid molecule, it is administered in sufficient amounts to transduce the targeted cells and to provide sufficient levels of gene transfer and expression to reduce or inhibit activation of the NFkB, JNK/p38 pathway and provide a therapeutic benefit without undue adverse or with medically acceptable physiological effects, which can be determined by those skilled in the medical arts. Dosages of these therapeutic compositions will depend primarily on factors such as the condition being treated, the age, weight and health of the patient, and may thus vary among patients. For example, a therapeutically effective adult dosage of a viral vector or siRNA nanoparticle is generally in the range of from about 100 μ L to about 100 mL of a carrier containing concentrations of from about 1×10^5 to about 1×10^{15} particles, about 1×10^{11} to 1×10^{13} particles, or about 1×10^9 to 1×10^{12} virus particles.

In still another aspect, methods for use of the compositions described herein involve veterinary use for the treatment of inflammatory conditions in animals, e.g., for autoimmune diseases, reactive dermatitis, pruritis, alopecia and any other skin ailment for which the underlying pathogenesis involves alteration of NFkB, JNK/p38 pathway. Selection of administration routes, dosages and therapeutic regimens may be selected by a veterinarian.

In one aspect, a composition in a pharmaceutically acceptable carrier or excipient or formulation is provided. The composition includes: (a) a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain optionally linked covalently or non-covalently or by affinity to a selected protein transduction domain (PTD) or penetrating peptide (CPP); (b) an additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain, optionally linked to the same PTD or CPP of (a) or to an additional PTD or CPP, or (c) a combination of constructs (a) and (b) in any order.

In one embodiment of the composition, the first construct is linked covalently or non-covalently to one or more of the additional constructs in a single polypeptide. Additionally, the composition further includes an optional linker amino acid sequence interposed between each construct.

In one embodiment, the functional equivalent includes a chemically or recombinantly modified amino acid sequence of the effector protein, or a fragment of the naturally-occurring effector amino acid sequence, or of the derivative of said chemically or recombinantly modified amino acid sequence of the effector protein that shares the functional activity of the effector protein.

In one embodiment, the first effector protein is NleE or a functional equivalent thereof.

In one embodiment, the additional effector protein is one or more of NleC, NleD, NleB, NleH, YopM, YopE, YopH, YopJ, YopP, SspH1, OspG, OspF, IpaH9.8, IpaH1.4, IpaH2.5, IpaH4.5, IpaH7.8 and SlrP. In one embodiment, the PTD or CPP is the naturally occurring PTD of the selected effector protein. In one embodiment, the PTD is a sequence of YopM, of SspH1, or of *Shigella* IpaH protein. In one embodiment, the PTD is amino acids 1-50 SEQ ID NO: 2.

In one embodiment, the PTD or CPP is one or more of Poly-Arg, Tat and VP22, df Tat, a cyclic CPPs, IMT-P8, seven arginine (R7) and Streptolysin O (SLO)-mediated systems, elastin like polypeptide, CPP-adaptor system, 1,2-Benzisothiazolin-3-one (BIT) and Tat, activatable cell-penetrating peptides, LDP12, M918, BR2, POD, native protein independent of R11-CPP, Poly-arginine/Tat and Tat-PTD, Pep-1, Cady-2, R8, azo-R8, Penetratin, HR9 and IR9 peptides, or pVEC.

In one embodiment, each targeted functional domain of each effector protein in the construct is expressed in the same cell. In one embodiment, each construct has a non-overlapping redundant role in inhibiting NFkB, JNK and p38 pathways when present in a mammalian cell. In one embodiment, the formulation comprises ingredients suitable for application to, and absorption through, the cells of the skin. In one embodiment, the formulation comprises CAGE solvent. In one embodiment, the first construct is the YopM PTD fused to NleE or the SspH1 PTD fused to NleE or the IpaH PTD fused to NleE. In one embodiment, the additional construct comprises YopM, optionally associated with the YopM PTD. In one embodiment, the composition further includes at least one linker between the first and additional constructs.

In one embodiment, the first effector is NleE and the additional effector is NleC. In another embodiment, the first effector is NleE and the additional effector is NleD. In another embodiment, the first effector is NleE and the additional effector is NleB. In another embodiment, the first effector is NleE and the additional effector is NleH. In another embodiment, the first effector is NleE and the additional effector is YopB. In another embodiment, the first effector is NleE and the additional effector is YopH. In another embodiment, the first effector is NleE and the additional effector is YopJ. In another embodiment, the first effector is NleE and the additional effector is YopP. In another embodiment, the first effector is NleE and the additional effector is SspH1. In another embodiment, the first effector is NleE and the additional effector is OspG. In another embodiment, the first effector is NleE and the additional effector is OspF. In another embodiment, the first effector is NleE and the additional effector is IpaH9.8. In another embodiment, the first effector is NleE and the additional effector is IpaH1.4. In another embodiment, the first effector is NleE and the additional effector is IpaH2.5. In another embodiment, the first effector is NleE and the additional effector is IpaH4.5. In another embodiment, the first effector is NleE and the additional effector is IpaH7.8. In another embodiment, the first effector is NleE and the additional effector is SlrP. In any of the embodiments above, the constructs may be in any order.

In one embodiment, the composition further includes a skin cell targeting peptide.

In another aspect, a nucleic acid construct is provided. The nucleic acid sequence includes a nucleic acid sequence encoding the polypeptide of any of the compositions described above.

In one embodiment, the nucleic acid construct further includes regulatory sequences necessary to express the polypeptide in a suitable cell.

In one embodiment, the nucleic acid construct is a DNA, RNA, a plasmid or a recombinant vector or virus.

In one embodiment, the nucleic acid construct is in a pharmaceutically acceptable carrier or excipient or formulation.

In another aspect, a method is provided for treating, preventing or ameliorating an NFkB, JNK or p38 mediated inflammatory disorder in a subject, or a subject at risk from developing such disorder. The method includes administering to the subject any of the compositions or nucleic acid constructs described above.

In one embodiment of the method, the composition is administered in an amount sufficient to down-regulate the innate inflammatory response in a targeted cell or tissue of the subject mediated by the pathologic or non-pathologic activation of intracellular NFkB, JNK or P38 signal transduction pathway.

In one embodiment, the method reduces symptoms, and inhibits progression, of damage to the subject's cells or tissue caused by inflammation.

In one embodiment of the method, a composition containing multiple constructs targets multiple targeted functional domains that are expressed in the same cell and each construct plays multiple, non-overlapping redundant roles in inhibiting NFkB, JNK and p38 pathways.

In one embodiment of the method, the presence of multiple constructs in the compositions produces a synergistic therapeutic effect.

EXAMPLES

Example 1: Preparation of Recombinant Constructs

Sequences. The native YopM, NleE, OSPZ, and IpaH7.8 nucleotide sequences were codon optimized for efficient bacterial expression of the encoded polypeptides. The optimized nucleotide sequences were synthesized de novo as long single stranded nucleotide chains. The overlapping and complementary single strand DNA sequences were annealed and the mixture subject to polymerize chain reaction (PCR) to generate full-length double-stranded (ds) DNA. The ds DNA was gel isolated and cloned into plasmid pQE60. The sequence of the inserted DNA was confirmed by nucleotide sequence analysis.

TAT-NleE wild type and NleE Mutant R107A. Both Vector and Amplicon were digested using BamH1 restriction enzyme overnight at 37° C. The Vector was then phosphatase treated for 1 hour then heat inactivated prior to ligation. All material was run through agarose gels and gene cleaned using a Qiagen gel extraction kit. Ligation reaction was performed at 16° C. overnight, and half of the volume was used to transform the DNA into XL-1 blue competent cells. Colonies from the subsequent transformation were picked and mini-preps were performed using a Qiagen Mini-Prep Kit. Resulting DNA was digested using BamH1 for 1 hr at 37° C. to test for the presence of the insert. 675 bp Positive clones were then digested using EcoR5 (internal to the insert) and external Hind3 site for 1 hr at 37° C. to test the orientation of the insert. Positive orientation yielded a drop-out that was 175 bp in size. Positive clones were then transformed into S9-competent cells to test protein expression rates by mini-induction. A clone was chosen from that group, and used for future maxi-induction purifications under native conditions. A schematic of the construct (based

upon the backbone plasmid pQE60 available from Qiagen Inc, Catalogue number: #32903) is shown in FIG. 3.

TAT-*Shigella* OSPZ. Both Vector and Amplicon were digested using BamH1 restriction enzyme overnight at 37°C. The Vector was then phosphatase treated for 1 hr then the enzyme was heat inactivated prior to ligation. All material was electrophoresed through agarose gels and the DNA isolated from the gel using a Qiagen DNA gel extraction kit. Ligation reaction was performed at 16°C overnight, and half of the volume was used to transform the DNA into XL-1 blue competent cells. Colonies from the subsequent transformation were picked and mini-preps were performed using a Qiagen Mini-Prep Kit. Resulting DNA was digested using BamH1 37°C for 1 hr to test for the presence of the 900 bp insert. Positive clones were then digested using EcoR5 (internal to the insert) and Hind3 located after the HIS-tag 37°C 1 hr to test the orientation of the insert. Positive orientation yielded a drop-out DNA fragment that was 200 bp in size. Positive clones were then transformed into S9-competent cells to test protein expression rates by mini-induction. A clone was chosen from that group, and used for future maxi-induction of proteins and purification of protein under native conditions. A schematic of the construct is shown in FIG. 4.

YopM PTD-NleE wild type and Mutant R107A. Both Vector and Amplicon were digested using BamH1 restriction enzyme overnight at 37°C. The Vector was then phosphatase treated for 1 hr and heat inactivated prior to ligation. All material was run through agarose gels and gene cleaned using a Qiagen gel extraction kit. Ligation reaction was performed at 16°C overnight, and half of the volume was used to transform the DNA into XL-1 blue competent cells. Colonies from the subsequent transformation were picked and mini-preps were performed using a Qiagen Mini-Prep Kit. Resulting DNA was digested using BamH1 at 37°C for 1 hr to test for the presence of the insert. Positive clones were then digested using Bgl2 (internal to the insert) and Bgl2 at 37°C for 1 hr to test the orientation of the insert. Positive orientation yielded a drop-out that was 275 bp in size. Positive clones were then transformed into S9-competent cells to test protein expression rates by mini-induction. A clone was chosen from that group, and used for future maxi-induction purifications under native conditions. A schematic of the construct is shown in FIG. 5.

YopM PTD-NleE wild type and Mutant R107A, No Linker, PAPA Linker, and GSGS Linker. A QE60 vector containing a codon optimized YopM sequence was purchased from Epoch Life Science Inc. The plasmid was digested using Cla1 (internal to YopM) and Bgl2 (located outside of the cloning site). NleE was then digested using Cla1 and BamH1. Note that after ligation, the BamH1/Bgl2 site on the vector was lost. Because the vector was opened at a site that was internal to YopM, the ligation of NleE (along with its mutant R107A counterpart) lead to the fusion of the two inserts. This site is located downstream to the PTD, thus leaving it intact. Variations of this construct were made using either PAPA or GSGS linkers. These were added to the beginning of NleE during PCR by having the bridges be part of the forward primer sequence. Screening of these various constructs were done via PCR of mini-preps using NleE Cla1 FOR and TNHis Bam REV primers designed for NleE. Positive clones were then sent for sequencing to verify the clones prior to transformation into S9-cells for protein expression. A schematic of the construct is shown in FIG. 6.

Schematics of the constructs used are shown in FIG. 1 and FIG. 2.

Purification of recombinant polypeptides. The transformed SG13009 [pREP4] were plated on an agar plate containing both ampicillin and kanamycin. A single colony was used to inoculate a 200 mL culture, shaking at 200 RPM at 37° C. overnight. A volume of the overnight culture was added to 1 L of 2YT medium containing ampicillin and kanamycin and grown as described above in an incubator shaker at 200 RPM 37° C. to an OD 600 nm of 0.6-0.8. Then, 1 mL of 1M Isopropyl β-D-1-thiogalactopyranoside (IPTG) was added to the culture and growth was continued overnight with shaking at 200 RPM at 20° C. The following morning, cells were collected by centrifugation at 6,000 RPM at 4° C. The pelleted cells were resuspended in 40 mL Sonication Buffer and sonicated on ice for two cycles of 4-5 minutes each. The sonification buffer had a pH of 7.5 and 1 L of sonification buffer included one bottle of PBS, 300 mM NaCl, 10% Glycerol, 5 mM Imidazole. The remainder of the volume was filled with ddH₂O. The buffer was then filtered and 1 ml 0.1M PMSF and 1 ml (2 mg/ml) each of Aprotinin, Leupeptine, and Pepstatin were added fresh. The sonicate was centrifuged 12,000 RPM at 4° C. for 30 minutes to separate lysed cells from soluble protein.

The supernatant was collected and incubated with pre-equilibrated Ni-NTA beads (available from BioRad) for 1 hour at 4° C. by rotation. The binding capacity of polypeptides for Ni-NTA beads was about 5-10 mg/ml. Following the binding step, bound materials were centrifuged 4,000 RPM 4° C. for 10 minutes to pellet the Ni-NTA beads and separate from unwanted materials or "flow-through". Beads were then washed twice in 40 mL wash buffer by rotation in 4° C. for 30 minutes per wash cycle. The wash buffer had a pH of 7.5 and 1 L of wash buffer included one bottle of PBS, 300 mM NaCl, 10% Glycerol, and 20 mM Imidazole. The remainder of the volume was filled with ddH₂O. In addition, 1 ml 0.1 M PMSF was added fresh. The washed Ni-Beads were loaded on to a disposable gravity drip column and washed with residual Wash Buffer.

Bound proteins were eluted with elution buffer, using 2× elution buffer per 1 mL of packed volume of beads, i.e., 2 ml elution buffer for 1 ml of Ni-NTA beads. The elution buffer had a pH of 7.5 and 100 ml of elution buffer contained one bottle of PBS, 300 mM NaCl, 10% Glycerol, and 500 mM Imidazole. The remainder of the volume was filled with ddH₂O. The elution buffer was filtered before use. Eluate was collected in 2 mL fractions. The protein content of the fractions was analyzed using Bradford reagent. Individual fractions were dialyzed overnight in dialysis buffer at 4° C. The dialysis buffer had a pH of 7.5 and 4 L of elution buffer included one bottle of PBS, 300 mM NaCl, and 10% Glycerol. The remainder of the volume was then filled with ddH₂O. The following were freshly added to the elution buffer: 1 ml 0.1 M PMSF and 1 ml 1 M DTT. In the morning, the dialysis buffer was discarded and replaced with fresh buffer and dialysis was continued for an additional 3 hours. After dialysis, the individual fractions were pooled and centrifuged at 12,000 RPM at 4° C. for 20 minutes to remove any precipitation. The soluble protein was concentrated in the final concentration was determined using Bradford reagent.

Purified polypeptides were analyzed by SDS-polyacrylamide gel electrophoresis (SDS-PAGE) analysis using known volumes of bovine serum albumin (BSA) as standards. Concentrated samples were aliquoted and stored at -80° C. Proteins were used directly after thawing on ice. An SDS-PAGE analysis of purified recombinant chimeric effec-

tor proteins is shown in FIG. 7. Each lane contained 2 µg of protein. Proteins were visualized by Coomassie blue staining.

Example 3: Cytokine Release

The effect of the bacterial effector proteins on cytokine release was analyzed in a cell-based assay using the human monocyte cell line, THP-1. Cytokine release was measured in vitro using the LEGENDplex Multi-Analyte Flow Assay Kit (BioLegend, Cat no. 740118) according to the supplier's directions. For culturing THP-1 cells, RPMI with glutamine and 20% FBS was used. These cells were incubated at 37° C. in humidified air 5% CO₂ atmosphere. Finally, cell medium was changed every 2-3 days, and the cells were passaged at 95% confluence. To allow the cells to adapt to their environment, 3×10⁵ cells were added to a 24-well plate the night before. The next day, protein was added to these cells for a period of 3 hours. The protein was then washed out with PBS. After cells were stimulated overnight with LPS stimulation, a concentration of 1 µg/mL was used. The next day, cells were washed again and cytokine were measured. All 13 inflammatory cytokines were measured using the LEGENDplex Human Inflammation Plane (13-plex) assay from Biolegend.

We analyzed the effect of effector protein constructs on the release of the pro-inflammatory cytokines IL-1beta, TNF-alpha, IL-6, MCP-1, IL 23 and IL-8. The effector protein constructs are shown in the table below:

Construct name	Protein transduction domain	Effector 1	Linker	Effector 2	SEQ ID NO
YopM Full-length	YopM	YopM			3
TAT-NleE	TAT	EPEC NleE	—	—	28
TAT Shigella OSPZ	TAT	Shigella OSPZ			34
YopM PTD-NleE	YopM	EPEC NleE	—	—	37
YopM PTD-NleE (No linker)	YopM	YopM (L-rich)	—	EPEC NleE	10
YopM PTD-NleE (GSGS linker)	YopM	YopM (L-rich)	GSGS	EPEC NleE	16
YopM PTD-NleE (PAPA linker)	YopM	YopM (L-rich)	PAPA	EPEC NleE	13
YopM PTD-YopM (L-rich)	YopM	YopM (L-rich)	—	—	7
IpaH 7.8		IpaH 7.8			
IpaH 9.8		IpaH 9.8			

The results of these experiments are shown in FIGS. 8, 9, 10, 11 and 12. As shown in FIGS. 8, 9, 10, 11 and 12, the effector proteins reduced the levels of cytokines released into the medium relative to control samples that did not receive effector proteins. This effect was dose-dependent, with higher doses of effector proteins resulting in a steeper reduction in levels of released cytokines. These dose-dependent effects are shown in FIG. 8 with respect to IL-6. Similar dose-dependent effects were observed for IL-1beta, TNF-alpha, IL-6, MCP-1, IL 23, and IL-8.

The effector fusion proteins that included more than one effector protein domain produced a steeper reduction in the level of all released cytokines than did native effector proteins. As shown in FIG. 9, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NleE; YopM PTD-

YopM (L-rich)-GSGS-NleE; YopM PTD-YopM (L-rich)-PAPA NleE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of TNF alpha released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

As shown in FIG. 10, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NleE; YopM PTD-YopM (L-rich)-GSGS-NleE; YopM PTD-YopM (L-rich)-PAPA NleE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of IL-6 released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

As shown in FIG. 11, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NleE; YopM PTD-YopM (L-rich)-GSGS-NleE; YopM PTD-YopM (L-rich)-PAPA NleE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of MCP-1 released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

As shown in FIG. 12, the paired effector fusion proteins YopM PTD-YopM (L-rich)-NleE; YopM PTD-YopM (L-rich)-GSGS-NleE; YopM PTD-YopM (L-rich)-PAPA NleE; and YopM PTD-YopM (L-rich) produced a greater reduction in levels of IL-23 released into the medium shown than did the full length YopM (rYopM) or *Shigella* OSPZ. This reduction was observed for effector fusion proteins that did not include a linker sequence and for effector proteins that included either a GSGS or a PAPA linker sequence.

Similar results were observed for IL-8. Taken together, the results shown in FIGS. 8, 9, 10, 11, and 12 indicate that fusion proteins that target multiple inflammatory pathways are more effective at reducing an inflammatory response than are proteins that target a single inflammatory pathway.

Example 4: Caspase Activity

The effect of YopM on caspase activity was measured using R&D System's Caspase-1/ICE Colorimetric Assay Kit (K111-100) according to the supplier's instructions. THP-1 cells were incubated with rYopM or a truncated YopM (YopMo) and then stimulated with LPS (lipopolysaccharide) and ATP to induce activation of caspase 1. More specifically, 2×10⁶ THP-1 cells were seeded in 6-well cell culture plates in triplicate. The cells were incubated for 2 hours with rYopM (25 µg/ml) or a truncated YopM (YopMo) and then LPS was added (1 µg/mL) and the cells were incubated for an additional 4 hours. Finally, ATP (5 mM) was added and the cells were incubated for 45 minutes. To pellet the cells, centrifugation was carried out at 250×g for 10 min. The cell pellet (2×10⁶ cells) was resuspended in 50 µl lysis buffer (Caspase 1/ICE Colorimetric Assay Kit R & D Systems) and incubated on ice for 10 min. Cell debris were pelleted at 10,000×g for 1 min. The colorimetric assay was carried out in 96-well plates. For this purpose, 50 µl cell lysate/well were placed and mixed with 50 µl 2× reaction buffer (Caspase 1/ICE Colorimetric Assay Kit R & D Systems). After addition of 5 µl of the substrate Ac-YVAD-pNA, incubation was carried out at 37° C. for 3 h. Subsequently, the absorbance at 405 nm was measured.

The results of this experiment are shown in FIG. 13. As shown in FIG. 13, a dose-dependent reduction of caspase 1 activity was observed in THP-1 cells that had been treated with rYopM. A dose-dependent reduction of caspase 1 activity was also observed in THP-1 cells that had been treated with the truncated YopM (YopMo). These data showed that the truncated YopM (YopMo) that contained only the L-rich region of the YopM retained the caspase 1 reducing function of the full length YopM.

Examples 5: Effector Polypeptide Uptake

Confocal microscopy. Cell uptake of effector polypeptides was analyzed using confocal microscopy. We analyzed single cell uptake of the following fusion protein constructs:

Construct name	Protein transduction domain	Effector 1	Linker	Effector 2	SEQ ID NO
TAT-NleE	TAT	EPEC NLeE	—	—	28
YopM PTD-NleE (No linker)	YopM	YopM (L-rich)	—	EPEC NLeE	10
YopM PTD-NleE (GSGS linker)	YopM	YopM (L-rich)	GSGS	EPEC NLeE	16
YopM PTD-NleE (PAPA linker)	YopM	YopM (L-rich)	PAPA	EPEC NLeE	13

Fusion proteins were labeled with FITC using the Sigma-Aldrich FluoroTag™ FITC Conjugation Kit according to the supplier's instructions as follows. Protein and FITC were dissolved in carbonate-bicarbonate buffer. The FITC was slowly added to the protein with stirring, then covered with foil and stirred for two hours at room temperature. The conjugate was separated from free FITC on a G-25 column in the fractions were collected. Fractions containing conjugate were pooled and the F/P ratio of conjugate was determined spectrophotometrically. The labeled protein was stabilized with 1% bovine serum albumin and 0.1% sodium azide and stored at 0-5° C.

HaCat cells were cultured using DMEM High glucose medium with 10% FBS. These cells were incubated at 37° C. in humidified air 5% CO₂ atmosphere. Finally, cell medium was changed every 2-3 days, and the cells were passaged at 95% confluence. Cells were grown to appropriate confluency. Next, appropriate coated coverslips were added to well plates and the cells were seeded in the well plates containing the coverslips. For uptake analysis, polypeptides were added to the cells to a final concentration of 50 µg/mL and incubated for two hours. The protein with the cells were incubated and the protein was aspirated after the incubation period was complete. The cells were fixed with 4% formaldehyde and washed once again. The membrane was permeabilized with 0.2% triton-x-100 and washed again. The cells were blocked for 30 minutes and incubated in primary antibody (either Cy5 or Actin) overnight. The antibody was then aspirated, washed, and incubated in appropriate secondary antibody for 30 minutes. The cells were then washed and DAPI solution was applied for 5 minutes. The cells were then washed and mounted using Prolong Gold Anti-Fade Reagent. The mounted cells were allowed to cure overnight and then imaged. The DAPI was imaged using a wavelength of 490 nm. The Cy5 and Actin were imaged using 594 nm wavelength.

Detection of protein from fixed cells was performed using Nikon 80i upright fluorescent microscope. A filter cube specific for the relative green range (~488 nm) was used to detect FITC labelled protein within cells.

The results of this experiment are shown in FIGS. 14, 15, 16, and 17. As shown in FIG. 14, incubation of cells with QE12-TAT-NleE resulted in nuclear fluorescence. As shown in FIGS. 15 and 16, both QE60-YopM PTD-NleE (GSGS linker) and QE60-YopM PTD-NleE (PAPA linker), respectively, produced punctate cytoplasmic fluorescence that was not observed in untreated control cells (FIG. 17). Taken together, these data show that the fusion polypeptides were taken up by individual cells.

Inverted microscopy. Uptake of QE12-TAT-NleE was analyzed by inverted microscopy as follows. A segment of skin was isolated from a shaved section of a mouse, and the tissue was preserved in a saline solution to maintain the cell viability within the tissue. The tissue was stained with 50 µg of FITC labeled TAT-NleE that was applied to the top of the sample and allowed to diffuse over a 2 hour period. Afterwards, the tissue was fixed with paraformaldehyde and stained with DAPI. The skin segment was then sliced in 10 µm increments, and the top slice was observed in an inverted microscope was used with varying objectives. As shown in FIG. 18, the TAT-NleE fusion polypeptide entered intact mouse skin and penetrated through multiple cell layers.

Two-photon microscopy. Penetration of TAT-NleE- into mouse skin was analyzed by two-photon microscopy. A 1 cmx1 cm segment of mouse skin was stained with 50 µg of FITC-labeled TAT-NleE for two hours and counter stained with DAPI. the sample was analyzed by two-photon microscopy. More specifically, mouse hair was removed post-euthanasia and a 1 cmx1 cm segment of skin was removed and submerged in 1xPBS pH 7.4 to maintain tissue viability. Afterwards, the PBS was removed, and 50 µg FITC-TAT-NleE was applied to the top of the skin and allowed to incubate at room temperature for two hours. Afterwards, the tissue was washed three times for five minutes each with 1xPBS pH 7.4 to remove all waste. From there the tissue was permeabilized with Triton-x and fixed with 4% PFA. DAPI was then added to solution of PBS at a concentration of 1:1000 and incubated in the tissue for 3-5 minutes. Then the washing period was repeated (three times for five minutes each in 1 ml of PBS). The tissue was then mounted on a dish submerged in PBS ready for 2-photon imaging.

A 3 dimensional image is shown in FIG. 19 and a 10 µm slice of the stained tissue is shown in FIG. 20. As shown in FIGS. 19 and 20, the TAT-NleE penetrated the skin into the epidermis, but not the dermis.

Example 6: Methylase Activity

Methylase activity of recombinant proteins that included NleE effectors was analyzed using the MTase glo assay from Promega. We assayed the methylase activity of wild-type NleE, mutant NleE R107A and fusion proteins YopM PTD-NleE (GSGS linker), YopM PTD-NleE (PAPA linker). We also analyzed the methylase activity of NleE from *Shigella*.

The in vitro methylase assay was developed using as a base, the MTase glo assay from Promega Inc. The primary assay buffer for enzyme activity was (25 mM Tris, pH 8, 50 mM NaCl, 1 mM EDTA, 3 mM MgCl₂, 0.1% BSA, 0.005% Tween20, 5 mM DTT). To determine the effect of various vehicles and buffers on NleE a 40 µL aliquot of vehicle was spiked with 1 µL of 20 mg/mL NleE (0.5 mg/mL final). After 30 mins, 5 µL of spiked vehicle was diluted into 5000 µL of primary assay buffer—(termed 'SPIKED' in graphs).

To determine the effect of diluted vehicle on NLeE activity, 5 uL of vehicle diluted to 5000 uL of assay buffer was used. 1000 uL was then removed and used for the no enzyme controls. 1 uL of 0.5 mg/mL NLEE was then added to 1000 uL of diluted vehicle and the assay commenced. The methyltransferase assay itself was performed in 4x1:2 serial dilutions of each sample generated above in assay buffer and 2.5 uL added to 384-well assay plate. Reactions were initiated at various time points by the addition of 2.5 uL of 2x MTaseGlo, 1 uM GST-TAB2 protein substrate and 20 uM SAM in assay buffer. After the time course, reactions were terminated by the addition of 5 uL Methyltransferase Detection Reagent and the multiwell plates read for quantitative O.D. using a standard plate reader.

The results of this analysis are shown in FIGS. 21, 22 and 23. As shown in FIG. 21, both fusion constructs retained the methylase functionality of native NLeE. As shown in FIG. 22, the presence of an R107A mutation in NLeE abolished

The experimental setup was exactly as described above except that the buffer controls contained an equivalent amount of Formula A, Formula B or Formula C as controls. The results of this experiment are summarized in FIG. 24. As shown in FIG. 24a, NLeE methylase activity was retained in the presence of all three formulations. As shown in FIG. 24b, NLeE was stable in the presence of each of the formulations, although the degree of stability varied.

Example 8: How Specific is NleE for Inhibiting NFkB?

We cloned expressed and tested 64 other human C4 ZF containing proteins, among them the sequences identified in the table below. But few of them were modified by NleE. Thus, the specificity of NleE is extremely high.

CAN15_HUMAN/7-26	WsCvr..CtflNpagqrqCsiC	
CAN15_HUMAN/48-67	WpCar..CtfrNlglkeaCevC	
CAN15_HUMAN/147-166	WaCpr..CtllhNtpvassCvcC	
CAN15_HUMAN/344-363	WsCak..CtfrNptvaprCsaC	
CAN15_HUMAN/416-435	WaCpa..CtllNalrakiCaaC	
EWS_HUMAN/522-543	WqCpnpgCgnqNfawrteCnqC	Oncogene RNA Binder
FUS_HUMAN/426-447	WkCpnptCenmNfswrneCnqC	Oncogene Annealing
HOIL1_HUMAN/197-216	WqCpg..CtflNkptrpgCemC	Binds Ub Chains
MDM2_HUMAN/303-322	WkCts..CnemNpplpshCnrC	P53 regulation
MDM4_HUMAN/304-323	WqCte..CkklNspkskryCfrC	P53 regulation
NEIL3_HUMAN/321-340	WtCvv..CtllNkpsskaCdaC	Endonuclease8 Repair
NPL4_HUMAN/584-603	WaCqh..CtfinNqpgtghCemC	Ub Chaperone Erad
NRP1_YEAST/359-378	WnCps..CgfsNfqrtaCfrC	Rna Binder No Fcn
NRP1_YEAST/585-604	WkCst..CtyhNfaknvvClrC	
NU153_HUMAN/662-681	WqCdt..CllqNkvtdnkCiaC	NUP153 Nuc Pore
NU153_HUMAN/726-745	WdCdt..ClvqNkpeakCvaC	
NU153_HUMAN/797-816	WeCsv..CcvSNaednkCvsC	
NU153_HUMAN/855-874	WdCel..ClvqNkadstkClaC	
RBM10_HUMAN/217-236	WlCnk..CgvqNfkrrekCfkC	RNA Binding
RBM5_HUMAN/185-204	WlCnk..CclnNfkrklrCfrC	RNA Binding
RBP2_HUMAN/1356-1375	WhCns..CslkNastakkCvsC	RANBP2 Huge Nuc Pore
RBP2_HUMAN/1419-1438	WdCsi..ClvrNeptvsrCiaC	
RBP2_HUMAN/1483-1502	WdCsa..ClvqNegsstkCaaC	
RBP2_HUMAN/1547-1566	WdCss..ClvrNeatratCvaC	
RBP2_HUMAN/1610-1629	WdCsv..ClvrNeasatkCiaC	
RBP2_HUMAN/1669-1688	WdCsv..ClvrNeasatkCiaC	
RBP2_HUMAN/1728-1747	WdCsv..ClvrNeasatkCiaC	
RBP2_HUMAN/1785-1804	WdCsv..CcvqNessslkCvaC	
RBP56_HUMAN/358-379	WvCpnpsCgnmNfarnsCnqC	
RNF31_HUMAN/304-323	WhCaa..CamlNepwavlCvaC	
RNF31_HUMAN/354-373	WaCqs..CtfeNeaavilCsiC	
RNF31_HUMAN/413-432	WyCih..CtfeNsspgwvCvmC	
RYBP_HUMAN/25-44	WdCsv..CtfrNsacafkCsiC	
SHRPN_HUMAN/352-371	WsCps..CtflNapdrpgCemC	
TAB2_HUMAN/668-687	WnCta..CtflNhpallrCeqC	NFKB pathway
TAB3_HUMAN/687-706	WnCds..CtflNhpalnrCeqC	NFKB pathway
TX13A_HUMAN/380-399	WdCpw..CnavNfsrrdtCfdC	

methylase activity. As shown in FIG. 23, the *Shigella* NleE showed methylase activity at levels comparable to that seen for the EPEC NleE

Example 7: Effect of Formulations on NleE Methylation Activity

We analyzed the effective three different commercial formulations/delivery agents on NleE methylation activity in vitro. These were designated: Formula A, Formula B, and Formula C

A cysteine methylation assay was performed and shows that NleE does not utilize zinc fingers of known highly related zinc fingers. NleE targets NFKB signaling at a key step in TAK1 kinase activation. The CYS-4 Zinc Finger of TAB2 was required to bind free K63 Ub chains to activate TAK1 kinase, which thereby triggers NFKB signaling. The TAB2/3 Cys 4 Zinc-Finger binds K63 linked Ub chains: The metal chelated structure of the ZF required binding. NleE targets NFKB signaling by methylating a single Cysteine in the C4 ZF TAB2 of TAB2 abolishing K63 Ub sensing. Cys

673 in TAB2 is the target of NleE. This highly specific activity of NleE is repurposed in the compositions described herein to shut down the NFkB pathway and reduce inflammation.

Example 9: Penetration of Recombinant NleE into the Skin in an Ex Vivo Pig Skin Model

A key claim and potential stumbling block to use of NleE as a skin anti-inflammatory compound is to show that this globular folded protein of MW 38 kDa is able to penetrate the dermis and epidermis in order to access cells in which the NFkB system is aberrantly activated. A recombinant construct was prepared by fusing the amino acid sequence of NleE to an N-terminal histidine, and covalently labelling that peptide with FITC dye, resulting in the peptide His-NleE-FITC. --This construct was mixed with the eutectic, choline-based solvent CAGE, as described above at a concentration above at a concentration of 1-10 mg/ml. The amount of peptide in CAGE or PBS was 1 mg/ml. The final CAGE concentration was 90% v/v. The mixture was applied to live pig skin being perfused in a Franz diffusion chamber at 37 degrees C. for 24 or 48 hours. The Franz diffusion chamber is described by Technical Brief 2009 vol. 10, Particle Sciences-Drug Development Services (Bethlehem, PA). After these time periods, skin was harvested, fixed and sections followed by microscopy to visualize the FITC dye. The results (FIG. 25) show that CAGE-solubilized NleE FITC was able to penetrate the stratum corneum, epidermis and dermis and that virtually every cellular compartment of the skin. Up to 20-30 cell layers were contacted.-

In comparison to published reports by Banerjee et. al 2017, particularly FIG. 1 using CAGE to transport the globular proteins ovalbumin and bovine serum albumin, the NleE penetrates as well if not better than these test proteins. NleE solubilized in PBS and applied to the pig skin showed no transduction past the stratum corneum.

In summary, this pre-clinical data provides support that recombinant NleE protein is stable in the eutectic solvent CAGE and can be delivered to multi layers of the skin locally in a topical manner. It also provides support for using NleE protein and the combinations described herein that

contain an intrinsic PTD domain which renders the protein capable of penetrating cells and tissues and turning off inflammatory cytokines. The ability of these combinations of proteins to penetrate, in an apparently active form and inhibit production of a multitude of inflammatory cytokines, is reflected in its therapeutic effect on psoriatic lesions in a mouse skin model. These compositions delivered in a cream topically, in therapeutic levels are not anticipated to appreciably enter the regional lymphatics.

These observations strongly support the use of recombinant bacterial effector proteins containing PTDs (Table 1) are useful as a platform for inhibiting various accessible inflammatory disease processes in the body.

Thus, methods and compositions for treating inflammation related to activation of the NFkB, JNK and p38 pathways, e.g., inflammatory conditions of skin, include a composition comprising a first construct comprising a selected immunomodulatory effector protein or functional equivalent thereof that targets a first functional domain; and at least one additional construct comprising a different effector protein or a functional equivalent thereof that targets an additional functional domain. The first construct is optionally linked to a selected protein transduction domain (PTD) or penetrating peptide (CPP) or to a targeting moiety. The additional construct is optionally linked to the same PTD or CPP of the first construct or to an additional PTD or CPP. In one embodiment, a combination of these constructs is provided. In another embodiment the first construct is fused to one or more additional constructs in a single polypeptide. The composition further comprise an optional linker amino acid sequence interposed between each construct and an optional targeting moiety.

Each and every patent, patent application, and publication, including websites cited throughout specification are incorporated herein by reference. Similarly, the SEQ ID NOs which are reference herein and which appear in the appended Sequence Listing are incorporated by reference. While the invention has been described with reference to particular embodiments, it will be appreciated that modifications can be made without departing from the spirit of the invention. Such modifications are intended to fall within the scope of the appended claims.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 93

<210> SEQ ID NO 1

<211> LENGTH: 224

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
NleE sequence

<400> SEQUENCE: 1

Met Ile Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr
1 5 10 15

Lys Tyr Ala Glu His Val Val Lys Asn Ile Tyr Pro Glu Ile Lys His
20 25 30

Asp Tyr Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser
35 40 45

Gly Ile Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn
50 55 60

Glu Lys Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys

-continued

65	70	75	80
Pro Glu Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu	85	90	95
Gly Ser Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln Phe	100	105	110
Ser Val Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu	115	120	125
Pro Asp Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu	130	135	140
Tyr Ile Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg	145	150	155
Ala Asn Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln	165	170	175
Glu Leu Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met	180	185	190
Ile Ile Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu	195	200	205
Ile Asp Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu	210	215	220

<210> SEQ ID NO 2

<211> LENGTH: 409

<212> TYPE: PRT

<213> ORGANISM: Yersinia pestis

<400> SEQUENCE: 2

Met Phe Ile Asn Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro	1	5	10	15
Leu Arg His Ser Ser Asn Leu Thr Glu Met Pro Val Glu Ala Glu Asn	20	25	30	
Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ser Glu Trp Glu Arg	35	40	45	
Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg	50	55	60	
Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn	65	70	75	80
Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu	85	90	95	
Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser	100	105	110	
Leu Lys Ser Leu Leu Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp	115	120	125	
Leu Pro Pro Leu Leu Glu Tyr Leu Gly Val Ser Asn Asn Gln Leu Glu	130	135	140	
Lys Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Val	145	150	155	160
Asp Asn Asn Ser Leu Lys Lys Leu Pro Asp Leu Pro Pro Ser Leu Glu	165	170	175	
Phe Ile Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln	180	185	190	
Asn Leu Pro Phe Leu Thr Ala Ile Tyr Ala Asp Asn Asn Ser Leu Lys	195	200	205	
Lys Leu Pro Asp Leu Pro Leu Ser Leu Glu Ser Ile Val Ala Gly Asn	210	215	220	

-continued

```

Asn Ile Leu Glu Glu Leu Pro Glu Leu Gln Asn Leu Pro Phe Leu Thr
225                230                235                240

Thr Ile Tyr Ala Asp Asn Asn Leu Leu Lys Thr Leu Pro Asp Leu Pro
                245                250                255

Pro Ser Leu Glu Ala Leu Asn Val Arg Asp Asn Tyr Leu Thr Asp Leu
                260                265                270

Pro Glu Leu Pro Gln Ser Leu Thr Phe Leu Asp Val Ser Glu Asn Ile
                275                280                285

Phe Ser Gly Leu Ser Glu Leu Pro Pro Asn Leu Tyr Tyr Leu Asn Ala
290                295                300

Ser Ser Asn Glu Ile Arg Ser Leu Cys Asp Leu Pro Pro Ser Leu Glu
305                310                315                320

Glu Leu Asn Val Ser Asn Asn Lys Leu Ile Glu Leu Pro Ala Leu Pro
                325                330                335

Pro Arg Leu Glu Arg Leu Ile Ala Ser Phe Asn His Leu Ala Glu Val
                340                345                350

Pro Glu Leu Pro Gln Asn Leu Lys Gln Leu His Val Glu Tyr Asn Pro
                355                360                365

Leu Arg Glu Phe Pro Asp Ile Pro Glu Ser Val Glu Asp Leu Arg Met
370                375                380

Asn Ser Glu Arg Val Val Asp Pro Tyr Glu Phe Ala His Glu Thr Thr
385                390                395                400

Asp Lys Leu Glu Asp Asp Val Phe Glu
                405

```

```

<210> SEQ ID NO 3
<211> LENGTH: 367
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

```

```

<400> SEQUENCE: 3

```

```

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1          5          10          15

Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20        25        30

Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
35        40        45

Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50        55        60

Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65        70        75        80

Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
85        90        95

Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
100       105       110

Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
115       120       125

Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
130       135       140

Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Val
145       150       155       160

Asp Asn Asn Ser Leu Lys Lys Leu Pro Asp Leu Pro Pro Ser Leu Glu
165       170       175

```

-continued

Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Ser Glu Leu Gln
 180 185 190
 Asn Leu Pro Phe Leu Thr Glu Ile His Ala Asp Asn Asn Ser Leu Lys
 195 200 205
 Thr Leu Pro Asp Leu Pro Pro Ser Leu Lys Thr Leu Asn Val Arg Glu
 210 215 220
 Asn Tyr Leu Thr Asp Leu Pro Glu Leu Pro Gln Ser Leu Thr Phe Leu
 225 230 235 240
 Asp Val Ser Asp Asn Ile Phe Ser Gly Leu Ser Glu Leu Pro Pro Asn
 245 250 255
 Leu Tyr Tyr Leu Asp Ala Ser Ser Asn Gly Ile Arg Ser Leu Cys Asp
 260 265 270
 Leu Pro Pro Ser Leu Val Glu Leu Asp Val Arg Asp Asn Gln Leu Ile
 275 280 285
 Glu Leu Pro Ala Leu Pro Pro His Leu Glu Arg Leu Ile Ala Ser Leu
 290 295 300
 Asn His Leu Ala Glu Val Pro Glu Leu Pro Gln Asn Leu Lys Gln Leu
 305 310 315 320
 His Val Glu His Asn Ala Leu Arg Glu Phe Pro Asp Ile Pro Glu Ser
 325 330 335
 Val Glu Asp Leu Arg Met Asp Ser Glu Arg Val Thr Asp Thr Tyr Glu
 340 345 350
 Phe Ala His Glu Thr Thr Asp Lys Leu Glu Asp Asp Val Phe Glu
 355 360 365

<210> SEQ ID NO 4
 <211> LENGTH: 1104
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 4

atgtttataa ctccaagaaa tgtatctaata acttttttgc aagaaccatt acgtcattct 60
 tctgatttaa ctgagatgcc ggttgaagca gaaaatgtta aatctaagac tgaatattat 120
 aatgcatggg cggtatggga acgaaatgcc cctccgggga atggtgaaca gagggaaatg 180
 gcggtttcaa gggtacgcga ttgcctggac cgacaagccc atgagctaga actaaataat 240
 ctggggctga gttctttgcc ggaattacct ccgcatttag agagttagt ggcgctcatgt 300
 aattctctta cagaattacc ggaattgccg cagagcctga aatcacttca agttgataat 360
 aacaatctga aggcattatc cgatttacct ccttcactgg aatttcttgc tgctggtaat 420
 aatcagctgg aagaattgcc agagttgcaa aactcgtcct tcttgaaaa tattgatgtt 480
 gataacaatt cactgaaaaa actacctgat ttacctcctt cactggaatt tcttgctgct 540
 ggtaataatc agctggaaga attgtcagag ttacaaaact tgccattctt gactgagatt 600
 catgctgata acaattcact gaaaacatta cccgatttac cccttccct gaaaacactt 660
 aatgtcagag aaaattatct aactgatctg ccagaattac cgcagagttt aaccttctta 720
 gatgtttctg ataatatctt tcttgatta tcggaattgc caccaaaact gtattatctc 780
 gatgcatcca gcaatggaat aagatcctta tgcgatttac cccttccct ggtagaactt 840
 gatgtcagag ataatcagtt gatcgaaact ccagcgttac ctccacactt agaacgttta 900
 atcgcttcac ttaatcatct tgctgaagta cctgaattgc cgcaaaacct gaaacagctc 960

-continued

```

cacgtagagc acaacgctct aagagagttt cccgatatac ctgagtcagt ggaagatctt 1020
cggtgggact ctgaacgtgt aactgataca tatgaatttg ctcatgagac tacagacaaa 1080
cttgaagatg atgtatttga gtag                                     1104

```

```

<210> SEQ ID NO 5
<211> LENGTH: 86
<212> TYPE: PRT
<213> ORGANISM: Yersinia pestis

```

```

<400> SEQUENCE: 5

```

```

Met Phe Ile Asn Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1          5          10          15
Leu Arg His Ser Ser Asn Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20        25        30
Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ser Glu Trp Glu Arg
35        40        45
Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50        55        60
Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65        70        75        80
Leu Gly Leu Ser Ser Leu
85

```

```

<210> SEQ ID NO 6
<211> LENGTH: 258
<212> TYPE: DNA
<213> ORGANISM: Yersinia pestis

```

```

<400> SEQUENCE: 6

```

```

atgttcacca ctccacgcaa tgtatctaac acctttctgc aggaaccgct gcgtcattct 60
agcgacctga ccgaaatgcc agttgaagcg gagaacgtga aatctaagac tgaatactac 120
aacgcgtggg cagtatggga gcgcaatgca ccaccaggta acggtgaaca gcgtgaaatg 180
gcagtaagcc gtctgcgtga ttgcctggat cgccaggctc acgagctgga gctgaacaac 240
ctgggtctgt ctagcctg                                     258

```

```

<210> SEQ ID NO 7
<211> LENGTH: 131
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

```

```

<400> SEQUENCE: 7

```

```

Met Phe Ile Asn Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1          5          10          15
Leu Arg His Ser Ser Asn Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20        25        30
Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ser Glu Trp Glu Arg
35        40        45
Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50        55        60
Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65        70        75        80
Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
85        90        95

```

-continued

Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
100 105 110

Leu Lys Ser Leu Leu Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
115 120 125

Leu Pro Pro
130

<210> SEQ ID NO 8

<211> LENGTH: 393

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 8

```
atgtttatta acccgcgcaa cgtgagcaac acctttctgc aggaaccgct gcgccatagc    60
agcaacctga ccgaaatgcc ggtggaagcg gaaaacgtga aaagcaaaac cgaatattat    120
aacgcgtgga gcgaatggga acgcaacgcg ccgccgggca acggcgaaca gcgcgaaatg    180
gcggtgagcc gcctgcgcga ttgctgggat cgccaggcgc atgaactgga actgaacaac    240
ctgggcctga gcagcctgcc ggaactgccg ccgcactctgg aaagcctggt ggcgagctgc    300
aacagcctga ccgaaactgcc ggaactgccg cagagcctga aaagcctgct ggtggataac    360
aacaacctga aagcgctgag cgatctgccg ccg                                393
```

<210> SEQ ID NO 9

<211> LENGTH: 390

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 9

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1 5 10 15

Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20 25 30

Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
35 40 45

Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50 55 60

Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65 70 75 80

Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
85 90 95

Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
100 105 110

Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
115 120 125

Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
130 135 140

Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Ile
145 150 155 160

Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
165 170 175

-continued

Ala	Glu	His	Val	Val	Lys	Asn	Ile	Tyr	Pro	Lys	Ile	Lys	His	Asp	Tyr	180	185	190
Phe	Asn	Glu	Ser	Pro	Asn	Ile	Tyr	Asp	Lys	Lys	Tyr	Ile	Ser	Gly	Ile	195	200	205
Thr	Arg	Gly	Val	Ala	Glu	Leu	Lys	Gln	Glu	Glu	Phe	Val	Asn	Glu	Lys	210	215	220
Ala	Arg	Arg	Phe	Ser	Tyr	Met	Lys	Thr	Met	Tyr	Ser	Val	Cys	Pro	Glu	225	230	235
Ala	Phe	Glu	Pro	Ile	Ser	Arg	Asn	Glu	Ala	Ser	Thr	Pro	Glu	Gly	Ser	245	250	255
Trp	Leu	Thr	Val	Ile	Ser	Gly	Lys	Arg	Pro	Met	Gly	Gln	Phe	Ser	Val	260	265	270
Asp	Ser	Leu	Tyr	Asn	Pro	Asp	Leu	His	Ala	Leu	Cys	Glu	Leu	Pro	Asp	275	280	285
Ile	Cys	Cys	Lys	Ile	Phe	Pro	Lys	Glu	Asn	Asn	Asp	Phe	Leu	Tyr	Ile	290	295	300
Val	Val	Val	Tyr	Arg	Asn	Asp	Ser	Pro	Leu	Gly	Glu	Gln	Arg	Ala	Asn	305	310	315
Arg	Phe	Ile	Glu	Leu	Tyr	Asn	Ile	Lys	Arg	Asp	Ile	Met	Gln	Glu	Leu	325	330	335
Asn	Tyr	Glu	Leu	Pro	Glu	Leu	Lys	Ala	Val	Lys	Ser	Glu	Met	Ile	Ile	340	345	350
Ala	Arg	Glu	Met	Gly	Glu	Ile	Phe	Ser	Tyr	Met	Pro	Gly	Glu	Ile	Asp	355	360	365
Ser	Tyr	Met	Lys	Tyr	Ile	Asn	Asn	Lys	Leu	Ser	Lys	Ile	Glu	Gly	Ser	370	375	380
His	His	His	His	His	His											385	390	

<210> SEQ ID NO 10
 <211> LENGTH: 384
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 10

Met	Phe	Ile	Thr	Pro	Arg	Asn	Val	Ser	Asn	Thr	Phe	Leu	Gln	Glu	Pro	1	5	10	15
Leu	Arg	His	Ser	Ser	Asp	Leu	Thr	Glu	Met	Pro	Val	Glu	Ala	Glu	Asn	20	25	30	
Val	Lys	Ser	Lys	Thr	Glu	Tyr	Tyr	Asn	Ala	Trp	Ala	Val	Trp	Glu	Arg	35	40	45	
Asn	Ala	Pro	Pro	Gly	Asn	Gly	Glu	Gln	Arg	Glu	Met	Ala	Val	Ser	Arg	50	55	60	
Leu	Arg	Asp	Cys	Leu	Asp	Arg	Gln	Ala	His	Glu	Leu	Glu	Leu	Asn	Asn	65	70	75	80
Leu	Gly	Leu	Ser	Ser	Leu	Pro	Glu	Leu	Pro	Pro	His	Leu	Glu	Ser	Leu	85	90	95	
Val	Ala	Ser	Cys	Asn	Ser	Leu	Thr	Glu	Leu	Pro	Glu	Leu	Pro	Gln	Ser	100	105	110	
Leu	Lys	Ser	Leu	Gln	Val	Asp	Asn	Asn	Asn	Leu	Lys	Ala	Leu	Ser	Asp	115	120	125	
Leu	Pro	Pro	Ser	Leu	Glu	Phe	Leu	Ala	Ala	Gly	Asn	Asn	Gln	Leu	Glu	130	135	140	

-continued

Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Ile
 145 150 155 160
 Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
 165 170 175
 Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
 180 185 190
 Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
 195 200 205
 Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
 210 215 220
 Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
 225 230 235 240
 Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
 245 250 255
 Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val
 260 265 270
 Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
 275 280 285
 Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
 290 295 300
 Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
 305 310 315 320
 Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu
 325 330 335
 Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile
 340 345 350
 Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp
 355 360 365
 Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 370 375 380

<210> SEQ ID NO 11
 <211> LENGTH: 1173
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 11

atgttcacga ctccacgcaa tgtatctaac acctttctgc aggaaccgct gcgtcattct	60
agcgacctga ccgaaatgcc agttgaagcg gagaacgtga aatctaagac tgaatactac	120
aacgcgtggg cagtatggga gcgcaatgca ccaccaggta acggtgaaca gcgtgaaatg	180
gcagtaagcc gtctgcgtga ttgcctggat cgccaggctc acgagctgga gctgaacaac	240
ctgggtctgt ctagcctgcc agagctccca ccacatctgg aaagcctggt ggctagctgt	300
aactctctga ctgaactgcc agagctgcca caaagcctga aatccctgca ggtggacaac	360
aacaacctga aagcgctgtc cgatctgcca ccgtctctgg agtttctggc agctggtaac	420
aaccaactgg aagaactgcc agagctgcag aactcctcct ttctgaagat catcgatatt	480
aatcctgtta ctaatactca ggcggtgtcc cctataaata ctaaatatgc tgaacatgtg	540
gtgaaaaata tttaccgcaa aattaacat gattacttta atgaatcacc caatatatat	600
gataagaagt atatatccgg tataaccaga ggagtagctg aactaaaaca ggaagaattt	660

-continued

```

gttaacgaga aagccagacg gttttcttat atgaagacta tgtattctgt atgtccagaa 720
gcgtttgaac ctatttccag aaatgaagcc agtacaccgg aaggaagctg gctaacagtt 780
atatccggaa aagcccaat ggggcagttt tctgtagata gtttatacaa tcttgattta 840
catgcattat gtgagcttcc ggacatttgt tgtaagatct tccctaaaga aaataatgat 900
ttttataca tagttgttgt gtacagaaat gacagccctc taggagaaca acgggcaaat 960
agatttatag aattatataa tataaaaaga gatatcatgc aggaattaaa ttatgagtta 1020
ccagagttaa aggcagtaaa atctgaaatg attatcgcac gtgaaatggg agaaatcttt 1080
agctacatgc ctggggaaat agacagttat atgaaataca taaataataa actttctaaa 1140
attgagggat ctcacacca tcaccatcac taa 1173

```

```

<210> SEQ ID NO 12
<211> LENGTH: 406
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

```

```

<400> SEQUENCE: 12

```

```

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1      5      10      15
Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20     25     30
Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
35     40     45
Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50     55     60
Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65     70     75     80
Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
85     90     95
Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
100    105    110
Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
115    120    125
Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
130    135    140
Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Pro
145    150    155    160
Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Ile
165    170    175
Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
180    185    190
Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
195    200    205
Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
210    215    220
Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
225    230    235    240
Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
245    250    255
Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
260    265    270

```

-continued

Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val
 275 280 285
 Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
 290 295 300
 Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
 305 310 315 320
 Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
 325 330 335
 Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu
 340 345 350
 Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile
 355 360 365
 Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp
 370 375 380
 Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 385 390 395 400
 His His His His His His
 405

<210> SEQ ID NO 13
 <211> LENGTH: 400
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 13

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
 1 5 10 15
 Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
 20 25 30
 Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
 35 40 45
 Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
 50 55 60
 Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
 65 70 75 80
 Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
 85 90 95
 Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
 100 105 110
 Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
 115 120 125
 Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
 130 135 140
 Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Pro
 145 150 155 160
 Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Ile
 165 170 175
 Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
 180 185 190
 Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
 195 200 205
 Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile

-continued

210	215	220
Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys		
225	230	235 240
Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu		
	245	250 255
Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser		
	260	265 270
Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val		
	275	280 285
Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp		
	290	295 300
Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile		
305	310	315 320
Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn		
	325	330 335
Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu		
	340	345 350
Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile		
	355	360 365
Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp		
	370	375 380
Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser		
385	390	395 400

<210> SEQ ID NO 14

<211> LENGTH: 1221

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 14

```

atgttcacatca ctccacgcaa tgtatctaac acctttctgc aggaaccgct gcgtcattct      60
agcgacctga ccgaaatgcc agtgaagcg gagaacgtga aatctaagac tgaatactac      120
aacgcgtggg cagtatggga gcgcaatgca ccaccaggtg acggtgaaca gcgtgaaatg      180
gcagtaagcc gtctgcgtga ttgcttgat cgccaggctc acgagctgga gctgaacaac      240
ctgggtctgt ctagctgcc agagctccca ccacatctgg aaagcctggt ggctagctgt      300
aactctctga ctgaactgcc agagctgcca caaagcctga aatccctgca ggtggacaac      360
aacaacctga aagcgtgtgc cgaatcgcca ccgtctctgg agtttctggc agctggtaac      420
aaccaactgg aagaactgcc agagctgcag aactcctcct ttctgaagat catcgatcca      480
gcaccagcac cagcaccagc accagcacca gcaccagcac cagcaattaa tctgttact      540
aatactcagg gcgtgtcccc tataaatact aaatatgctg aacatgtggt gaaaaatatt      600
taccgaaaa ttaaacatga ttactttaat gaatcaccca atatatatga taagaagtat      660
atatccggta taaccagagg agtagctgaa ctaaacacagg aagaatttgt taacgagaaa      720
gccagacggt tttcttatat gaagactatg tattctgtat gtccagaagc gtttgaacct      780
atttccagaa atgaagccag tacaccggaa ggaagctggc taacagtat atccgaaaaa      840
cgcccaatgg ggcagttttc ttagatagt ttatacaatc ctgatttaca tgcattatgt      900
gagcttccgg acatttgttg taagatcttc cctaaagaaa ataatgattt ttatacata      960
gtgtgtgtgt acagaaatga cagccctcta ggagaacaac gggcaaatag atttatagaa     1020

```

-continued

```

ttatataata taaaagaga tatcatgcag gaattaaatt atgagttacc agagttaaag 1080
gcagtaaaat ctgaaatgat tatcgacagt gaaatgggag aaatcttttag ctacatgcct 1140
ggggaaatag acagtttatat gaaatacata aataataaac tttctaaaat tgagggatct 1200
catcaccatc accatcacta a 1221

```

```

<210> SEQ ID NO 15
<211> LENGTH: 406
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 15

```

```

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1      5      10      15
Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20     25     30
Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
35     40     45
Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50     55     60
Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65     70     75     80
Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
85     90     95
Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
100    105    110
Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
115    120    125
Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
130    135    140
Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Gly
145    150    155    160
Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Ile
165    170    175
Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
180    185    190
Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
195    200    205
Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
210    215    220
Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
225    230    235    240
Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
245    250    255
Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
260    265    270
Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val
275    280    285
Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
290    295    300
Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
305    310    315    320

```

-continued

Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
 325 330 335

Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu
 340 345 350

Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile
 355 360 365

Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp
 370 375 380

Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 385 390 395 400

His His His His His His
 405

<210> SEQ ID NO 16
 <211> LENGTH: 400
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 16

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
 1 5 10 15

Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
 20 25 30

Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
 35 40 45

Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
 50 55 60

Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
 65 70 75 80

Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
 85 90 95

Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
 100 105 110

Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
 115 120 125

Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
 130 135 140

Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Gly
 145 150 155 160

Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Ile
 165 170 175

Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
 180 185 190

Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
 195 200 205

Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
 210 215 220

Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
 225 230 235 240

Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
 245 250 255

Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser

-continued

260	265	270
Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val		
275	280	285
Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp		
290	295	300
Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile		
305	310	315
Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn		
325	330	335
Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu		
340	345	350
Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile		
355	360	365
Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp		
370	375	380
Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser		
385	390	395
		400
<210> SEQ ID NO 17		
<211> LENGTH: 1221		
<212> TYPE: DNA		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide		
<400> SEQUENCE: 17		
atgttcac	ca ctccacgcaa	tgtatctaac acctttctgc aggaaccgct gcgtcattct 60
agcgacctga	ccgaaatgcc	agtggaagcg gagaacgtga aatctaagac tgaatactac 120
aacgcgtggg	cagtatggga	gcgcaatgca ccaccaggta acggtgaaca gcgtgaaatg 180
gcagtaagcc	gtctgcgtga	ttgcctggat cgcaggctc acgagctgga gctgaacaac 240
ctgggtctgt	ctagcctgcc	agagctccca ccacatctgg aaagcctggg ggctagctgt 300
aactctctga	ctgaactgcc	agagctgcca caaagcctga aatccctgca ggtggacaac 360
aacaacctga	aagcgtgtgc	cgatctgcca ccgtctctgg agtttctggc agctggtaac 420
aaccaactgg	aagaactgcc	agagctgcag aactcctcct ttctgaagat catcgatggg 480
agcggtagcg	gtagcggtag	cggtagcggg agcggtagcg gtagcattaa tcctgttact 540
aatactcagg	gcgtgtcccc	tataaatact aaatatgctg aacatgtggg gaaaaatatt 600
taccgaaaa	ttaaacatga	ttactttaat gaatcaccca atatatatga taagaagtat 660
atatccggta	taaccagagg	agtagctgaa ctaaaacagg aagaatttgt taacgagaaa 720
gccagacggg	tttcttatat	gaagactatg tattctgtat gtccagaagc gtttgaacct 780
atttccagaa	atgaagccag	tacaccggaa ggaagctggc taacagttat atccgaaaa 840
gccccaatgg	ggcagttttc	tgtagatagt ttatacaatc ctgatttaca tgcattatgt 900
gagcttccgg	acattttgtg	taagatcttc cctaaagaaa ataagtattt ttatatacata 960
gttggtgtgt	acagaaatga	cagccctcta ggagaacaac gggcaaatag atttatagaa 1020
ttatataata	taaaaagaga	tatcatgcag gaattaaatt atgagttacc agagttaaag 1080
gcagtaaaat	ctgaaatgat	tatgcacagt gaaatgggag aaatctttag ctacatgcct 1140
ggggaaatag	acagttatat	gaaatacata aataataaac tttctaaaat tgagggatct 1200
catcaccatc	accatcacta	a 1221

-continued

```

<210> SEQ ID NO 18
<211> LENGTH: 390
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 18

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1      5      10      15
Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20     25     30
Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
35     40     45
Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50     55     60
Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65     70     75     80
Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
85     90     95
Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
100    105   110
Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
115    120   125
Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
130    135   140
Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Ile
145    150   155   160
Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
165    170   175
Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
180    185   190
Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
195    200   205
Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
210    215   220
Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
225    230   235   240
Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
245    250   255
Trp Leu Thr Val Ile Ser Gly Lys Ala Pro Met Gly Gln Phe Ser Val
260    265   270
Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
275    280   285
Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
290    295   300
Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
305    310   315   320
Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu
325    330   335
Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile
340    345   350
Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp
355    360   365

```

-continued

Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 370 375 380

His His His His His His
 385 390

<210> SEQ ID NO 19
 <211> LENGTH: 384
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 19

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
 1 5 10 15
 Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
 20 25 30
 Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
 35 40 45
 Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
 50 55 60
 Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
 65 70 75 80
 Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
 85 90 95
 Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
 100 105 110
 Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
 115 120 125
 Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
 130 135 140
 Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Ile
 145 150 155 160
 Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
 165 170 175
 Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
 180 185 190
 Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
 195 200 205
 Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
 210 215 220
 Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
 225 230 235 240
 Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
 245 250 255
 Trp Leu Thr Val Ile Ser Gly Lys Ala Pro Met Gly Gln Phe Ser Val
 260 265 270
 Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
 275 280 285
 Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
 290 295 300
 Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
 305 310 315 320
 Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu

-continued

325	330	335	
Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile			
340	345	350	
Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp			
355	360	365	
Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser			
370	375	380	

<210> SEQ ID NO 20
 <211> LENGTH: 1173
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 20

atgttcac	ca	ctccacg	caa	tgtatcta	ac	acctttct	gc	aggaaccg	ct	gcgtcatt	ct	60
agcgacct	ga	ccgaaatg	cc	agtggaag	cg	gagaacgt	ga	aatctaag	ac	tgaatact	ac	120
aacgcgtg	ggg	cagtatgg	ga	gcgcaatg	ca	ccaccagg	ta	acggtgaa	ca	gcgtgaa	atg	180
gcagtaag	cc	gtctgcgt	ga	ttgcctgg	at	cgccaggc	tc	acgagctg	ga	gctgaaca	ac	240
ctgggtct	gt	ctagcctg	cc	agagctcc	ca	ccacatct	gg	aaagcctg	gt	ggctagct	gt	300
aactctct	ga	ctgaactg	cc	agagctgc	ca	caaagcct	ga	aatccctg	ca	ggtgga	caac	360
aacaacct	ga	aagcgctg	tc	cgatctgc	ca	ccgtctct	gg	agtttctg	gc	agctgg	taac	420
aaccaact	gg	aagaactg	cc	agagctgc	ag	aactcctc	ct	ttctgaag	at	catcgat	att	480
aatcctgt	ta	ctaatact	ca	ggcgctgt	cc	cctataaa	ta	ctaaatat	gc	tgaacat	gtg	540
gtgaaaaa	ta	tttaccgg	aa	aattaaac	at	gattactt	ta	atgaatca	cc	caatatat	at	600
gataaga	agt	atatatcc	gg	tataacc	aga	ggagtagc	tg	aactaaa	aca	ggaaga	attt	660
gttaacg	aga	aagccag	acg	gttttctt	at	atgaagac	ta	tgtattct	gt	atgtcc	agaa	720
gcgtttga	aac	ctatttcc	ag	aaatgaag	cc	agtacacc	gg	aaggaagc	tg	gctaac	agtt	780
atatccgg	aa	aagcccca	at	ggggcagt	tt	tctgtaga	ta	gtttata	caa	tcctgat	tta	840
catgcatt	at	gtgagctt	cc	ggacattt	gt	tgtaagat	ct	tcctaa	aaga	aaataat	gat	900
ttttata	ca	tagttgtt	gt	gtacagaa	at	gacagcc	ctc	taggaga	aaca	acgggca	aat	960
agatttat	ag	aattatata	aa	tataaaa	aga	gatatcat	gc	aggaatta	aaa	ttatgag	tta	1020
ccagagt	taa	aggcagta	aaa	atctgaaa	tg	attatcg	cac	gtgaaatg	ggg	agaaatc	ttt	1080
agctacat	gc	ctggggaa	aat	agacagt	tat	atgaaata	ca	taaataa	ataa	actttct	aaa	1140
attgaggg	gat	ctcatcac	ca	tcaccat	cac	taa						1173

<210> SEQ ID NO 21
 <211> LENGTH: 406
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 21

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro			
1	5	10	15
Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn			
20	25	30	

-continued

Val	Lys	Ser	Lys	Thr	Glu	Tyr	Tyr	Asn	Ala	Trp	Ala	Val	Trp	Glu	Arg
	35						40					45			
Asn	Ala	Pro	Pro	Gly	Asn	Gly	Glu	Gln	Arg	Glu	Met	Ala	Val	Ser	Arg
	50					55					60				
Leu	Arg	Asp	Cys	Leu	Asp	Arg	Gln	Ala	His	Glu	Leu	Glu	Leu	Asn	Asn
	65				70					75					80
Leu	Gly	Leu	Ser	Ser	Leu	Pro	Glu	Leu	Pro	Pro	His	Leu	Glu	Ser	Leu
			85						90					95	
Val	Ala	Ser	Cys	Asn	Ser	Leu	Thr	Glu	Leu	Pro	Glu	Leu	Pro	Gln	Ser
			100					105						110	
Leu	Lys	Ser	Leu	Gln	Val	Asp	Asn	Asn	Asn	Leu	Lys	Ala	Leu	Ser	Asp
	115						120					125			
Leu	Pro	Pro	Ser	Leu	Glu	Phe	Leu	Ala	Ala	Gly	Asn	Asn	Gln	Leu	Glu
	130					135					140				
Glu	Leu	Pro	Glu	Leu	Gln	Asn	Ser	Ser	Phe	Leu	Lys	Ile	Ile	Asp	Gly
	145				150					155					160
Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Ile
			165					170							175
Asn	Pro	Val	Thr	Asn	Thr	Gln	Gly	Val	Ser	Pro	Ile	Asn	Thr	Lys	Tyr
			180					185						190	
Ala	Glu	His	Val	Val	Lys	Asn	Ile	Tyr	Pro	Lys	Ile	Lys	His	Asp	Tyr
	195						200					205			
Phe	Asn	Glu	Ser	Pro	Asn	Ile	Tyr	Asp	Lys	Lys	Tyr	Ile	Ser	Gly	Ile
	210					215					220				
Thr	Arg	Gly	Val	Ala	Glu	Leu	Lys	Gln	Glu	Glu	Phe	Val	Asn	Glu	Lys
	225				230				235						240
Ala	Arg	Arg	Phe	Ser	Tyr	Met	Lys	Thr	Met	Tyr	Ser	Val	Cys	Pro	Glu
			245					250						255	
Ala	Phe	Glu	Pro	Ile	Ser	Arg	Asn	Glu	Ala	Ser	Thr	Pro	Glu	Gly	Ser
			260					265					270		
Trp	Leu	Thr	Val	Ile	Ser	Gly	Lys	Ala	Pro	Met	Gly	Gln	Phe	Ser	Val
	275						280					285			
Asp	Ser	Leu	Tyr	Asn	Pro	Asp	Leu	His	Ala	Leu	Cys	Glu	Leu	Pro	Asp
	290					295					300				
Ile	Cys	Cys	Lys	Ile	Phe	Pro	Lys	Glu	Asn	Asn	Asp	Phe	Leu	Tyr	Ile
	305				310					315					320
Val	Val	Val	Tyr	Arg	Asn	Asp	Ser	Pro	Leu	Gly	Glu	Gln	Arg	Ala	Asn
			325						330					335	
Arg	Phe	Ile	Glu	Leu	Tyr	Asn	Ile	Lys	Arg	Asp	Ile	Met	Gln	Glu	Leu
			340					345					350		
Asn	Tyr	Glu	Leu	Pro	Glu	Leu	Lys	Ala	Val	Lys	Ser	Glu	Met	Ile	Ile
	355						360					365			
Ala	Arg	Glu	Met	Gly	Glu	Ile	Phe	Ser	Tyr	Met	Pro	Gly	Glu	Ile	Asp
	370					375					380				
Ser	Tyr	Met	Lys	Tyr	Ile	Asn	Asn	Lys	Leu	Ser	Lys	Ile	Glu	Gly	Ser
	385				390				395						400
His	His	His	His	His	His										
					405										

<210> SEQ ID NO 22

<211> LENGTH: 400

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

polypeptide

<400> SEQUENCE: 22

```

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1           5           10           15

Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20           25           30

Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
35           40           45

Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50           55           60

Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65           70           75           80

Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
85           90           95

Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
100          105          110

Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
115          120          125

Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
130          135          140

Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Gly
145          150          155          160

Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Ile
165          170          175

Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
180          185          190

Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
195          200          205

Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
210          215          220

Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
225          230          235          240

Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
245          250          255

Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
260          265          270

Trp Leu Thr Val Ile Ser Gly Lys Ala Pro Met Gly Gln Phe Ser Val
275          280          285

Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
290          295          300

Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
305          310          315          320

Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
325          330          335

Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu
340          345          350

Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile
355          360          365

Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp
370          375          380

Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
385          390          395          400

```

-continued

```

<210> SEQ ID NO 23
<211> LENGTH: 1221
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 23

atgttcacatca ctccacgcaa tgtatctaac acctttctgc aggaaccgct gcgtcattct      60
agcgacctga cggaaatgcc agttgaagcg gagaacgtga aatctaagac tgaatactac      120
aacgcgtggg cagtatggga gcgcaatgca ccaccaggta acggtgaaca gcgtgaaatg      180
gcagtaagcc gtctgcgtga ttgcctggat cgccaggctc acgagctgga gctgaacaac      240
ctgggtctgt ctgacctgcc agagctccca ccacatctgg aaagcctggg ggctagctgt      300
aactctctga ctgaactgcc agagctgcca caaagcctga aatccctgca ggtggacaac      360
aacaacctga aagcgctgtc cgatctgcca ccgtctctgg agtttctggc agctggtaac      420
aaccaactgg aagaactgcc agagctgcag aactcctcct ttctgaagat catcgatggg      480
agcggtagcg gtagcggtag cggtagcggg agcggtagcg gtagcattaa tcctgttact      540
aatactcagg gcgtgtcccc tataaatact aaatatgctg aacatgtggg gaaaaatatt      600
taccggaaaa ttaaacatga ttactttaat gaatcaccca atatatatga taagaagtat      660
atatccggta taaccagagg agtagctgaa ctaaaacagg aagaatttgt taacgagaaa      720
gccagacggg tttcttatat gaagactatg tattctgtat gtccagaagc gtttgaacct      780
atttccagaa atgaagccag tacaccggaa ggaagctggc taacagttaa atccggaaaa      840
gccccaatgg ggcagttttc ttagatagtg ttatacaatc ctgatttaca tgcattatgt      900
gagcttccgg acattttgtg taagatcttc cctaaagaaa ataattgattt ttatacata      960
gttgttgtgt acagaaatga cagccctcta ggagaacaac gggcaaatag atttatagaa     1020
ttatataata taaaaagaga tatcatgcag gaattaaatt atgagttacc agagttaaag     1080
gcagtaaaat ctgaaatgat tatcgcacgt gaaatgggag aaatctttag ctacatgcct     1140
ggggaaatag acagttatat gaaatacata aataataaac tttctaaaaa tgagggatct     1200
catcaccatc accatcacta a                                     1221

```

```

<210> SEQ ID NO 24
<211> LENGTH: 406
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 24

Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln Glu Pro
1           5           10          15

Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20          25          30

Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
35          40          45

Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
50          55          60

Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
65          70          75          80

```

Leu	Gly	Leu	Ser	Ser	Leu	Pro	Glu	Leu	Pro	Pro	His	Leu	Glu	Ser	Leu	
				85					90					95		
Val	Ala	Ser	Cys	Asn	Ser	Leu	Thr	Glu	Leu	Pro	Glu	Leu	Pro	Gln	Ser	
				100					105					110		
Leu	Lys	Ser	Leu	Gln	Val	Asp	Asn	Asn	Asn	Leu	Lys	Ala	Leu	Ser	Asp	
				115					120					125		
Leu	Pro	Pro	Ser	Leu	Glu	Phe	Leu	Ala	Ala	Gly	Asn	Asn	Gln	Leu	Glu	
				130					135					140		
Glu	Leu	Pro	Glu	Leu	Gln	Asn	Ser	Ser	Phe	Leu	Lys	Ile	Ile	Asp	Pro	
				145					150					155		
Ala	Pro	Ala	Pro	Ala	Pro	Ala	Pro	Ala	Pro	Ala	Pro	Ala	Pro	Ala	Ile	
				165					170					175		
Asn	Pro	Val	Thr	Asn	Thr	Gln	Gly	Val	Ser	Pro	Ile	Asn	Thr	Lys	Tyr	
				180					185					190		
Ala	Glu	His	Val	Val	Lys	Asn	Ile	Tyr	Pro	Lys	Ile	Lys	His	Asp	Tyr	
				195					200					205		
Phe	Asn	Glu	Ser	Pro	Asn	Ile	Tyr	Asp	Lys	Lys	Tyr	Ile	Ser	Gly	Ile	
				210					215					220		
Thr	Arg	Gly	Val	Ala	Glu	Leu	Lys	Gln	Glu	Glu	Phe	Val	Asn	Glu	Lys	
				225					230					235		
Ala	Arg	Arg	Phe	Ser	Tyr	Met	Lys	Thr	Met	Tyr	Ser	Val	Cys	Pro	Glu	
				245					250					255		
Ala	Phe	Glu	Pro	Ile	Ser	Arg	Asn	Glu	Ala	Ser	Thr	Pro	Glu	Gly	Ser	
				260					265					270		
Trp	Leu	Thr	Val	Ile	Ser	Gly	Lys	Ala	Pro	Met	Gly	Gln	Phe	Ser	Val	
				275					280					285		
Asp	Ser	Leu	Tyr	Asn	Pro	Asp	Leu	His	Ala	Leu	Cys	Glu	Leu	Pro	Asp	
				290					295					300		
Ile	Cys	Cys	Lys	Ile	Phe	Pro	Lys	Glu	Asn	Asn	Asp	Phe	Leu	Tyr	Ile	
				305					310					315		
Val	Val	Val	Tyr	Arg	Asn	Asp	Ser	Pro	Leu	Gly	Glu	Gln	Arg	Ala	Asn	
				325					330					335		
Arg	Phe	Ile	Glu	Leu	Tyr	Asn	Ile	Lys	Arg	Asp	Ile	Met	Gln	Glu	Leu	
				340					345					350		
Asn	Tyr	Glu	Leu	Pro	Glu	Leu	Lys	Ala	Val	Lys	Ser	Glu	Met	Ile	Ile	
				355					360					365		
Ala	Arg	Glu	Met	Gly	Glu	Ile	Phe	Ser	Tyr	Met	Pro	Gly	Glu	Ile	Asp	
				370					375					380		
Ser	Tyr	Met	Lys	Tyr	Ile	Asn	Asn	Lys	Leu	Ser	Lys	Ile	Glu	Gly	Ser	
				385					390					395		
His	His	His	His	His	His											
				405												

<400> SEQUENCE: 25

Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala Glu Asn
20 25 30

-continued

Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp Glu Arg
 35 40 45
 Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val Ser Arg
 50 55 60
 Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu Asn Asn
 65 70 75 80
 Leu Gly Leu Ser Ser Leu Pro Glu Leu Pro Pro His Leu Glu Ser Leu
 85 90 95
 Val Ala Ser Cys Asn Ser Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser
 100 105 110
 Leu Lys Ser Leu Gln Val Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp
 115 120 125
 Leu Pro Pro Ser Leu Glu Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu
 130 135 140
 Glu Leu Pro Glu Leu Gln Asn Ser Ser Phe Leu Lys Ile Ile Asp Pro
 145 150 155 160
 Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Pro Ala Ile
 165 170 175
 Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
 180 185 190
 Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
 195 200 205
 Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
 210 215 220
 Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
 225 230 235 240
 Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
 245 250 255
 Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
 260 265 270
 Trp Leu Thr Val Ile Ser Gly Lys Ala Pro Met Gly Gln Phe Ser Val
 275 280 285
 Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
 290 295 300
 Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
 305 310 315 320
 Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
 325 330 335
 Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu
 340 345 350
 Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile
 355 360 365
 Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp
 370 375 380
 Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 385 390 395 400

<210> SEQ ID NO 26

<211> LENGTH: 1221

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

-continued

<400> SEQUENCE: 26

```

atgttcatca ctccacgcaa tgtatctaac acctttctgc aggaaccgct gcgtcattct    60
agcgacctga cggaaatgcc agttgaagcg gagaacgtga aatctaagac tgaatactac    120
aacgcgtggg cagtatggga gcgcaatgca ccaccaggta acggtgaaca gcgtgaaatg    180
gcagtaagcc gtctgcgtga ttgcctggat cgccaggctc acgagctgga gctgaacaac    240
ctgggtctgt ctgacctgcc agagctccca ccacatctgg aaagcctggg ggctagctgt    300
aactctctga ctgaactgcc agagctgcca caaagcctga aatccctgca ggtggacaac    360
aacaacctga aagcgctgtc cgatctgcca ccgtctctgg agtttctggc agctggtaac    420
aaccaactgg aagaactgcc agagctgcag aactcctcct ttctgaagat catcgatcca    480
gcaccagcac cagcaccagc accagcacca gcaccagcac cagcaattaa tcctgttact    540
aatactcagg gcgtgtcccc tataaatact aaatatgtct aacatgtggt gaaaaatatt    600
taccggaaaa ttaaacatga ttactttaat gaatcaccca atatatatga taagaagtat    660
atatccggta taaccagagg agtagctgaa ctaaaacagg aagaatttgt taacgagaaa    720
gccagacggt tttcttatat gaagactatg tattctgtat gtccagaagc gtttgaacct    780
atttccagaa atgaagccag tacaccggaa ggaagctggc taacagtat atccggaaaa    840
gccccaatgg ggcagttttc ttagatagtg ttatacaatc ctgatttaca tgcattatgt    900
gagcttccgg acattttgtg taagatcttc cctaaagaaa ataatgattt ttatacata    960
gttgttgtgt acagaaatga cagccctcta ggagaacaac gggcaaatag atttatagaa   1020
ttatataata taaaagaga tatcatgcag gaattaaatt atgagttacc agagttaaag   1080
gcagtaaaat ctgaaatgat tatcgcacgt gaaatgggag aaatctttag ctacatgcct   1140
ggggaaatag acagttatat gaaatacata aataataaac tttctaaaat tgagggatct   1200
catcaccatc accatcacta a                                     1221

```

<210> SEQ ID NO 27

<211> LENGTH: 252

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 27

```

Gly Ser Met Arg Lys Lys Arg Arg Gln Arg Arg Arg Ile Asn Pro Val
1           5           10          15

Thr Asn Thr Gln Ile Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro
20          25          30

Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys
35          40          45

Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn Tyr Asp Lys Lys Tyr
50          55          60

Ile Ser Gly Ile Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe
65          70          75          80

Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser
85          90          95

Val Cys Pro Glu Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr
100         105         110

Pro Glu Gly Ser Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly
115         120         125

```

-continued

Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys
130 135 140

Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp
145 150 155 160

Phe Leu Tyr Ile Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu
165 170 175

Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile
180 185 190

Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser
195 200 205

Glu Met Ile Ile Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro
210 215 220

Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys
225 230 235 240

Ile Glu Gly Ser Arg Ser His His His His His His
245 250

<210> SEQ ID NO 28
 <211> LENGTH: 246
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 28

Gly Ser Met Arg Lys Lys Arg Arg Gln Arg Arg Ile Asn Pro Val
1 5 10 15

Thr Asn Thr Gln Ile Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro
20 25 30

Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys
35 40 45

Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn Tyr Asp Lys Lys Tyr
50 55 60

Ile Ser Gly Ile Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe
65 70 75 80

Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser
85 90 95

Val Cys Pro Glu Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr
100 105 110

Pro Glu Gly Ser Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly
115 120 125

Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys
130 135 140

Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp
145 150 155 160

Phe Leu Tyr Ile Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu
165 170 175

Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile
180 185 190

Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser
195 200 205

Glu Met Ile Ile Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro
210 215 220

Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys
225 230 235 240

-continued

Ile Glu Gly Ser Arg Ser
245

<210> SEQ ID NO 29
<211> LENGTH: 762
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 29

```

ggatccatga ggaagaagcg gagacagcga cgaagaatta atcctgttac taatactcag      60
attaatcctg ttactaatac tcagggcgctg tccctataa atactaaata tgctgaacat      120
gtggtgaaaa atatttaccg gaaaattaaa catgattact ttaatgaatc acccaatata      180
tatgataaga agtatatata cggtataacc agaggagtag ctgaactaaa acaggaagaa      240
tttggttaacg agaagaccag acggttttct tatatgaaga ctatgtattc tgtatgtcca      300
gaagcgtttg aacctatttc cagaaatgaa gccagtacac cggaaggaag ctggctaaca      360
gttatatccg gaaaacgccc aatggggcag ttttctgtag atagtttata caatcctgat      420
ttacatgcat tatgtgagct tccggacatt tgttgtaaga tcttcocctaa agaaaataat      480
gattttttat acatagttgt tgtgtacaga aatgacagcc ctctaggaga acaacgggca      540
aatagattta tagaattata taatataaaa agagatatca tgcaggaatt aaattatgag      600
ttaccagagt taaaggcagt aaaatctgaa atgattatcg cacgtgaaat gggagaaatc      660
tttagctaca tgcttgggga aatagacagt tatatgaaat acataaataa taaactttct      720
aaaattgagg gatccagatc tcatcaccat caccatcact aa                          762

```

<210> SEQ ID NO 30
<211> LENGTH: 252
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 30

```

Gly Ser Met Arg Lys Lys Arg Arg Gln Arg Arg Arg Ile Asn Pro Val
1          5          10          15

Thr Asn Thr Gln Ile Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro
20          25          30

Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys
35          40          45

Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn Tyr Asp Lys Lys Tyr
50          55          60

Ile Ser Gly Ile Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe
65          70          75          80

Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser
85          90          95

Val Cys Pro Glu Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr
100         105         110

Pro Glu Gly Ser Trp Leu Thr Val Ile Ser Gly Lys Ala Pro Met Gly
115         120         125

Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys
130         135         140

```

-continued

Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp
 145 150 155 160
 Phe Leu Tyr Ile Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu
 165 170 175
 Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile
 180 185 190
 Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser
 195 200 205
 Glu Met Ile Ile Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro
 210 215 220
 Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys
 225 230 235 240
 Ile Glu Gly Ser Arg Ser His His His His His His
 245 250

<210> SEQ ID NO 31
 <211> LENGTH: 246
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 31

Gly Ser Met Arg Lys Lys Arg Arg Gln Arg Arg Ile Asn Pro Val
 1 5 10 15
 Thr Asn Thr Gln Ile Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro
 20 25 30
 Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys
 35 40 45
 Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn Tyr Asp Lys Lys Tyr
 50 55 60
 Ile Ser Gly Ile Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe
 65 70 75 80
 Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser
 85 90 95
 Val Cys Pro Glu Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr
 100 105 110
 Pro Glu Gly Ser Trp Leu Thr Val Ile Ser Gly Lys Ala Pro Met Gly
 115 120 125
 Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys
 130 135 140
 Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp
 145 150 155 160
 Phe Leu Tyr Ile Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu
 165 170 175
 Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile
 180 185 190
 Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser
 195 200 205
 Glu Met Ile Ile Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro
 210 215 220
 Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys
 225 230 235 240
 Ile Glu Gly Ser Arg Ser
 245

-continued

```

<210> SEQ ID NO 32
<211> LENGTH: 762
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 32

ggatccatga ggaagaagcg gagacagcga cgaagaatta atcctgttac taatactcag      60
attaatcctg ttactaatac tcagggcgctg tccctataa atactaaata tgctgaacat      120
gtggtgaaaa atatttaccc gaaaattaaa catgattact ttaatgaatc acccaatata      180
tatgataaga agtatatatc cgggtataacc agaggagtag ctgaactaaa acaggaagaa      240
tttggttaacg agaagccag acggttttct tatatgaaga ctatgtattc tgtatgtcca      300
gaagcgtttg aacctatttc cagaaatgaa gccagtacac cggaaggaag ctggctaaca      360
gttatatccg gaaaagcccc aatggggcag ttttctgtag atagtttata caatcctgat      420
ttacatgcat tatgtgagct tccggacatt tgttgtaaga tcttcocctaa agaaaataat      480
gattttttat acatagttgt tgtgtacaga aatgacagcc ctctaggaga acaacgggca      540
aatagattta tagaattata taatataaaa agagatatca tgcaggaatt aaattatgag      600
ttaccagagt taaaggcagt aaaatctgaa atgattatcg cacgtgaaat gggagaaatc      660
tttagctaca tgcttgggga aatagacagt tatatgaaat acataaataa taaactttct      720
aaaattgagg gatccagatc tcatcaccat caccatcact aa                          762

```

```

<210> SEQ ID NO 33
<211> LENGTH: 257
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 33

Gly Ser Met Arg Lys Lys Arg Arg Gln Arg Arg Arg Ser Ile Glu Ile
1          5          10          15

Lys Met Ile Ser Pro Ile Lys Asn Ile Lys Asn Val Phe Pro Ile Asn
20         25         30

Thr Ala Asn Thr Glu Tyr Ile Val Arg Asn Ile Tyr Pro Arg Val Glu
35         40         45

His Gly Tyr Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile
50         55         60

Ser Gly Ile Thr Arg Ser Met Ala Gln Leu Lys Ile Glu Glu Phe Ile
65         70         75         80

Asn Glu Lys Ser Arg Arg Leu Asn Tyr Met Lys Thr Met Tyr Ser Pro
85         90         95

Cys Pro Glu Asp Phe Gln Pro Ile Ser Arg Asp Glu Ala Ser Thr Pro
100        105        110

Glu Gly Ser Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln
115        120        125

Phe Ser Val Asp Ser Leu Tyr His Pro Asp Leu His Ala Leu Cys Glu
130        135        140

Leu Pro Glu Ile Ser Cys Lys Ile Phe Pro Lys Glu Asn Ser Asp Phe
145        150        155        160

```

Leu	Tyr	Ile	Ile	Val	Val	Phe	Arg	Asn	Asp	Ser	Pro	Gln	Gly	Glu	Leu
				165					170					175	
Arg	Ala	Asn	Arg	Phe	Ile	Glu	Leu	Tyr	Asp	Ile	Lys	Arg	Glu	Ile	Met
				180				185					190		
Gln	Val	Leu	Arg	Asp	Glu	Ser	Pro	Glu	Leu	Lys	Ser	Ile	Lys	Ser	Glu
		195					200					205			
Ile	Ile	Ile	Ala	Arg	Glu	Met	Gly	Glu	Leu	Phe	Ser	Tyr	Ala	Ser	Glu
						215					220				
Glu	Ile	Asp	Ser	Tyr	Ile	Lys	Gln	Met	Asn	Asp	Arg	Leu	Ser	Gln	Ile
225					230					235					240
Lys	Ala	Arg	Met	Pro	Val	Thr	Gly	Ser	Arg	Ser	His	His	His	His	His
				245					250					255	

His

```
<210> SEQ ID NO 34
<211> LENGTH: 251
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

<400> SEQUENCE: 34

Gly 1	Ser	Met	Arg	Lys 5	Lys	Arg	Arg	Gln	Arg 10	Arg	Arg	Ser	Ile	Glu	Ile 15
Lys	Met	Ile	Ser 20	Pro	Ile	Lys	Asn	Ile 25	Lys	Asn	Val	Phe	Pro 30	Ile	Asn
Thr	Ala	Asn	Thr 35	Glu	Tyr	Ile	Val 40	Arg	Asn	Ile	Tyr	Pro 45	Arg	Val	Glu
His	Gly 50	Tyr	Phe	Asn	Glu	Ser 55	Pro	Asn	Ile	Tyr	Asp 60	Lys	Lys	Tyr	Ile
Ser 65	Gly	Ile	Thr	Arg	Ser 70	Met	Ala	Gln	Leu	Lys 75	Ile	Glu	Glu	Phe	Ile 80
Asn	Glu	Lys	Ser 85	Arg	Arg	Leu	Asn	Tyr	Met 90	Lys	Thr	Met	Tyr	Ser 95	Pro
Cys	Pro	Glu	Asp 100	Phe	Gln	Pro	Ile	Ser	Arg 105	Asp	Glu	Ala	Ser 110	Thr	Pro
Glu	Gly	Ser	Trp 115	Leu	Thr	Val	Ile 120	Ser	Gly	Lys	Arg	Pro 125	Met	Gly	Gln
Phe 130	Ser	Val	Asp	Ser	Leu	Tyr 135	His	Pro	Asp	Leu	His 140	Ala	Leu	Cys	Glu
Leu 145	Pro	Glu	Ile	Ser	Cys 150	Lys	Ile	Phe	Pro	Lys 155	Glu	Asn	Ser	Asp	Phe 160
Leu	Tyr	Ile	Ile 165	Val	Val	Phe	Arg	Asn	Asp 170	Ser	Pro	Gln	Gly	Glu 175	Leu
Arg	Ala	Asn	Arg 180	Phe	Ile	Glu	Leu	Tyr	Asp 185	Ile	Lys	Arg	Glu 190	Ile	Met
Gln	Val	Leu	Arg 195	Asp	Glu	Ser	Pro 200	Glu	Leu	Lys	Ser 205	Ile	Lys	Ser	Glu
Ile 210	Ile	Ile	Ala	Arg	Glu	Met 215	Gly	Glu	Leu	Phe	Ser 220	Tyr	Ala	Ser	Glu
Glu 225	Ile	Asp	Ser	Tyr 230	Ile	Lys	Gln	Met	Asn	Asp 235	Arg	Leu	Ser	Gln	Ile 240
Lys	Ala	Arg	Met 245	Pro	Val	Thr	Gly	Ser	Arg	Ser					

-continued

```

<210> SEQ ID NO 35
<211> LENGTH: 774
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 35

ggatccatga ggaagaagcg gagacagcga cgaagatcca tcgagatcaa gatgatctct      60
ccgatcaaga acatcaagaa cgtattcccg atcaacactg cgaataccga gtacatcggt      120
cgcaacatct acccacgcgt agaacacggc tacttcaacg agagcccaaa catctacgac      180
aagaagtaca tcagcgggtat cactcgctct atgggtcaac tgaagatcga agagttcatc      240
aacgagaagt cccgtcgctt gaactacatg aagaccatgt actctccgtg tccggaagat      300
ttccaaccga tctctcgtga tgaagctagc actccagaag gcagctgggt gaccgtgatc      360
tctggcaaac gtccgatggg tcagttctcc gttgactctc tgtaccatcc agacctgcac      420
gctctgtgcg aactgccaga gatttcttgc aagatctttc cgaaagagaa ctctgatttc      480
ctgtacatca tcgttgtgtt ccgcaacgat tctccacaag gtgaactgcg tgctaaccgc      540
ttcatcgaac tgtagcatat caagcgtgag atcatgcagg tgctgcgcga cgagtctcca      600
gaactgaaga gcatacaatc cgagatcadc attgcccgtg aaatgggcga actgttctct      660
tacgcatctg aagaaatcga ctcttacatc aagcagatga acgacgtctt gtcccagatc      720
aaggctcgta tgccggtaac cggatccaga tctcatcacc atcaccatca ctaa      774

```

```

<210> SEQ ID NO 36
<211> LENGTH: 322
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 36

Gly Ser Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln
1             5             10            15

Glu Pro Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala
20            25            30

Glu Asn Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp
35            40            45

Glu Arg Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val
50            55            60

Ser Arg Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu
65            70            75            80

Asn Asn Leu Gly Leu Ser Ser Leu Met Ile Asn Pro Val Thr Asn Thr
85            90            95

Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys
100           105           110

Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn
115           120           125

Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu
130           135           140

Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr
145           150           155           160

Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser

```

-continued

Arg	Asn	Glu	Ala	Ser	Thr	Pro	Glu	Gly	Ser	Trp	Leu	Thr	Val	Ile	Ser
	180							185					190		
Gly	Lys	Arg	Pro	Met	Gly	Gln	Phe	Ser	Val	Asp	Ser	Leu	Tyr	Asn	Pro
	195					200						205			
Asp	Leu	His	Ala	Leu	Cys	Glu	Leu	Pro	Asp	Ile	Cys	Cys	Lys	Ile	Phe
	210				215					220					
Pro	Lys	Glu	Asn	Asn	Asp	Phe	Leu	Tyr	Ile	Val	Val	Val	Tyr	Arg	Asn
	225				230					235					240
Asp	Ser	Pro	Leu	Gly	Glu	Gln	Arg	Ala	Asn	Arg	Phe	Ile	Glu	Leu	Tyr
			245					250						255	
Asn	Ile	Lys	Arg	Asp	Ile	Met	Gln	Glu	Leu	Asn	Tyr	Glu	Leu	Pro	Glu
		260					265						270		
Leu	Lys	Ala	Val	Lys	Ser	Glu	Met	Ile	Ile	Ala	Arg	Glu	Met	Gly	Glu
		275					280						285		
Ile	Phe	Ser	Tyr	Met	Pro	Gly	Glu	Ile	Asp	Ser	Tyr	Met	Lys	Tyr	Ile
	290					295					300				
Asn	Asn	Lys	Leu	Ser	Lys	Ile	Glu	Gly	Ser	Arg	Ser	His	His	His	His
	305				310					315					320

His His

<210> SEQ ID NO 37
 <211> LENGTH: 316
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 37

Gly	Ser	Met	Phe	Ile	Thr	Pro	Arg	Asn	Val	Ser	Asn	Thr	Phe	Leu	Gln
1			5					10						15	
Glu	Pro	Leu	Arg	His	Ser	Ser	Asp	Leu	Thr	Glu	Met	Pro	Val	Glu	Ala
		20					25						30		
Glu	Asn	Val	Lys	Ser	Lys	Thr	Glu	Tyr	Tyr	Asn	Ala	Trp	Ala	Val	Trp
	35					40					45				
Glu	Arg	Asn	Ala	Pro	Pro	Gly	Asn	Gly	Glu	Gln	Arg	Glu	Met	Ala	Val
	50					55					60				
Ser	Arg	Leu	Arg	Asp	Cys	Leu	Asp	Arg	Gln	Ala	His	Glu	Leu	Glu	Leu
	65			70					75					80	
Asn	Asn	Leu	Gly	Leu	Ser	Ser	Leu	Met	Ile	Asn	Pro	Val	Thr	Asn	Thr
		85						90						95	
Gln	Gly	Val	Ser	Pro	Ile	Asn	Thr	Lys	Tyr	Ala	Glu	His	Val	Val	Lys
		100						105					110		
Asn	Ile	Tyr	Pro	Lys	Ile	Lys	His	Asp	Tyr	Phe	Asn	Glu	Ser	Pro	Asn
	115						120					125			
Ile	Tyr	Asp	Lys	Lys	Tyr	Ile	Ser	Gly	Ile	Thr	Arg	Gly	Val	Ala	Glu
	130					135					140				
Leu	Lys	Gln	Glu	Glu	Phe	Val	Asn	Glu	Lys	Ala	Arg	Arg	Phe	Ser	Tyr
	145				150					155					160
Met	Lys	Thr	Met	Tyr	Ser	Val	Cys	Pro	Glu	Ala	Phe	Glu	Pro	Ile	Ser
			165						170					175	
Arg	Asn	Glu	Ala	Ser	Thr	Pro	Glu	Gly	Ser	Trp	Leu	Thr	Val	Ile	Ser
		180						185					190		

Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro

-continued

195	200	205	
Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe			
210	215	220	
Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn			
225	230	235	240
Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr			
	245	250	255
Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu			
	260	265	270
Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu			
	275	280	285
Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile			
	290	295	300
Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser Arg Ser			
305	310	315	
 <210> SEQ ID NO 38			
<211> LENGTH: 969			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide			
 <400> SEQUENCE: 38			
ggatccatgt tcatcactcc acgcaatgta tctaacacct ttctgcagga accgctgcgt	60		
cattctagcg acctgaccga aatgccagtt gaagcggaga acgtgaaatc taagactgaa	120		
tactacaacg cgtgggcagt atgggagcgc aatgcaccac caggtaacgg tgaacagcgt	180		
gaaatggcag taagccgtct gcgtgattgc ctggatcgcc aggctcacga gctggagctg	240		
aacaacctgg gtctgtctag cctgatgatt aatcctgtta ctaatactca gggcgtgtcc	300		
cctataaata ctaaatatgc tgaacatgtg gtgaaaaata tttacccgaa aattaaacat	360		
gattacttta atgaatcacc caatatatat gataagaagt atatatccgg tataaccaga	420		
ggagtagctg aactaaaaca ggaagaattt gttaacgaga aagccagacg gttttcttat	480		
atgaagacta tgtattctgt atgtccagaa gcgtttgaac ctatttccag aaatgaagcc	540		
agtacaccgg aaggaagctg gctaacagtt atatccggaa aacgcccaat ggggcagttt	600		
tctgtagata gtttatacaa tcttgattta catgcattat gtgagcttcc ggacatttgt	660		
tgtaagatct tccctaaaga aaataatgat tttttataca tagttgttgt gtacagaaat	720		
gacagccctc taggagaaca acgggcacaa agatttatag aattatataa tataaaaaga	780		
gatatcatgc aggaattaaa ttatgagtta ccagagttaa aggcagttaa atctgaaatg	840		
attatcgcac gtgaaatggg agaaatcttt agctacatgc ctggggaaat agacagttat	900		
atgaaataca taaataataa actttctaaa attgagggat ccagatctca tcaccatcac	960		
catcactaa	969		

<210> SEQ ID NO 39
 <211> LENGTH: 322
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <400> SEQUENCE: 39

-continued

Gly Ser Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln
 1 5 10 15
 Glu Pro Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala
 20 25 30
 Glu Asn Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp
 35 40 45
 Glu Arg Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val
 50 55 60
 Ser Arg Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu
 65 70 75 80
 Asn Asn Leu Gly Leu Ser Ser Leu Met Ile Asn Pro Val Thr Asn Thr
 85 90 95
 Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys
 100 105 110
 Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn
 115 120 125
 Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu
 130 135 140
 Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr
 145 150 155 160
 Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser
 165 170 175
 Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
 180 185 190
 Gly Lys Ala Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro
 195 200 205
 Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe
 210 215 220
 Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn
 225 230 235 240
 Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr
 245 250 255
 Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu
 260 265 270
 Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu
 275 280 285
 Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile
 290 295 300
 Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser Arg Ser His His His His
 305 310 315 320
 His His

<210> SEQ ID NO 40
 <211> LENGTH: 316
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide
 <400> SEQUENCE: 40

Gly Ser Met Phe Ile Thr Pro Arg Asn Val Ser Asn Thr Phe Leu Gln
 1 5 10 15
 Glu Pro Leu Arg His Ser Ser Asp Leu Thr Glu Met Pro Val Glu Ala
 20 25 30

-continued

Glu Asn Val Lys Ser Lys Thr Glu Tyr Tyr Asn Ala Trp Ala Val Trp
 35 40 45
 Glu Arg Asn Ala Pro Pro Gly Asn Gly Glu Gln Arg Glu Met Ala Val
 50 55 60
 Ser Arg Leu Arg Asp Cys Leu Asp Arg Gln Ala His Glu Leu Glu Leu
 65 70 75 80
 Asn Asn Leu Gly Leu Ser Ser Leu Met Ile Asn Pro Val Thr Asn Thr
 85 90 95
 Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys
 100 105 110
 Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn
 115 120 125
 Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu
 130 135 140
 Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr
 145 150 155 160
 Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser
 165 170 175
 Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
 180 185 190
 Gly Lys Ala Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro
 195 200 205
 Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe
 210 215 220
 Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn
 225 230 235 240
 Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr
 245 250 255
 Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu
 260 265 270
 Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu
 275 280 285
 Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile
 290 295 300
 Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser Arg Ser
 305 310 315

<210> SEQ ID NO 41

<211> LENGTH: 969

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 41

```

ggatccatgt tcactactcc acgcaatgta tctaacacct ttctgcagga accgctgcgt      60
cattctagcg acctgaccga aatgccagtt gaagcggaga acgtgaaatc taagactgaa      120
tactacaacg cgtgggcagt atgggagcgc aatgcaccac caggtaacgg tgaacagcgt      180
gaaatggcag taagccgtct gcgtgattgc ctggatcgcc aggcctcacga gctggagctg      240
aacaacctgg gtctgtctag cctgatgatt aatcctgtta ctaatactca gggcgtgtcc      300
cctataaata ctaaatatgc tgaacatgtg gtgaaaaata tttacccgaa aattaaacat      360
gattacttta atgaatcacc caatatatat gataagaagt atatatccgg tataaccaga      420

```

-continued

```

ggagtagctg aactaaaaca ggaagaattt gttaacgaga aagccagacg gttttcttat 480
atgaagacta tgtattctgt atgtccagaa gcgtttgaac ctatttccag aaatgaagcc 540
agtacaccgg aaggaagctg gctaacagtt atatccggaa aagccccaat ggggcagttt 600
tctgtagata gtttatacaa tcctgattta catgcattat gtgagcttcc ggacatttgt 660
tgtaagatct tccctaaaga aaataatgat tttttataca tagttgttgt gtacagaaat 720
gacagccctc taggagaaca acgggcaaat agatttatag aattatataa tataaaaaga 780
gatatcatgc aggaattaaa ttatgagtta ccagagttaa aggcagtaaa atctgaaatg 840
attatcgcac gtgaaatggg agaaatcttt agctacatgc ctggggaaat agacagttat 900
atgaaataca taaataataa acttttctaaa attgagggat ccagatctca tcaccatcac 960
catcactaa 969

```

```

<210> SEQ ID NO 42
<211> LENGTH: 240
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 42

```

```

Met Thr Lys Ile Thr Leu Ser Pro Gln Asn Phe Arg Ile Gln Lys Gln
 1             5             10            15
Glu Thr Thr Leu Leu Lys Glu Lys Ser Thr Glu Lys Asn Ser Leu Ala
      20             25            30
Lys Ser Ile Leu Ala Val Lys Asn His Phe Ile Glu Leu Arg Ser Lys
      35             40            45
Leu Ser Glu Arg Phe Ile Ser His Lys Asn Thr Glu Ser Ser Ala Thr
      50             55            60
His Phe His Arg Gly Ser Ala Ser Glu Gly Arg Ala Val Leu Thr Asn
      65             70            75            80
Lys Val Val Lys Asp Phe Met Leu Gln Thr Leu Asn Asp Ile Asp Ile
      85             90            95
Arg Gly Ser Ala Ser Lys Asp Pro Ala Tyr Ala Ser Gln Thr Arg Glu
      100            105           110
Ala Ile Leu Ser Ala Val Tyr Ser Lys Asn Lys Asp Gln Cys Cys Asn
      115            120           125
Leu Leu Ile Ser Lys Gly Ile Asn Ile Ala Pro Phe Leu Gln Glu Ile
      130            135           140
Gly Glu Ala Ala Lys Asn Ala Gly Leu Pro Gly Thr Thr Lys Asn Asp
      145            150           155           160
Val Phe Thr Pro Ser Gly Ala Gly Ala Asn Pro Phe Ile Thr Pro Leu
      165            170           175
Ile Ser Ser Ala Asn Ser Lys Tyr Pro Arg Met Phe Ile Asn Gln His
      180            185           190
Gln Gln Ala Ser Phe Lys Ile Tyr Ala Glu Lys Ile Ile Met Thr Glu
      195            200           205
Val Ala Pro Leu Phe Asn Glu Cys Ala Met Pro Thr Pro Gln Gln Phe
      210            215           220
Gln Leu Ile Leu Glu Asn Ile Ala Asn Lys Tyr Ile Gln Asn Thr Pro
      225            230           235           240

```

```

<210> SEQ ID NO 43
<211> LENGTH: 1011

```

-continued

```

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 43

tttctaattc atcaatcaga tggacatagc atttgctata aaaaataaaa gtattcctgc      60
tatctatata taaatgagtt atgtacatat aaaaggagca ttaccgtgac aaaaataaact    120
ttatttcccc ataactttag aatccaaaaa caggaagcca caccactaaa agaaaaatca    180
accgagaaaa attcttttagc aaaagtatt ctgcgagtaa aaaatcactt catcaaatta    240
aattcaaaat tatcggaacg ttttatttcg cataagaaca ctgaatcttc tgcaacacac    300
tttcaccgag gaagcgcacg tgagggccgg gcagtgttga caaataaagt cgttaaaaac    360
tttatgcttc aaacgctcca tgatatagat attagaggta gcgcgagtaa agaccccgca    420
tacgccagcc agaccctgta agctatacta tcggcagttt acagcaagta taaagatcag    480
tattgtaact tgctcatcag caaaggaatc gacatagcgc cttttcttaa ggaaattggc    540
gaggctgcgc aaaatgcagg tctgcccgga gcaaccaaga atgacgtttt tagcccaagc    600
ggcgcaggag ccaatccttt tataactccg ttgattacat cagcatacag taagtatcca    660
catatgttta ccagtcaaca tcagaaggca tcctttaaca tctatgcgga gaagatcatt    720
atgacagaag ttgtaccgct gtttaatgag tgtgctatgc cgactccaca gcaattccaa    780
caaatactag aaaacattgc taataaatat atccaaaaca ctccctgaac acagaaacac    840
caaaaaatat gcgagcctct tcttgattaa tatgaaccaa tagtatccat aatttttccc    900
aggaactaac tctggagcta aaccgtcatt taccagtgtc aaaattatac actcaaccat    960
caaaataata gccattgctg ctatataaca tatagcagca atctctacta c              1011

```

```

<210> SEQ ID NO 44
<211> LENGTH: 561
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

```

```

<400> SEQUENCE: 44

Met Gln Ile Gln Ser Phe Tyr His Ser Ala Ser Leu Lys Thr Gln Glu
1      5      10      15
Ala Phe Lys Ser Leu Gln Lys Thr Leu Tyr Asn Gly Met Gln Ile Leu
20     25     30
Ser Gly Gln Gly Lys Ala Pro Ala Lys Ala Pro Asp Ala Arg Pro Glu
35     40     45
Ile Ile Val Leu Arg Glu Pro Gly Ala Thr Trp Gly Asn Tyr Leu Gln
50     55     60
His Gln Lys Ala Ser Asn His Ser Leu His Asn Leu Tyr Asn Leu Gln
65     70     75     80
Arg Asp Leu Leu Thr Val Ala Ala Thr Val Leu Gly Lys Gln Asp Pro
85     90     95
Val Leu Thr Ser Met Ala Asn Gln Met Glu Leu Ala Lys Val Lys Ala
100    105    110
Asp Arg Pro Ala Thr Lys Gln Glu Glu Ala Ala Ala Lys Ala Leu Lys
115    120    125
Lys Asn Leu Ile Glu Leu Ile Ala Ala Arg Thr Gln Gln Gln Asp Gly
130    135    140

```

-continued

Leu	Pro	Ala	Lys	Glu	Ala	His	Arg	Phe	Ala	Ala	Val	Ala	Phe	Arg	Asp	145	150	155	160
Ala	Gln	Val	Lys	Gln	Leu	Asn	Asn	Gln	Pro	Trp	Gln	Thr	Ile	Lys	Asn	165	170	175	
Thr	Leu	Thr	His	Asn	Gly	His	His	Tyr	Thr	Asn	Thr	Gln	Leu	Pro	Ala	180	185	190	
Ala	Glu	Met	Lys	Ile	Gly	Ala	Lys	Asp	Ile	Phe	Pro	Ser	Ala	Tyr	Glu	195	200	205	
Gly	Lys	Gly	Val	Cys	Ser	Trp	Asp	Thr	Lys	Asn	Ile	His	His	Ala	Asn	210	215	220	
Asn	Leu	Trp	Met	Ser	Thr	Val	Ser	Val	His	Glu	Asp	Gly	Lys	Asp	Lys	225	230	235	240
Thr	Leu	Phe	Cys	Gly	Ile	Arg	His	Gly	Val	Leu	Ser	Pro	Tyr	His	Glu	245	250	255	
Lys	Asp	Pro	Leu	Leu	Arg	His	Val	Gly	Ala	Glu	Asn	Lys	Ala	Lys	Glu	260	265	270	
Val	Leu	Thr	Ala	Ala	Leu	Phe	Ser	Lys	Pro	Glu	Leu	Leu	Asn	Lys	Ala	275	280	285	
Leu	Ala	Gly	Glu	Ala	Val	Ser	Leu	Lys	Leu	Val	Ser	Val	Gly	Leu	Leu	290	295	300	
Thr	Ala	Ser	Asn	Ile	Phe	Gly	Lys	Glu	Gly	Thr	Met	Val	Glu	Asp	Gln	305	310	315	320
Met	Arg	Ala	Trp	Gln	Ser	Leu	Thr	Gln	Pro	Gly	Lys	Met	Ile	His	Leu	325	330	335	
Lys	Ile	Arg	Asn	Lys	Asp	Gly	Asp	Leu	Gln	Thr	Val	Lys	Ile	Lys	Pro	340	345	350	
Asp	Val	Ala	Ala	Phe	Asn	Val	Gly	Val	Asn	Glu	Leu	Ala	Leu	Lys	Leu	355	360	365	
Gly	Phe	Gly	Leu	Lys	Ala	Ser	Asp	Ser	Tyr	Asn	Ala	Glu	Ala	Leu	His	370	375	380	
Gln	Leu	Leu	Gly	Asn	Asp	Leu	Arg	Pro	Glu	Ala	Arg	Pro	Gly	Gly	Trp	385	390	395	400
Val	Gly	Glu	Trp	Leu	Ala	Gln	Tyr	Pro	Asp	Asn	Tyr	Glu	Val	Val	Asn	405	410	415	
Thr	Leu	Ala	Arg	Gln	Ile	Lys	Asp	Ile	Trp	Lys	Asn	Asn	Gln	His	His	420	425	430	
Lys	Asp	Gly	Gly	Glu	Pro	Tyr	Lys	Leu	Ala	Gln	Arg	Leu	Ala	Met	Leu	435	440	445	
Ala	His	Glu	Ile	Asp	Ala	Val	Pro	Ala	Trp	Asn	Cys	Lys	Ser	Gly	Lys	450	455	460	
Asp	Arg	Thr	Gly	Met	Met	Asp	Ser	Glu	Ile	Lys	Arg	Glu	Ile	Ile	Ser	465	470	475	480
Leu	His	Gln	Thr	His	Met	Leu	Ser	Ala	Pro	Gly	Ser	Leu	Pro	Asp	Ser	485	490	495	
Gly	Gly	Gln	Lys	Ile	Phe	Gln	Lys	Val	Leu	Leu	Asn	Ser	Gly	Asn	Leu	500	505	510	
Glu	Ile	Gln	Lys	Gln	Asn	Thr	Gly	Gly	Ala	Gly	Asn	Lys	Val	Met	Lys	515	520	525	
Asn	Leu	Ser	Pro	Glu	Val	Leu	Asn	Leu	Ser	Tyr	Gln	Lys	Arg	Val	Gly	530	535	540	
Asp	Glu	Asn	Ile	Trp	Gln	Ser	Val	Lys	Gly	Ile	Ser	Ser	Leu	Ile	Thr	545	550	555	560

-continued

Ser

<210> SEQ ID NO 45
 <211> LENGTH: 1253
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

 <400> SEQUENCE: 45

 tgccctgcaaa ggaggctcgc cgccttgccg cagcagattt taagagtgcc caggtcaagc 60
 agcttaataa tcaaccgtgg cagaccataa aaaatacact aacgcataat gggcatcaat 120
 ataccagcac gcaggttcct gccgcagaga tgaaaatcgg cgcacaggat atttttctta 180
 aagcctatca gggaaagggc gtatgcagtt gggatacca aaatattcat cagctacta 240
 acctgtggat gtccacaata agtgtacatg aggacggta agataaacg ctttttagtg 300
 ggatacgtca tggcgtgctt tcaccctatc atgtggaaga tccgcttctg cgtcagaccg 360
 gcgctgaaag cagagccaaa gaagtattaa ctgcagcgtc ctttagtaaa cctgagttgc 420
 ttaccagggc cttaaagggc gaagcggtaa gcctgaaact ggtatctgtc tgtttactca 480
 ccgcgtcgaa tgttctaggc caggagggaa caatggtcaa ggagcaaatg agagcatggc 540
 aatcgttgac ccagccggga aaaatgattc atttaaaaat ccgcaatgac gatggcgaac 600
 tacagacggt aaaaaataag ccggaagtcg ccgcgttcaa tgtgggtgtc aatgagttag 660
 cactcaagtt cggttttggc cttaaggcat ctgatagtta taatatcgag gcgctacagc 720
 agttattagg caatgattta cgcctgagg ccagaccagg cggttgggtt ggcgagtggc 780
 tggcgcgata tccggataac gatgaaagcg taaatacatt agcacgccag attaaagata 840
 tctggcaaaa taagctgcat cacaaagatg gcggcgaacc ctataaatta gcacaacgcc 900
 ttgcaatggt agccaatgaa atcgatgtgg taccgcctg gaattgtaaa agcggcaaaag 960
 atcgcacagg aatgatggac tcagaaacca agcgagaagc catttctttc catcagaccc 1020
 atactttgag ttctccaggc agccttcctg atcgcagcgg acagcaaatc ttccaaaaag 1080
 tattacttaa tagcggaaac ctggaaattc aaaaacaaaa tacaagcggg gcgggaaaca 1140
 aagtaataaa aaactgtcgc ccagaagtgc ttaatctttc ctatcataaa cgaattggag 1200
 atgaaaaatc ctggcaatcg gtaaaaggaa tttctacatt aatcatttct tga 1253

<210> SEQ ID NO 46
 <211> LENGTH: 240
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 46

Met Thr Asn Ile Thr Leu Ser Thr Gln His Tyr Arg Ile His Arg Ser
 1 5 10 15

 Asp Val Glu Pro Val Lys Glu Lys Thr Thr Glu Lys Asp Ile Phe Ala
 20 25 30

 Lys Ser Ile Thr Ala Val Arg Asn Ser Phe Ile Ser Leu Ser Thr Ser
 35 40 45

 Leu Ser Asp Arg Phe Ser Leu His Gln Gln Thr Asp Ile Pro Thr Thr
 50 55 60

 His Phe His Arg Gly Asn Ala Ser Glu Gly Arg Ala Val Leu Thr Ser

-continued

65	70	75	80
Lys Thr Val Lys Asp Phe Met Leu Gln Lys Leu Asn Ser Leu Asp Ile			
	85	90	95
Lys Gly Asn Ala Ser Lys Asp Pro Ala Tyr Ala Arg Gln Thr Cys Glu			
	100	105	110
Ala Ile Leu Ser Ala Val Tyr Ser Asn Asn Lys Asp Gln Cys Cys Lys			
	115	120	125
Leu Leu Ile Ser Lys Gly Val Ser Ile Thr Pro Phe Leu Lys Glu Ile			
	130	135	140
Gly Glu Ala Ala Gln Asn Ala Gly Leu Pro Gly Glu Ile Lys Asn Gly			
145	150	155	160
Val Phe Thr Pro Gly Gly Ala Gly Ala Asn Pro Phe Val Val Pro Leu			
	165	170	175
Ile Ala Ser Ala Ser Ile Lys Tyr Pro His Met Phe Ile Asn His Asn			
	180	185	190
Gln Gln Val Ser Phe Lys Ala Tyr Ala Glu Lys Ile Val Met Lys Glu			
	195	200	205
Val Thr Pro Leu Phe Asn Lys Gly Thr Met Pro Thr Pro Gln Gln Phe			
	210	215	220
Gln Leu Thr Ile Glu Asn Ile Ala Asn Lys Tyr Leu Gln Asn Ala Ser			
225	230	235	240

<210> SEQ ID NO 47

<211> LENGTH: 1316

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 47

taagtgcag aagaacaaaa tccatcagga aaataaaatt tataaatatc aatgagtaaa	60
aatggttggt gagaagggtg ctattttttg aaagcaagaa atataaacaa agtgtagcta	120
tgcatagtta tctaaaagga gaactaccgt gactaacata acactatcca cccagcacta	180
cagaatccat agaagtgcg ttgaaccagt aaaagaaaaa acaacggaga aggacatttt	240
tgcaaaaagt attactgccg ttagaaatag ctttatcagc ctgtcgacga gtctgtcaga	300
tcgttttagc ctgcatcaac aaacagacat accgactacc cattttcatc gtgggaacgc	360
ttctgagggt agggcggtat taaccagtaa aactgttaaa gattttatgc tgcaaaagct	420
caatagtctg gatatacaag gtaatgcgag taaagatccg gcctatgctc gtcagacatg	480
cgaagccata ttatcagccg tgtacagtaa taataaagat caatgttgta aattactcat	540
cagtaaaggg gtcagtatta cccctttttt gaaagagata ggagaggctg cgcagaatgc	600
ggggctacct ggggagataa aaaatggcgt atttactcca ggtggggcag gggcgaatcc	660
ttttgtcgtc cccctcattg ctcccgagag tattaaatat ccgcatatgt ttataaatca	720
taatcagcag gtatctttta aagcgtatgc tgagaaaatc gttatgaaag aggttacgcc	780
gctgtttaat aaggggacga tgccaacgcc acaacaattt cagttaacta tagaaaaatat	840
tgccaataag tatcttcaga atgctcctg atggtagtaa atacaatact gctattaacc	900
atatgaatta atagcagatt ttctgaaact atatattcgt ttttatggcg acgcttttaa	960
caaaacgtcg tcacttttat tttaataataa aaatcaaatg gataacctta aaacttaaaa	1020
taattgttat tactctaaat acctgcacaa ccagacagct tttttgttaa cgctgcgtgt	1080

-continued

```

tgtaaagtct atgatacccg cgcacggatg atgtctcgat aaggattaat ggtctgtaa 1140
ctgtcaggag aggttatgcc gcctttaaaa aaaattgtgc tgcgcctggt tgtgggggag 1200
atggtcgcca cggtaacgac acctgcattg gcgctgggtt gccttgagga tcaactcgct 1260
aaagagtgtg ctatatcgtg tgctgaggtg atgtgggtta tgtctcgaga tcggat 1316

```

<210> SEQ ID NO 48

<211> LENGTH: 900

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 48

```

Met Pro Met Trp Ala Gly Gly Val Gly Ser Pro Arg Arg Gly Met Ala
1          5          10          15
Pro Ala Ser Thr Asp Asp Leu Phe Ala Arg Lys Leu Arg Gln Pro Ala
20        25        30
Arg Pro Pro Leu Thr Pro His Thr Phe Glu Pro Arg Pro Val Arg Gly
35        40        45
Pro Leu Leu Arg Ser Gly Ser Asp Ala Gly Glu Ala Arg Pro Pro Thr
50        55        60
Pro Ala Ser Pro Arg Ala Arg Ala His Ser His Glu Glu Ala Ser Arg
65        70        75        80
Pro Ala Ala Thr Ser Thr Arg Leu Phe Thr Asp Pro Leu Ala Leu Leu
85        90        95
Gly Leu Pro Ala Glu Glu Pro Glu Pro Ala Phe Pro Pro Val Leu Glu
100       105       110
Pro Arg Trp Phe Ala His Tyr Asp Val Gln Ser Leu Leu Phe Asp Trp
115       120       125
Ala Pro Arg Ser Gln Gly Met Gly Ser His Ser Glu Ala Ser Ser Gly
130       135       140
Thr Leu Ala Ser Ala Glu Asp Gln Ala Ala Ser Ser Asp Leu Leu His
145       150       155       160
Gly Ala Pro Gly Phe Val Cys Glu Leu Gly Gly Glu Gly Glu Leu Gly
165       170       175
Leu Gly Gly Pro Ala Ser Pro Pro Val Pro Pro Ala Leu Pro Asn Ala
180       185       190
Ala Val Ser Ile Leu Glu Glu Pro Gln Asn Arg Thr Ser Ala Tyr Ser
195       200       205
Leu Glu His Ala Asp Leu Gly Ala Gly Tyr Tyr Arg Lys Tyr Phe Tyr
210       215       220
Gly Lys Glu His Gln Asn Phe Phe Gly Met Asp Glu Ser Leu Gly Pro
225       230       235       240
Val Ala Val Ser Leu Arg Arg Glu Glu Lys Glu Gly Ser Gly Gly Gly
245       250       255
Thr Leu His Ser Tyr Arg Val Ile Val Arg Thr Thr Gln Leu Arg Thr
260       265       270
Leu Arg Gly Thr Ile Ser Glu Asp Ala Leu Pro Pro Gly Pro Pro Arg
275       280       285
Gly Leu Ser Pro Arg Lys Leu Leu Glu His Val Ala Pro Gln Leu Ser
290       295       300
Pro Ser Cys Leu Arg Leu Gly Ser Ala Ser Pro Lys Val Pro Arg Thr
305       310       315       320

```

-continued

Leu	Leu	Thr	Leu	Asp 325	Glu	Gln	Val	Leu	Ser 330	Phe	Gln	Arg	Lys	Val 335	Gly
Ile	Leu	Tyr	Cys 340	Arg	Ala	Gly	Gln	Gly 345	Ser	Glu	Glu	Glu	Met 350	Tyr	Asn
Asn	Gln	Glu	Ala 355	Gly	Pro	Ala	Phe	Met 360	Gln	Phe	Leu	Thr 365	Leu	Leu	Gly
Asp	Val	Val	Arg 370	Leu	Lys	Gly 375	Phe	Glu	Ser	Tyr	Arg 380	Ala	Gln	Leu	Asp
Thr 385	Lys	Thr	Asp	Ser 390	Thr	Gly	Thr	His	Ser	Leu 395	Tyr	Thr	Thr	Tyr	Gln 400
Asp	His	Glu	Ile 405	Met	Phe	His	Val	Ser 410	Thr	Met	Leu	Pro	Tyr	Thr 415	Pro
Asn	Asn	Gln	Gln 420	Gln	Leu	Leu	Arg	Lys 425	Arg	His	Ile	Gly 430	Asn	Asp	Ile
Val	Thr	Ile 435	Val	Phe	Gln	Glu	Pro 440	Gly	Ser	Lys	Pro	Phe 445	Cys	Pro	Thr
Thr 450	Ile	Arg	Ser	His	Phe	Gln 455	His	Val	Phe	Leu	Val 460	Val	Arg	Ala	His
Thr 465	Pro	Cys	Thr	Pro	His 470	Thr	Thr	Tyr	Arg	Val 475	Ala	Val	Ser	Arg	Thr 480
Gln	Asp	Thr	Pro	Ala 485	Phe	Gly	Pro	Ala	Leu 490	Pro	Ala	Gly	Gly	Gly 495	Pro
Phe	Ala	Ala	Asn 500	Ala	Asp	Phe	Arg	Ala 505	Phe	Leu	Leu	Ala	Lys 510	Ala	Leu
Asn	Gly	Glu	Gln 515	Ala	Ala	Gly	His 520	Ala	Arg	Gln	Phe	His 525	Ala	Met	Ala
Thr 530	Arg	Thr	Arg	Gln	Gln	Tyr 535	Leu	Gln	Asp	Leu	Ala 540	Thr	Asn	Glu	Val
Thr 545	Thr	Thr	Ser	Leu	Asp 550	Ser	Ala	Ser	Arg	Phe 555	Gly	Leu	Pro	Ser	Leu 560
Gly	Gly	Arg	Arg	Arg 565	Ala	Ala	Pro	Arg	Gly 570	Pro	Gly	Ala	Glu	Leu 575	Gln
Ala	Ala	Gly	Ser 580	Leu	Val	Trp	Gly	Val 585	Arg	Ala	Ala	Pro	Gly 590	Ala	Arg
Val	Ala	Ala 595	Gly	Ala	Gln	Ala	Ser 600	Gly	Pro	Glu	Gly	Ile 605	Glu	Val	Pro
Cys 610	Leu	Leu	Gly	Ile	Ser	Ala 615	Glu	Ala	Leu	Val	Leu 620	Val	Ala	Pro	Arg
Asp 625	Gly	Arg	Val	Val	Phe 630	Asn	Cys	Ala	Cys	Arg 635	Asp	Val	Leu	Ala	Trp 640
Thr	Phe	Ser	Glu	Gln 645	Gln	Leu	Asp	Leu	Tyr 650	His	Gly	Arg	Gly	Glu 655	Ala
Ile	Thr	Leu	Arg 660	Phe	Asp	Gly	Ser	Pro 665	Gly	Gln	Ala	Val	Gly 670	Glu	Val
Val	Ala 675	Leu	Gln	Leu	Val	Ser	Arg 680	Gly	Cys	Glu	Thr 685	Arg	Glu	Leu	
Ala 690	Leu	Pro	Arg	Asp	Gly	Gln 695	Gly	Arg	Leu	Gly	Phe 700	Glu	Val	Asp	Ala
Glu 705	Gly	Phe	Val	Thr	His 710	Val	Glu	Arg	Phe	Thr 715	Phe	Ala	Glu	Thr	Ala 720
Gly	Leu	Arg	Pro	Gly 725	Ala	Arg	Leu	Leu	Arg 730	Val	Cys	Gly	Gln	Thr	Leu 735
Pro	Ser	Leu	Arg	Pro	Glu	Ala	Ala	Ala	Gln	Leu	Leu	Arg	Ser	Ala	Pro

-continued

740	745	750
Lys Val Cys Val Thr Val Leu Pro Pro Asp Glu Ser Gly Arg Pro Arg		
755	760	765
Arg Ser Phe Ser Glu Leu Tyr Thr Leu Ser Leu Gln Glu Pro Ser Arg		
770	775	780
Arg Gly Ala Pro Asp Pro Val Gln Asp Glu Val Gln Gly Val Thr Leu		
785	790	795
Leu Pro Thr Thr Lys Gln Leu Leu His Leu Cys Leu Gln Asp Gly Gly		
805	810	815
Ser Pro Pro Gly Pro Gly Asp Leu Ala Glu Glu Arg Thr Glu Phe Leu		
820	825	830
His Ser Gln Asn Ser Leu Ser Pro Arg Ser Ser Leu Ser Asp Glu Ala		
835	840	845
Pro Val Leu Pro Asn Thr Thr Pro Asp Leu Leu Leu Ala Thr Thr Ala		
850	855	860
Lys Pro Ser Val Pro Ser Ala Asp Ser Glu Thr Pro Leu Thr Gln Asp		
865	870	875
Arg Pro Gly Ser Pro Ser Gly Ser Glu Asp Lys Gly Asn Pro Ala Pro		
885	890	895
Glu Leu Arg Ala		
900		

<210> SEQ ID NO 49
 <211> LENGTH: 720
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 49

atggttacaa gtgtaaggac tcagccccc gtcataatgc caggtatgca gaccgagatc	60
aaaacgcagg ccacgaatct tgcggcgaat ctttccgcag tcagagaaag tgccacaacg	120
acgctgtcag gggaaattaa aggcccgcaa ctggaagatt ttcccgcgt gatcaaacag	180
gcgagtctgg atgcattgtt taaatgcggg aaagacgctg aggcgttaaa agaagttttt	240
accaattcaa ataatgtcgc cgtaagaaa gcgataatgg agtttgccgg gctctttcgt	300
tcagcgctca acgccaccag tgattctcct gaggcgaaga cgctactgat gaagggtggg	360
gcagagtata ccgcgcaaat cataaaagat ggctgaaag aaaagtcagc ttttgggcca	420
tggctgccag aaacaaagaa agcgaagcg aagctggaaa acctggaaaa gcagctgtta	480
gatattatca aaaataacac tgggtgtgaa ttaagtaaat tatcgacgaa tcttgttatg	540
caggaggtga tgccctatat tgccagctgc attgaacata actttggctg tacgtagat	600
ccgttaacc gcagcaatct tacgcacctt gttgacaaag cggcggcgaa ggcggttgag	660
gcgcttgata tgtgccacca aaaattaacg caagagcagg gtaccagcgt aggacgggaa	720

<210> SEQ ID NO 50
 <211> LENGTH: 907
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 50

Met Trp Asn Val Ser Lys Ser Ser Asn Asn Leu Gly Ala Tyr Lys Leu

-continued

1	5	10	15
Pro Leu Glu Ala Gln Thr Pro Pro Glu Lys Ile Ser Pro Phe Asp Ala	20	25	30
Met Ser Ala Ala Gln Pro Glu Gly Lys Ala Pro His Asp Gln Leu Gln	35	40	45
Asn Asp Gln Tyr Pro Ile Gln Gln Ala Glu Asp Arg Gly Arg His Leu	50	55	60
Val Glu Gln Ala Glu Ile Gln Ala His Val Gln His Cys His Ser Lys	65	70	75
Ala Pro Glu Ile Gly Asp Ala Thr Lys Thr Gln Ser Val Ser Glu Lys	85	90	95
Leu Gly Thr Ala Lys Asn Ser Ser Cys Asp Ala Ser Gln Ile Leu Ile	100	105	110
Gly Ser Lys Asn Asp Asp Phe His Lys Asn Lys Ala Gly Ser Asn Gly	115	120	125
Asp Ile Asn Lys Ser Ser Asp Pro Ser Ala Leu Arg Cys Ser Leu Ser	130	135	140
Pro Ala Pro Arg Arg Val Pro Lys Ser Lys Lys Ser Tyr Gly Ala Ala	145	150	155
Thr Ile Gly Gly Lys Val Tyr His Pro His Glu Lys Thr Asp Ser Thr	165	170	175
Ile Ala Asp Phe Leu Ser Arg Ser Leu Ser Asn Asn Ala Tyr Arg Ser	180	185	190
Glu Arg His Leu Arg Lys Arg Ala Leu Ala Tyr Leu Asn His Ile Ser	195	200	205
Ala Glu Lys Glu Ile Thr Ser Asn Ala Cys Phe Ala Met Lys Asp Val	210	215	220
Asn Ser Phe Ala His Lys Gln Ser Glu Trp Leu Cys His Leu Glu Arg	225	230	235
Ser Leu Trp Arg Asp Glu Pro Ala Leu Gln Phe His Asp Arg Gln Gln	245	250	255
Leu Gly Asn Glu Val Leu Gly Leu Lys Lys Pro Asp Asp Gln Ser Pro	260	265	270
Tyr Phe Lys Pro Arg Ala Trp Lys Ile Ser Asp Glu Ala Ala Ser Ala	275	280	285
Phe Ala Met Met Leu Lys Gly Glu Ser Gly Pro Phe Thr Gln Asp Gln	290	295	300
Val Lys Val Gly Phe Glu Ile Cys Gln Glu Gly Glu Leu Leu Ala Gly	305	310	315
Arg Leu Lys Ile Gln Pro Arg Met Ala Phe Arg Leu Lys Asn Arg His	325	330	335
Asp Ala Asn Arg Ser Gly Thr His Ser Val Lys Ser Leu Ser Gly Leu	340	345	350
Asp Leu Ser Ala Asp Val Gly Thr Asp Ile Arg Glu Phe Phe Gln Val	355	360	365
Pro Val Met Ser Gly Thr Ser Gly Thr Ser Ser Asp Val Val Ile Ala	370	375	380
Ala Arg Tyr Ala Ala Met His Ala Gly Leu Lys Trp Ser Ala Pro Glu	385	390	395
Leu Thr Ile Asp Gln Ala Lys His Ala Leu Ile Asp Leu Ser Met Asp	405	410	415
Phe Phe Arg Arg Asn Gly Pro Ala Val Val Met Ala Leu Arg Met Asn	420	425	430

-continued

Ser	Leu	Arg	Lys	Asn	Gln	Gly	Leu	Pro	Tyr	Lys	Glu	Val	Asp	Arg	Cys
	435						440					445			
Glu	Val	Phe	Thr	His	Ser	Tyr	Ala	Glu	Ile	His	Gly	Ala	Ile	Ser	Leu
	450					455					460				
Thr	Ile	Asp	Gly	Val	Asp	Pro	Ala	Asp	Lys	Val	Glu	Val	Lys	Asn	Arg
465					470					475					480
Leu	Tyr	Gly	Tyr	Thr	Leu	Asp	Ala	Lys	Ala	Thr	Leu	Met	Lys	Ile	Ala
			485						490					495	
Asp	Arg	Ser	Ile	Arg	Arg	Gly	Val	Arg	Ser	Lys	Val	Asp	Ile	Arg	Ser
			500					505					510		
Thr	Ser	Thr	Ser	Leu	Gln	Thr	Pro	Gln	Leu	Arg	Arg	Val	Leu	Glu	Lys
		515					520					525			
Lys	Lys	Ile	Val	Gln	Lys	Val	Ala	Glu	Leu	Tyr	Ser	Glu	Met	Gly	Lys
	530					535					540				
Ala	Gly	Asn	Ser	Ala	Thr	Leu	Lys	Glu	Ala	Ile	Thr	Lys	Ser	Ser	Val
545					550					555					560
Lys	Glu	Leu	Leu	Val	Asn	Asp	Lys	Pro	Val	Val	Ser	Arg	Asp	Tyr	Ala
			565						570					575	
Leu	Gly	Glu	Pro	Leu	Met	Val	Arg	Ser	Leu	Arg	Phe	Ser	His	Asp	His
			580					585					590		
Glu	Ala	Thr	Ser	Ser	Phe	Gly	Ser	Ala	Gly	Lys	Thr	Pro	Ala	Lys	Arg
		595					600					605			
Glu	Val	Asp	Thr	Leu	Cys	Asp	Asn	Ser	Thr	Ala	Phe	Asp	Ile	Val	Met
	610					615					620				
Thr	Pro	Phe	Ser	Val	Ile	Asn	Ala	Lys	Ala	Lys	Gly	Asp	Thr	Ile	Ser
625					630					635					640
Glu	Met	Lys	Val	Pro	His	Arg	Pro	Lys	Trp	Lys	Gly	Leu	Pro	Ser	Val
			645						650					655	
Leu	Tyr	Lys	Val	Thr	Ala	Ser	Val	Asp	Leu	Pro	Glu	Tyr	Ala	Val	Ala
			660					665					670		
Arg	Pro	Gly	Phe	Gly	Asp	Ile	His	Ser	Phe	Asn	Ser	Asn	Lys	Ala	Phe
		675					680					685			
Ser	Ser	Glu	Phe	Ser	Ser	Val	Arg	Asn	Ser	Leu	Ser	His	Ala	Glu	Lys
		690				695					700				
Met	Gly	Phe	Ile	Glu	Asn	Ser	Leu	Lys	Pro	Tyr	Ile	Lys	His	Asp	Pro
705					710					715					720
Asp	Arg	Glu	Ser	Phe	Asp	Phe	Lys	His	Ser	Ile	Asp	Glu	Leu	Ala	Asp
			725						730					735	
Ala	Gln	Cys	Met	Leu	Gln	Ser	Arg	Lys	Pro	Asn	Ser	Thr	Leu	Arg	His
			740					745					750		
Asn	Glu	Tyr	Cys	Ala	Lys	Leu	Glu	Leu	Trp	Asp	Ala	Lys	Ala	Ile	Glu
		755					760					765			
Val	Gly	Met	Ser	Arg	Pro	Val	Ala	Val	Ala	Thr	Leu	Ile	Glu	Phe	Asn
	770					775					780				
Leu	Glu	Met	Leu	Ser	Ala	Ala	Arg	Tyr	Ile	Glu	Asp	Glu	Gly	Tyr	Asp
785					790					795					800
Gly	Lys	Leu	Ile	Thr	Asn	Phe	Leu	Glu	Arg	Gln	Leu	Ser	Trp	Phe	Gly
			805						810					815	
Gln	Asn	Ala	Ala	Leu	Asn	Lys	Glu	Val	Thr	Leu	Lys	Lys	Leu	Trp	Gly
			820					825					830		
Leu	Pro	Phe	Asp	Glu	Arg	Lys	Ala	Val	Ala	Glu	Lys	Val	Cys	Glu	Ala
		835					840					845			

-continued

Leu Arg Gln Gly Val Ser Leu Cys Val Tyr Glu Lys Asn Val Glu Gly
850 855 860

Ser Arg Ile Arg Glu Leu Ser Leu Leu Asn Phe Asn Ala Tyr Asp Ile
865 870 875 880

Met Arg Gly Ile Glu Leu Phe Leu Ser Ser Lys Leu Leu Gln Pro Pro
885 890 895

Thr Gly Ala Gly Pro Thr Val Lys Ser Arg Leu
900 905

<210> SEQ ID NO 51
<211> LENGTH: 1080
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 51

```

accgctatcg ctgttcgggg gactgaaact agacgcgcat ttacagcgca ctgcttttgc      60
acataaccct gctgtacctg acgataattg cgcgttacgg attgcggtag gaatgaatta      120
aaatatTTTT tatcatttat gaccatgtgt tgagctttta ttataaaaaa gattttttga      180
gtagtaattc ttatatataa tcattccggag gtggttggtg gcctggctca atcattgagg      240
catatttttg caggcaatat attgaatctg aaaagttaaa gatgatattt tgggtgcagg      300
agctatcatg tggagggaaa agtatgctaa gtcctacgac tcgtaatatg ggggcgagtt      360
tatcgccctc gcctgacgtc agcgggggagc taaacaccga agcattgacc tgtattgttg      420
agcgtctgga aagtgaattt atagatggca gctggattca tatcagttac gaggaaaccg      480
atctcgaaat gatgcctttt cttgttgcac aggccataaa gaagtatcca gagttaaate      540
ttaaatTTGT tatgtcagtc catgagcttg ttctctctat aaaggagacc agaattggaag      600
gcgttgaaat tgcccgattt ctcgtaaata tgggaagttc aggtatccat atttcagtcg      660
tcgatttttag agttatggac ggaaagacat cgggtgattt gttcgaacca gcagcgtgta      720
gcgcttttgg acctgcactg gcgttgagga ccaaagcagc tcttgaacgt gaacaactgc      780
ctgattgtta ttttgcctat gtcgagctgg acattcaacg aagctcttct gaatgcggtg      840
tttttagcct ggcgctcgcc aaaaaacttc agcttgaatt tatgaactta gtaaaaattc      900
atgaagataa tatttgtgaa cgtctgtgtg gtgaagaacc ttttctcccg tccgataaag      960
cagaccgcta tctgcccgtg agtttttaca aacatactca aggcgcacaa cgattaaatg     1020
aatatgtgga ggccaatccg gcggcgggaa gcagtatagt aaacaaaaag aatgaaacgc     1080

```

<210> SEQ ID NO 52
<211> LENGTH: 700
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 52

Met Phe Asn Ile Arg Asn Thr Gln Pro Ser Val Ser Met Gln Ala Ile
1 5 10 15

Ala Gly Ala Ala Ala Pro Glu Ala Ser Pro Glu Glu Ile Val Trp Glu
20 25 30

Lys Ile Gln Val Phe Phe Pro Gln Glu Asn Tyr Glu Glu Ala Gln Gln
35 40 45

-continued

Cys	Leu	Ala	Glu	Leu	Cys	His	Pro	Ala	Arg	Gly	Met	Leu	Pro	Asp	His
50						55					60				
Ile	Ser	Ser	Gln	Phe	Ala	Arg	Leu	Lys	Ala	Leu	Thr	Phe	Pro	Ala	Trp
65					70					75					80
Glu	Glu	Asn	Ile	Gln	Cys	Asn	Arg	Asp	Gly	Ile	Asn	Gln	Phe	Cys	Ile
			85						90					95	
Leu	Asp	Ala	Gly	Ser	Lys	Glu	Ile	Leu	Ser	Ile	Thr	Leu	Asp	Asp	Ala
			100					105					110		
Gly	Asn	Tyr	Thr	Val	Asn	Cys	Gln	Gly	Tyr	Ser	Glu	Ala	His	Asp	Phe
		115					120					125			
Ile	Met	Asp	Thr	Glu	Pro	Gly	Glu	Glu	Cys	Thr	Glu	Phe	Ala	Glu	Gly
	130					135					140				
Ala	Ser	Gly	Thr	Ser	Leu	Arg	Pro	Ala	Thr	Thr	Val	Ser	Gln	Lys	Ala
145					150						155				160
Ala	Glu	Tyr	Asp	Ala	Val	Trp	Ser	Lys	Trp	Glu	Arg	Asp	Ala	Pro	Ala
			165						170					175	
Gly	Glu	Ser	Pro	Gly	Arg	Ala	Ala	Val	Val	Gln	Glu	Met	Arg	Asp	Cys
			180					185					190		
Leu	Asn	Asn	Gly	Asn	Pro	Val	Leu	Asn	Val	Gly	Ala	Ser	Gly	Leu	Thr
		195					200					205			
Thr	Leu	Pro	Asp	Arg	Leu	Pro	Pro	His	Ile	Thr	Thr	Leu	Val	Ile	Pro
	210					215						220			
Asp	Asn	Asn	Leu	Thr	Ser	Leu	Pro	Glu	Leu	Pro	Glu	Gly	Leu	Arg	Glu
225					230					235					240
Leu	Glu	Val	Ser	Gly	Asn	Leu	Gln	Leu	Thr	Ser	Leu	Pro	Ser	Leu	Pro
			245						250					255	
Gln	Gly	Leu	Gln	Lys	Leu	Trp	Ala	Tyr	Asn	Asn	Trp	Leu	Ala	Ser	Leu
		260						265					270		
Pro	Thr	Leu	Pro	Pro	Gly	Leu	Gly	Asp	Leu	Ala	Val	Ser	Asn	Asn	Gln
		275					280					285			
Leu	Thr	Ser	Leu	Pro	Glu	Met	Pro	Pro	Ala	Leu	Arg	Glu	Leu	Arg	Val
	290					295					300				
Ser	Gly	Asn	Asn	Leu	Thr	Ser	Leu	Pro	Ala	Leu	Pro	Ser	Gly	Leu	Gln
305				310						315					320
Lys	Leu	Trp	Ala	Tyr	Asn	Asn	Arg	Leu	Thr	Ser	Leu	Pro	Glu	Met	Ser
			325						330					335	
Pro	Gly	Leu	Gln	Glu	Leu	Asp	Val	Ser	His	Asn	Gln	Leu	Thr	Arg	Leu
			340					345					350		
Pro	Gln	Ser	Leu	Thr	Gly	Leu	Ser	Ser	Ala	Ala	Arg	Val	Tyr	Leu	Asp
		355					360					365			
Gly	Asn	Pro	Leu	Ser	Val	Arg	Thr	Leu	Gln	Ala	Leu	Arg	Asp	Ile	Ile
	370					375					380				
Gly	His	Ser	Gly	Ile	Arg	Ile	His	Phe	Asp	Met	Ala	Gly	Pro	Ser	Val
385					390					395					400
Pro	Arg	Glu	Ala	Arg	Ala	Leu	His	Leu	Ala	Val	Ala	Asp	Trp	Leu	Thr
			405						410					415	
Ser	Ala	Arg	Glu	Gly	Glu	Ala	Ala	Gln	Ala	Asp	Arg	Trp	Gln	Ala	Phe
			420					425					430		
Gly	Leu	Glu	Asp	Asn	Ala	Ala	Ala	Phe	Ser	Leu	Val	Leu	Asp	Arg	Leu
	435						440					445			
Arg	Glu	Thr	Glu	Asn	Phe	Lys	Lys	Asp	Ala	Gly	Phe	Lys	Ala	Gln	Ile
	450					455					460				
Ser	Ser	Trp	Leu	Thr	Gln	Leu	Ala	Glu	Asp	Ala	Ala	Leu	Arg	Ala	Lys

-continued

465	470	475	480
Thr Phe Ala Met Ala Thr Glu Ala Thr Ser Thr Cys Glu Asp Arg Val			
	485	490	495
Thr His Ala Leu His Gln Met Asn Asn Val Gln Leu Val His Asn Ala			
	500	505	510
Glu Lys Gly Glu Tyr Asp Asn Asn Leu Gln Gly Leu Val Ser Thr Gly			
	515	520	525
Arg Glu Met Phe Arg Leu Ala Thr Leu Glu Gln Ile Ala Arg Glu Lys			
	530	535	540
Ala Gly Thr Leu Ala Leu Val Asp Asp Val Glu Val Tyr Leu Ala Phe			
	545	550	555
Gln Asn Lys Leu Lys Glu Ser Leu Glu Leu Thr Ser Val Thr Ser Glu			
	565	570	575
Met Arg Phe Phe Asp Val Ser Gly Val Thr Val Ser Asp Leu Gln Ala			
	580	585	590
Ala Glu Leu Gln Val Lys Thr Ala Glu Asn Ser Gly Phe Ser Lys Trp			
	595	600	605
Ile Leu Gln Trp Gly Pro Leu His Ser Val Leu Glu Arg Lys Val Pro			
	610	615	620
Glu Arg Phe Asn Ala Leu Arg Glu Lys Gln Ile Ser Asp Tyr Glu Asp			
	625	630	635
Thr Tyr Arg Lys Leu Tyr Asp Glu Val Leu Lys Ser Ser Gly Leu Val			
	645	650	655
Asp Asp Thr Asp Ala Glu Arg Thr Ile Gly Val Ser Ala Met Asp Ser			
	660	665	670
Ala Lys Lys Glu Phe Leu Asp Gly Leu Arg Ala Leu Val Asp Glu Val			
	675	680	685
Leu Gly Ser Tyr Leu Thr Ala Arg Trp Arg Leu Asn			
	690	695	700

<210> SEQ ID NO 53

<211> LENGTH: 2091

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 53

gccggaacgc tttaacgcgc ttcgtgaaaa gcaaatatcg gattatgaag acacgtaccg	60
gaagctgtat gacgaagtgc tgaaatcgtc cgggctggtc gacgataccg atgcagaacg	120
tactatcgga gtaagtgcga tggatagtgc gaaaaaagaa tttctggatg gcctgcgcgc	180
tcttgtggat gaggtgctgg gtagctatct gacagcccgc tggcgtctta actgagcacg	240
atattcaccg caccaggcga atgtggtgcg gtgaacaaag atattcctgg acaacaaca	300
tcagacagca ctgatgatgc acaggtgaaa caggggagac ttcttcagtc agggcgtagc	360
gcaactcaac cttttcgacg ataacgcgcc gcgcgccgat agtgcggaagt tgatggaagt	420
actggaccat cttaatgcaa aagacgggaa ggggacgctg tacttcgccg ggtaggggat	480
gtcgcaacag tgggctatga agcgagaaat gctttcacct cgggtatacga ccagattttc	540
cgatctgcc aatagtcagg aacgggtttg atcagctctt ccccttgatt ttccacattg	600
ccaacggcgc gcttcacggc gtgccaggta aatttatctg tcggcacggc accatcagta	660
attatctcct ctacctcctt cccgcttatg cctggcgca tccattttcg cgcggtgcca	720

-continued

```

ggtgacagaa cgagaggacg acggtcgtga atgtctacca gacctttatc agctgcggag 780
gtaacaatca ggaatccctc tgcgtcatcg ccgcgctcaa acggtgtact gccaatggca 840
gccatgaata tcgggttccc gtcctttctg tgaatgaaat acggtgtttt cttgtcgcct 900
tcctttcttc actcaaacca accatcagca aacacgatag ctccggccatg ttgccatagc 960
ggtttaaaca ttctgctggg ggcgcgagtc tcaacccgtg cattaatcag cggtgggtta 1020
tcccaccatc cgggcgcaaa tcccagaat accggatcca gatgcagttg ctgcgcgct 1080
tcactgagca gcagaacttt ggtaccgggc gccacgttgt accggcctat aggttcaggg 1140
tcataagcga tatcgcgctc ggcttcatcg gccagatatg ccagggtattc ttcgcgggtc 1200
tgtgcttggt caaagcgtcc acacatatga aaacctcagt cggtcagact gaaagtatag 1260
aagaggatat gcgagtggct gttccggtgt ttcttcgagc gacctcaacc aatgtagaag 1320
cttcactatt gggggttgcc actagtagca tcatgttgaa tgtactggcg tgaaaaaatt 1380
ggaatcttga agaaaactct tccccaaaac tataatcaac gttttgataa tcaatgagtt 1440
gtaaaagaca gttactggat ttttttgata gtaggaagaa tgataatttc aactttatca 1500
aatggttgat atgtttttgg caatgtaatg ctgcgccaca tgcagtgggt cgaagccgca 1560
gacctgattg ttaaaggat ggaaggcgcg attgccgca agaccgtgac ctatgacttt 1620
gaacgcctga tggaaggcgc taagctgctg aaatgtagtg agtttggtga cgcgattatc 1680
gcgaatatgt aataacgata attgttaaaa acaaaaacgg ggacttaacg tccccgtttt 1740
tattattagt attcgaacgg ttatcaaaac tttatcaaaa ctccctcaat tcagaccgca 1800
atagtagttc atcctttacc ccgatcgtca tgataacctg agacccccct gaaagctgat 1860
agttaccgat tttgattgta gtggttttct taataatgtg tgatttggtt attttttgaa 1920
agtgccttac cagatttatt gttattatca ggagaatttt ttaataaaaa gtgatggttt 1980
attgactaca atagtgggta ggtaagtat tttataaaat tattagcatg ctttatttgc 2040
tctttacaaa ggctgctaag aagtgttgaa atatcagcct gataaagaat a 2091

```

<210> SEQ ID NO 54

<211> LENGTH: 340

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 54

```

Met Asn Ile Cys Val Asn Ser Leu Tyr Arg Leu Ser Ile Pro Gln Phe
1             5             10             15

His Ser Leu Tyr Thr Glu Glu Val Ser Asp Glu Ala Leu Thr Leu Leu
20             25             30

Phe Ser Ala Val Glu Asn Gly Asp Gln Asn Cys Ile Asp Leu Leu Cys
35             40             45

Asn Leu Ala Leu Arg Asn Asp Asp Leu Gly His Arg Val Glu Lys Phe
50             55             60

Leu Phe Asp Leu Phe Ser Gly Lys Arg Thr Gly Ser Ser Asp Ile Asp
65             70             75             80

Lys Lys Ile Asn Gln Ala Cys Leu Val Leu His Gln Ile Ala Asn Asn
85             90             95

Asp Ile Thr Lys Asp Asn Thr Glu Trp Lys Lys Leu His Ala Pro Ser
100            105            110

Arg Leu Leu Tyr Met Ala Gly Ser Ala Thr Thr Asp Leu Ser Lys Lys

```

-continued

115	120	125
Ile Gly Ile Ala His Lys	Ile Met Gly Asp Gln	Phe Ala Gln Thr Asp
130	135	140
Gln Glu Gln Val Gly Val	Glu Asn Leu Trp Cys	Gly Ala Arg Met Leu
145	150	155
Ser Ser Asp Glu Leu Ala Ala	Ala Thr Gln Gly Leu Val	Gln Glu Ser
165	170	175
Pro Leu Leu Ser Val Asn Tyr	Pro Ile Gly Leu Ile His	Pro Thr Thr
180	185	190
Lys Glu Asn Ile Leu Ser Thr	Gln Leu Leu Glu Lys	Ile Ala Gln Ser
195	200	205
Gly Leu Ser His Asn Glu Val	Phe Leu Val Asn Thr	Gly Asp His Trp
210	215	220
Leu Leu Cys Leu Phe Tyr Lys	Leu Ala Glu Lys	Ile Lys Cys Leu Ile
225	230	235
Phe Asn Thr Tyr Tyr Asp Leu	Asn Glu Asn Thr Lys	Gln Glu Ile Ile
245	250	255
Glu Ala Ala Lys Ile Ala Gly	Ile Ser Glu Ser Asp	Glu Val Asn Phe
260	265	270
Ile Glu Met Asn Leu Gln Asn	Asn Val Pro Asn Gly	Cys Gly Leu Phe
275	280	285
Cys Tyr His Thr Ile Gln Leu	Leu Ser Asn Ala Gly	Gln Asn Asp Pro
290	295	300
Ala Thr Thr Leu Arg Glu Phe	Ala Glu Asn Phe Leu Thr	Leu Ser Val
305	310	315
Glu Glu Gln Ala Leu Phe Asn	Thr Gln Thr Arg Arg	Gln Ile Tyr Glu
325	330	335
Tyr Ser Leu Gln		
340		

<210> SEQ ID NO 55

<211> LENGTH: 954

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 55

gtgagcgtatg agacgcttgc gttgttgttt agcgccgtag agaacggtga tcagaattgt	60
attgatctgt tatgcaatct tgcgttacac aatgataacc tgggacatag agttgagaaa	120
tttctttttg aactctttag cggaaaaaga tcgggctcac cagatataga taaaaaatc	180
aatcaggctt gccttgattt acatcaaatt gccataacg atataacaaa aaataatact	240
gagtggaaaa agctacatac cccttcaga ttactttata tggccggttc cgcgacaacc	300
gacctttcta aaaaaataga aatagcacat aaaattatgg gcaaccagtt cgctcagaca	360
gataaagaac aggttgagtg tgaataatctt tgggtgtggtg tgcaaatgat gtcgtcagat	420
gagctggcag ctgcaacgca aggtctggtt caagaatcac cttttctctc ggtaaaactat	480
ccccattggac ttattcatcc taccacaaa gaaaatatat taagcaactca gctacttgaa	540
aagattgtct aatcaggatt atgtgaaaat gaaatcttc tgataaatac aggagatcac	600
tggcttctct gtttatttta taaacttgca gaaaaataa aatgtctcat atttaacagt	660
tatcatgatt taaatgaaaa tactaagcaa gagattatag aagcagcaa gattgcaggt	720

-continued

```

atatcagaaa gcgatgaggt taattttatc gaaattaatt tacagaataa tgtacccaac    780
ggctgtggtc tattttgtta ccatgcaatt caactcttat cgaatgccgg gcaaaacgat    840
cctgttacca cactacgaga atttgcggaa aatttcttaa cgcttccagt agaggaacaa    900
acactattta acacccaaac ccggcgacaa atatatgaat acagtctcca gtaa          954

```

```

<210> SEQ ID NO 56
<211> LENGTH: 189
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 56

```

```

Met Pro Gly Thr Ile Ser Ser Ser Gly Phe Gly Phe Ser Ile Ala Lys
 1             5             10             15
Gln Pro His Ser Ser Gly Gln Lys Thr Val Ile Asp Gly Phe Phe Leu
      20             25             30
Gly Thr Arg Lys Ile Ser Phe Ser Tyr Leu Arg Leu Glu Ser Glu Leu
      35             40             45
Met Gln Cys Ile Asn Leu Lys Asn Glu Gly Lys Met Asn Glu Trp Met
 50             55             60
Arg Glu Glu Cys Ile Cys Phe Val Ser Arg Asp Val Asn Lys Gln Leu
 65             70             75             80
Asp Ile Phe Ala Lys Asn Asn Gln Thr Thr Ile Pro Gly Cys Val Arg
      85             90             95
Glu Arg Val Phe Gln Arg Ala Ser Phe His Cys Gly Phe Ser Leu Asp
      100            105            110
Val Arg Cys Ala Gln Thr Ser Thr His His Met Ile Leu Asn Ser Leu
      115            120            125
Tyr Phe Gln Lys Lys Met Asp Thr Leu Phe Gly Ser Ala Asp Val Glu
      130            135            140
Val Arg Asn Gln Cys Val Arg Thr Ala Leu Ser Ser Leu Ala Asp Ile
      145            150            155            160
Phe Phe Glu Arg Asn Val Asn Ser Ile Asp Met Asn Lys Phe Arg Asp
      165            170            175
Lys Val Tyr Asp Ala Ile Val Gln Glu Ala Gln Arg Thr
      180            185

```

```

<210> SEQ ID NO 57
<211> LENGTH: 551
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

```

```

<400> SEQUENCE: 57

```

```

acagtaaact cctgtggttt tggatttagc attacaaaag cccctattc ttccggacaa    60
aaaccattta tagatggttt ctttttaggc gcgaggaaaa tctcattctc ttatcctcga    120
ctggaaagcg agttaatata gtgcattaac ctgaaaaatg aaggaaaaaa gaatgagtgg    180
atgaaggagg agggatatttg ttttgtttcg cgggatgtca ataaactcct ggatagtgtt    240
gctaaaaaca accagacaaa catacctgag ggggtccggg agcgggtttt tcagctcgca    300
agtttttatt gcggtttctc gttggatgca agatgcgccc agacatccac tcatcacatg    360
attttaaata gtcagtattt tcagaaaaaa atggatactc ttttgacttc agtagatata    420

```

-continued

```

aatgtcagaa atcagtgtgt ccgtacagcg ctaagtagtc tagcggatac tttttttgag 480
aataatgtta acaatataga tatgaataaa ttccgtgaca gggttcataa cactattgta 540
caggaggttc a 551

```

```

<210> SEQ ID NO 58
<211> LENGTH: 293
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 58

```

```

Met Leu Ser Pro Tyr Ser Val Asn Leu Gly Cys Ser Trp Asn Ser Leu
1          5          10          15

Thr Arg Asn Leu Thr Ser Pro Asp Asn Arg Val Leu Ser Ser Val Arg
20        25        30

Asp Ala Ala Val His Ser Asp Asn Gly Ala Gln Val Lys Val Gly Asn
35        40        45

Arg Thr Tyr Arg Val Val Ala Thr Asp Asn Lys Phe Cys Val Thr Arg
50        55        60

Glu Ser His Ser Gly Cys Phe Thr Asn Leu Leu His Arg Leu Gly Trp
65        70        75        80

Pro Lys Gly Glu Ile Ser Arg Lys Ile Glu Val Met Leu Asn Ala Ser
85        90        95

Pro Val Ser Ala Ala Met Glu Arg Gly Ile Val His Ser Asn Arg Pro
100       105       110

Asp Leu Pro Pro Val Asp Tyr Ala Pro Pro Glu Leu Pro Ser Val Asp
115       120       125

Tyr Asn Arg Leu Ser Val Pro Gly Asn Val Ile Gly Lys Gly Gly Asn
130       135       140

Ala Val Val Tyr Glu Asp Ala Glu Asp Ala Thr Lys Val Leu Lys Met
145       150       155       160

Phe Thr Thr Ser Gln Ser Asn Glu Glu Val Thr Ser Glu Val Arg Cys
165       170       175

Phe Asn Gln Tyr Tyr Gly Ala Gly Ser Ala Glu Lys Ile Tyr Gly Asn
180       185       190

Asn Gly Asp Ile Ile Gly Ile Arg Met Asp Lys Ile Asn Gly Glu Ser
195       200       205

Leu Leu Asn Ile Ser Ser Leu Pro Ala Gln Ala Glu His Ala Ile Tyr
210       215       220

Asp Met Phe Asp Arg Leu Glu Gln Lys Gly Ile Leu Phe Val Asp Thr
225       230       235       240

Thr Glu Thr Asn Ile Leu Tyr Asp Arg Ala Lys Asn Glu Phe Asn Pro
245       250       255

Ile Asp Ile Ser Ser Tyr Asn Val Ser Asp Arg Ser Trp Ser Glu Ser
260       265       270

Gln Ile Met Gln Ser Tyr His Gly Gly Lys Gln Asp Leu Ile Ser Val
275       280       285

Val Leu Ser Lys Ile
290

```

```

<210> SEQ ID NO 59
<211> LENGTH: 795
<212> TYPE: DNA

```

-continued

```

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 59

atgctgagcc cgtatagcgt gaacctgggc tgcagctgga acagcctgac ccgcaacctg    60
accagcccggtg ataaccgcgt gctgagcagc gtgcgcgatg cggcggtgca tagcgataac    120
ggcgcgccagg tgaagtggg caaccgcacc tatcgctggg tggcgaccga taacaaattt    180
tgcgtgaccc gcgaaagcca tagcggtgctc tttaaccaacc tgcgtgcatcg cctgggctgg    240
ccgaaaggcg aaattagccg caaaattgaa gtgatgctga acgcgagccc ggtgagcgcg    300
gcatggaac gcgcgattgt gcatagcaac cgcccgatc tgccgcgggt ggattatgcg    360
ccgcgggaac tgccgagcgt ggattataac cgctgagcg tgccgggcaa cgtgattggc    420
aaaggcggca acgcggtggt gtatgaagat gcggaagatg cgaccaaagt gctgaaaatg    480
tttaccacca gccagagcaa cgaagaagt accagcgaag tgcgctgctt taaccagtat    540
tatggcgcggt gcagcgcgga aaaaatttat ggcaacaacg gcgatattat tggcattcgc    600
atggataaaa ttaacggcga aagcctgctg aacattagca gcctgccggc gcaggcggaa    660
catgcgattt atgatatggt tgatcgctg gaacagaaag gcattctggt tgtggatacc    720
accgaaacca acattctgta tgatcgcgcg catggcgga aacaggatct gattagcgtg    780
gtgctgagca aaatt                                         795

```

```

<210> SEQ ID NO 60
<211> LENGTH: 336
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 60

Met Ile Pro Pro Leu Asn Arg Tyr Val Pro Ala Leu Ser Lys Asn Glu
1             5             10             15

Leu Val Lys Thr Val Thr Asn Arg Asp Ile Gln Phe Thr Ser Phe Asn
20            25            30

Gly Lys Asp Tyr Pro Leu Cys Phe Leu Asp Glu Lys Thr Pro Leu Leu
35            40            45

Phe Gln Trp Phe Glu Arg Asn Pro Ala Arg Phe Gly Lys Asn Asp Ile
50            55            60

Pro Ile Ile Asn Thr Glu Lys Asn Pro Tyr Leu Asn Asn Ile Ile Lys
65            70            75            80

Ala Ala Thr Ile Glu Lys Glu Arg Leu Ile Gly Ile Phe Val Asp Gly
85            90            95

Asp Phe Phe Pro Gly Gln Lys Asp Ala Phe Ser Lys Leu Glu Tyr Asp
100           105           110

Tyr Glu Asn Ile Lys Val Ile Tyr Arg Asn Asp Ile Asp Phe Ser Met
115           120           125

Tyr Asp Lys Lys Leu Ser Glu Ile Tyr Met Glu Asn Ile Ser Lys Gln
130           135           140

Glu Ser Met Pro Glu Glu Lys Arg Asp Cys His Leu Leu Gln Leu Leu
145           150           155           160

Lys Lys Glu Leu Ser Asp Ile Gln Glu Gly Asn Asp Ser Leu Ile Lys
165           170           175

```

-continued

Ser Tyr Leu Leu Asp Lys Gly His Gly Trp Phe Asp Phe Tyr Arg Asn
180 185 190

Met Ala Met Leu Lys Ala Gly Gln Leu Phe Leu Glu Ala Asp Lys Val
195 200 205

Gly Cys Tyr Asp Leu Ser Thr Asn Ser Gly Cys Ile Tyr Leu Asp Ala
210 215 220

Asp Met Ile Ile Thr Glu Lys Leu Gly Gly Ile Tyr Ile Pro Asp Gly
225 230 235 240

Ile Ala Val His Val Glu Arg Ile Asp Gly Arg Ala Ser Met Glu Asn
245 250 255

Gly Ile Ile Ala Val Asp Arg Asn Asn His Pro Ala Leu Leu Ala Gly
260 265 270

Leu Glu Ile Met His Thr Lys Phe Asp Ala Asp Pro Tyr Ser Asp Gly
275 280 285

Val Cys Asn Gly Ile Arg Lys His Phe Asn Tyr Ser Leu Asn Glu Asp
290 295 300

Tyr Asn Ser Phe Cys Asp Phe Ile Glu Phe Lys His Asp Asn Ile Ile
305 310 315 320

Met Asn Thr Ser Gln Phe Thr Gln Ser Ser Trp Ala Arg His Val Gln
325 330 335

<210> SEQ ID NO 61
<211> LENGTH: 540
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 61

atgttatctc cattaaatgt tcttcaattt aatttcagag gagagaccgc tttatcagat	60
agtgtctctc tccagactgt ttcttttctt ggaaaagatt attctatgga acccattgat	120
gaaaaaacac ccattctttt tcagtgggtt gaagcaaggc cagagcgata cggaaaaggt	180
gaagtaccga tattgaatac caaagagcat ccgtatttga gcaatattat aaatgctgca	240
aaaatagaaa atgagcgcgt aataggagta ctggtagacg gagactttac ttatgagcaa	300
agaaaagaat ttctcagtct tgaagatgaa catcaaaata taaagataat atatcgggaa	360
aatgttgatt tcagtatgta tgataaaaa ctgtctgata tttatcttga aaatattcat	420
gaacaagaat catatccagc gagttagaga gataattatc tgttaggctt attaagagaa	480
gagttaaaaa atattccata cggaaaggac tctttgattg aatcatatgc agaaaaaaga	540

<210> SEQ ID NO 62
<211> LENGTH: 330
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 62

Met Lys Ile Pro Ser Leu Gln Ser Asn Phe Asn Phe Ser Ala Pro Ala
1 5 10 15

Gly Tyr Ser Ala Pro Ile Ala Pro Asn Arg Ala Glu Asn Ala Tyr Ala
20 25 30

Asp Tyr Val Leu Asp Ile Gly Lys Arg Ile Pro Leu Ser Ala Ala Asp
35 40 45

-continued

Leu Ser Asn Val Tyr Glu Ser Val Ile Arg Ala Val His Asp Ser Arg
 50 55 60
 Ser Arg Leu Ile Asp Gln His Thr Val Asp Met Ile Gly Asn Thr Val
 65 70 75 80
 Leu Asp Ala Leu Ser Arg Ser Gln Thr Phe Arg Asp Ala Val Ser Tyr
 85 90 95
 Gly Ile His Asn Glu Lys Val His Ile Gly Cys Ile Lys Tyr Arg Asn
 100 105 110
 Glu Tyr Glu Leu Asn Glu Glu Ser Ser Val Lys Ile Asp Asp Ile Gln
 115 120 125
 Ser Leu Thr Cys Asn Glu Leu Tyr Glu Tyr Asp Val Gly Gln Glu Pro
 130 135 140
 Ile Phe Pro Ile Cys Glu Ala Gly Glu Asn Asp Asn Glu Glu Pro Tyr
 145 150 155 160
 Val Ser Phe Ser Val Ala Pro Asp Thr Asp Ser Tyr Glu Met Pro Ser
 165 170 175
 Trp Gln Glu Gly Leu Ile His Glu Ile Ile His His Val Thr Gly Ser
 180 185 190
 Ser Asp Pro Ser Gly Asp Ser Asn Ile Glu Leu Gly Pro Thr Glu Ile
 195 200 205
 Leu Ala Arg Arg Val Ala Gln Glu Leu Gly Trp Ser Val Pro Asp Phe
 210 215 220
 Lys Gly Tyr Ala Glu Pro Glu Arg Glu Ala His Leu Arg Leu Arg Asn
 225 230 235 240
 Leu Asn Ala Leu Arg Gln Ala Ala Met Arg His Glu Glu Asn Glu Arg
 245 250 255
 Ala Phe Phe Glu Arg Leu Gly Thr Ile Ser Asp Arg Tyr Glu Ala Ser
 260 265 270
 Pro Asp Phe Thr Glu Tyr Ser Ala Val Ser Asn Ile Gly Tyr Gly Phe
 275 280 285
 Ile Gln Gln His Asp Phe Pro Gly Leu Ala Ile Asn Asp Asn Leu Gln
 290 295 300
 Asp Ala Asn Gln Ile Gln Leu Tyr His Gly Ala Pro Tyr Ile Phe Thr
 305 310 315 320
 Phe Gly Asp Val Asp Lys His Asn Gln Arg
 325 330

<210> SEQ ID NO 63

<211> LENGTH: 1464

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 63

```

gaattcatcc ggtcaaacgg cttctttttg caggaaagga atatgagtta aaggtcattg      60
atgaaaaaac gcctattatc ctgctctact tgagggattg tcttttatgc acagaagagt      120
agatgctcat ccatattatg atggtttagg taaagggata aagaaatatt ttgattttac      180
tcaattacat gattacaatc atttttatga ctttattgag tttaaaccatc caaatattat      240
tatgaacaca agtcagtata caggcaggtc atggtaaatg gtttttacat agtttattct      300
gttgtaataa atgattagca tggtattagg tatcaacatg aaaattccct cactccagcc      360
cagcttcaac tttttcgccc cagcaggata ctctgctgcc gttgctccca atcgttcgga      420

```

-continued

```

caatgcctat gctgattacg tattggatat aggcaagcga ataccacttt ccgcggaaga 480
tttaggcaac ctatatgaaa atgtcattcg cgccgttcgt gacagccgta gcaagctcat 540
agatcagcat acggtcgata tgattggtaa cactatactt gatgctttga gccgatcaca 600
aacctttcgt gatgccgtaa gctatggcat tcataataag gaggtacaca ttggttgcat 660
taaatacaga aacgaatacg agctcaacgg agaatcccc gtcaaagttg atgatattca 720
atcactaacc tgtaccgaat tatatgaata cgatgtcggg caagaaccaa ttttaccat 780
ttgcgaggca ggagaaaacg ataacgaaga gccttatgtc agtttttagtg ttgcgccaga 840
tactgactct tatgagatgc catcgtggca ggaagggctg attcacgaga ttattcatca 900
tgtgactgga gctagcgatc cgtctggaga tagtaataata gagctaggac ccacggagat 960
tctcgacgt cgtgtcgtc aagagctggg atggactgtc cccgacttca taggatatgc 1020
agagccagat cgtgaagctc atcttagggg acgtaacctg aatgcccttc gacaggcggc 1080
catgcgacat gaagataatg agaggacttt cttcgaaagg ctgggtatga tcagtgatcg 1140
atatgaggcg agtcctgatt tcacagagta ttccgctgtg tctaacatag aatatggatt 1200
tatccagcaa catgattttc ccgggttggc tatcgacgat aatttacagg atgcaaatca 1260
gatccaactc tatcatggag caccttatat ctttacatc ggggatgtgg acaaacacaa 1320
tcagcgctga cgcgtctttg cagcgacaca aggtactac tcttgcatth taacggagtt 1380
gatgatggaa aatcgtgcaa ccttgtatgt aaaggcgaaa aaccaaattt tacggtagta 1440
agtgagcctg gcgggaatgg tacc 1464

```

<210> SEQ ID NO 64

<211> LENGTH: 558

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 64

```

Met Pro Ile Gly Asn Leu Gly His Asn Pro Asn Val Asn Asn Ser Ile
1      5              10              15

Pro Pro Ala Pro Pro Leu Pro Ser Gln Thr Asp Gly Ala Gly Gly Arg
20      25              30

Gly Gln Leu Ile Asn Ser Thr Gly Pro Leu Gly Ser Arg Ala Leu Phe
35      40              45

Thr Pro Val Arg Asn Ser Met Ala Asp Ser Gly Asp Asn Arg Ala Ser
50      55              60

Asp Val Pro Gly Leu Pro Val Asn Pro Met Arg Leu Ala Ala Ser Glu
65      70              75              80

Ile Thr Leu Asn Asp Gly Phe Glu Val Leu His Asp His Gly Pro Leu
85      90              95

Asp Thr Leu Asn Arg Gln Ile Gly Ser Ser Val Phe Arg Val Glu Thr
100     105              110

Gln Glu Asp Gly Lys His Ile Ala Val Gly Gln Arg Asn Gly Val Glu
115     120              125

Thr Ser Val Val Leu Ser Asp Gln Glu Tyr Ala Arg Leu Gln Ser Ile
130     135              140

Asp Pro Glu Gly Lys Asp Lys Phe Val Phe Thr Gly Gly Arg Gly Gly
145     150              155              160

Ala Gly His Ala Met Val Thr Val Ala Ser Asp Ile Thr Glu Ala Arg
165     170              175

```

-continued

Gln Arg Ile Leu Glu Leu Leu Glu Pro Lys Gly Thr Gly Glu Ser Lys
 180 185 190
 Gly Ala Gly Glu Ser Lys Gly Val Gly Glu Leu Arg Glu Ser Asn Ser
 195 200 205
 Gly Ala Glu Asn Thr Thr Glu Thr Gln Thr Ser Thr Ser Thr Ser Ser
 210 215 220
 Leu Arg Ser Asp Pro Lys Leu Trp Leu Ala Leu Gly Thr Val Ala Thr
 225 230 235 240
 Gly Leu Ile Gly Leu Ala Ala Thr Gly Ile Val Gln Ala Leu Ala Leu
 245 250 255
 Thr Pro Glu Pro Asp Ser Pro Thr Thr Thr Asp Pro Asp Ala Ala Ala
 260 265 270
 Ser Ala Thr Glu Thr Ala Thr Arg Asp Gln Leu Thr Lys Glu Ala Phe
 275 280 285
 Gln Asn Pro Asp Asn Gln Lys Val Asn Ile Asp Glu Leu Gly Asn Ala
 290 295 300
 Ile Pro Ser Gly Val Leu Lys Asp Asp Val Val Ala Asn Ile Glu Glu
 305 310 315 320
 Gln Ala Lys Ala Ala Gly Glu Glu Ala Lys Gln Gln Ala Ile Glu Asn
 325 330 335
 Asn Ala Gln Ala Gln Lys Lys Tyr Asp Glu Gln Gln Ala Lys Arg Gln
 340 345 350
 Glu Glu Leu Lys Val Ser Ser Gly Ala Gly Tyr Gly Leu Ser Gly Ala
 355 360 365
 Leu Ile Leu Gly Gly Gly Ile Gly Val Ala Val Thr Ala Ala Leu His
 370 375 380
 Arg Lys Asn Gln Pro Val Glu Gln Thr Thr Thr Thr Thr Thr Thr
 385 390 395 400
 Thr Thr Thr Ser Ala Arg Thr Val Glu Asn Lys Pro Ala Asn Asn Thr
 405 410 415
 Pro Ala Gln Gly Asn Val Asp Thr Pro Gly Ser Glu Asp Thr Met Glu
 420 425 430
 Ser Arg Arg Ser Ser Met Ala Ser Thr Ser Ser Thr Phe Phe Asp Thr
 435 440 445
 Ser Ser Ile Gly Thr Val Gln Asn Pro Tyr Ala Asp Val Lys Thr Ser
 450 455 460
 Leu His Asp Ser Gln Val Pro Thr Ser Asn Ser Asn Thr Ser Val Gln
 465 470 475 480
 Asn Met Gly Asn Thr Asp Ser Val Val Tyr Ser Thr Ile Gln His Pro
 485 490 495
 Pro Arg Asp Thr Thr Asp Asn Gly Ala Arg Leu Leu Gly Asn Pro Ser
 500 505 510
 Ala Gly Ile Gln Ser Thr Tyr Ala Arg Leu Ala Leu Ser Gly Gly Leu
 515 520 525
 Arg His Asp Met Gly Gly Leu Thr Gly Gly Ser Asn Ser Ala Val Asn
 530 535 540
 Thr Ser Asn Asn Pro Pro Ala Pro Gly Ser His Arg Phe Val
 545 550 555

<210> SEQ ID NO 65

<211> LENGTH: 1689

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 65

```

taaataaaaag gatatatgta tgcctattgg taaccttggg aataatgtaa atagcaatca    60
tttaattccc cctgogcgcg cactaccttc acaaacagac ggcgcggcac ggggagggaac    120
tggatcatcta attagctcta caggagcatt aggatctcgt tcattgtttt ctcccctgag    180
aaattctgtg gctgattctg tcgattccag agatattcca ggacttcctg caaacccatc    240
gaggcttgct gcagctacat ctgagacatg ctgcttgga ggatttgaag ttctccatga    300
taaggggcca cttgatactc ttaatcagca aattggaccc tctgcatttc gtgttgaagc    360
gcaggcagat ggtactcatg ccgctattgg agaaaaaat ggtttgaggg ttagcgttgc    420
attaagtcct caagaattgc aaagcttgca atctattgat attgagggga aaaacagatt    480
tgtttttacc gggggacgtg gcggtagtgg gcacccgatg gtcactgtcg catcagatat    540
cgcggaagct cgtatgaaaa tactggccaa attagaccca gacaatcatg gaggacgtca    600
acccaaggac gttgatacgc gttctgttgg tgttggcagc gcttcgggaa tagatgatgg    660
cgttgttagc gaaaccata cttcaacaac aaattccagc gttcgcctcag atcctaaatt    720
ctgggtttct gtcggcgcaa ttgctgctgg tttagcggga ctggcggcaa ctggtattgc    780
acaggcgttg gctttgacac cggaaccgga tgatcctaca accaccgatc ctgatcaggc    840
cgcaaatgct gcagaaagtg caacaaaaga tcagttaacg caagaagcat tcaagaaccc    900
tgagaaccag aaagttaaca tcgatgcgaa cggaatgct attcgcctcg gggaattaaa    960
agatgatatt gttgagcaaa tagcacaaca agctaaagag gctggtgagg tggccagaca   1020
gcaggctgtt gaaagcaatg cacaggcgca gcagcgatat gaggatcagt atgccagacg   1080
tcaggaggaa ttacagcttt catcggttat tggttacggc ctcagcagtg cattgattgg   1140
tgctggggga attggtgctg gtgtaacgac tgcgctccat agacgaaatc agccggcaga   1200
acagacaact actacaacaa cacatacggg agtgcagcag cagaccggag ggaatacccc   1260
agcacaaggt ggactgatg ccacaagac agaagatgct tctctgaata gacgtgattc   1320
gcaggggagt gttgcatcga cacactggtc agattcctct agcgaagtgg ttaatccata   1380
tgctgaagtt ggggagcctc ggaatagtct atcgactcgt cagcaagaag agcatattta   1440
cgatgaggtc gctgcagatc ctgtttatag cgtcattcag aatttttcac ggaatgctcc   1500
agttaccgga aggttaatgg gaagcccagg gcaaggatc caaagtactt atgcgcttct   1560
ggcaaacagc gctggattgc gtttaggtat gggaggatta acggggagtg gcgagagcgc   1620
agtaaatact gcaaatgcaa atgccgcacc aacgccggga ccagtacgtt tcgtttaaat   1680
atatctgtg                                     1689

```

<210> SEQ ID NO 66

<211> LENGTH: 358

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 66

```

Met Gln Asn Lys Ile Lys Gln Leu Lys Asn Tyr Ala Val Tyr Asp Asp
 1             5             10             15

Ile Glu Gly Phe Leu Ile Asn Lys Asp Ile Arg Ser Ser Ser Gly Asn
 20             25             30

```

-continued

Ser Asn Tyr Met Met Pro Ser Ser Thr Arg Arg Val Ser Asn Thr Arg
 35 40 45
 Lys Asn Tyr Glu Gly Asp Ile Lys Gln Phe Phe Ser Val Ile Lys Gly
 50 55 60
 Lys Asp Val Lys Ser Leu Val Pro Asp Asp Leu Val Val Ser Lys Ser
 65 70 75 80
 Glu Leu Ser Asn Tyr Val Lys Tyr Leu Gln Glu Lys Gly Leu Val Asn
 85 90 95
 Asn Ser Ile Asn Arg Lys Met Thr Ser Leu Lys Met Leu Tyr Thr Tyr
 100 105 110
 Leu Glu His Asp Tyr Lys Asp Tyr Ile Asp Leu Ser Val Phe Asn Thr
 115 120 125
 Val Glu Arg Leu Lys Thr Val Thr Lys Asn Trp Asp Lys Thr Thr Gln
 130 135 140
 Thr Glu Ala Glu Arg Ile Ala Gln Asp Met Tyr Ile Asn Glu Arg Gln
 145 150 155 160
 Lys Pro Leu Met Lys Lys Leu Phe Val Lys Phe Ala Ile Arg Thr Ser
 165 170 175
 Phe Arg Val Ser Ala Ile Leu Arg Val Arg Trp Lys Asp Ile Gln Leu
 180 185 190
 Asp Glu Ser Thr Gly His Tyr Ile Val Thr Val Ile Asp Lys Gly Ser
 195 200 205
 Gln Val Val Ser Thr Gly Ile Asn Gln Val Phe Tyr Glu Glu Leu Leu
 210 215 220
 Gln Leu Lys Glu Glu Asp Asp Ser Glu Thr Glu Leu Val Phe Gln Gly
 225 230 235 240
 Leu Ser Glu Gln Ser Leu Arg His Ser Leu Lys Arg Ser Lys Lys Arg
 245 250 255
 Leu Gly Ile Pro Pro Glu Arg Glu Leu Val Leu His Ser Phe Lys Gly
 260 265 270
 Val Gly Ile Asp Tyr Val Tyr Glu Asn Ser Gly His Asp Leu Leu Ala
 275 280 285
 Ala Lys Glu Gln Gly Asn His Lys Asn Thr Leu Thr Thr Glu Arg Tyr
 290 295 300
 Met Ser Arg Lys Ile Asn Ile Ala Asn Ser Ala Gly Val Thr Met Asp
 305 310 315 320
 Glu Lys Ile Asp Leu Asn Pro Leu Tyr Glu Ala Thr Gln Glu Asp Phe
 325 330 335
 Ile Ser Phe Phe Glu Asn Ala Asp Leu Val Thr Leu Lys Lys Phe Ile
 340 345 350
 Lys His Val Asn Glu Arg
 355

<210> SEQ ID NO 67

<211> LENGTH: 867

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 67

atgattgggc caatatcaca aataaacagc ttcggtggct tatcagaaaa agagaccgt 60

tctttaatca gtaatgaaga gcttaaaaat atcataatac agttggaaac tgatatagcg 120

-continued

```

gatggatcct ggttcataa aaattattca cgctggata tagaagtcac gccgcatta 180
gtaattcagg cgaacaataa atatccggaa atgaatctta atttgttac atctcccag 240
gacctttcga tagaaataaa aaatgtcata gaaaatggag ttggatcttc ccgcttcata 300
attaacatgg gggagggtgg aatacatctc agtgtaattg attacaaaca tataaatggg 360
aaaacatctc tgatattatt tgaaccagta aactttaata gtatggggcc agcgatactg 420
gcaataagta caaaaacggc cattgaacgt tatcaattac ctgattgccca tttttccatg 480
gtggaaatgg atattcagcg aagctcatct gaatgtggta ttttagttt ggcactggca 540
aaaaaacttt acaccgagag agatagcctg ttgaaaatac atgaagataa tataaaagg 600
atattaagtg atagtgaata tcttttacc cacaataagt tggatccgta tctccggta 660
actttttaca aacatactca aggtaaaaaa cgtcttaatg aatatttaaa tactaaccg 720
cagggagttg gtactgttgt taacaaaaaa aatgaaacca tctttaatag gtttgataac 780
aataaatcca ttatagatgg aaaggaatta tcagtttcgg tacataaaaa gagaatagct 840
gaatataaaa cacttctcaa agtataa 867

```

<210> SEQ ID NO 68

<211> LENGTH: 288

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 68

```

Met Ile Gly Pro Ile Ser Gln Ile Asn Ile Ser Gly Gly Leu Ser Glu
1           5           10          15
Lys Glu Thr Ser Ser Leu Ile Ser Asn Glu Glu Leu Lys Asn Ile Ile
20          25          30
Thr Gln Leu Glu Thr Asp Ile Ser Asp Gly Ser Trp Phe His Lys Asn
35          40          45
Tyr Ser Arg Met Asp Val Glu Val Met Pro Ala Leu Val Ile Gln Ala
50          55          60
Asn Asn Lys Tyr Pro Glu Met Asn Leu Asn Leu Val Thr Ser Pro Leu
65          70          75          80
Asp Leu Ser Ile Glu Ile Lys Asn Val Ile Glu Asn Gly Val Arg Ser
85          90          95
Ser Arg Phe Ile Ile Asn Met Gly Glu Gly Gly Ile His Phe Ser Val
100         105         110
Ile Asp Tyr Lys His Ile Asn Gly Lys Thr Ser Leu Ile Leu Phe Glu
115         120         125
Pro Ala Asn Phe Asn Ser Met Gly Pro Ala Met Leu Ala Ile Arg Thr
130         135         140
Lys Thr Ala Ile Glu Arg Tyr Gln Leu Pro Asp Cys His Phe Ser Met
145         150         155         160
Val Glu Met Asp Ile Gln Arg Ser Ser Ser Glu Cys Gly Ile Phe Ser
165         170         175
Phe Ala Leu Ala Lys Lys Leu Tyr Ile Glu Arg Asp Ser Leu Leu Lys
180         185         190
Ile His Glu Asp Asn Ile Lys Gly Ile Leu Ser Asp Gly Glu Asn Pro
195         200         205
Leu Pro His Asp Lys Leu Asp Pro Tyr Leu Pro Val Thr Phe Tyr Lys
210         215         220

```

-continued

His Thr Gln Gly Lys Lys Arg Leu Asn Glu Tyr Leu Asn Thr Asn Pro
225 230 235 240

Gln Gly Val Gly Thr Val Val Asn Lys Lys Asn Glu Thr Ile Val Asn
245 250 255

Arg Phe Asp Asn Asn Lys Ser Ile Val Asp Gly Lys Glu Leu Ser Val
260 265 270

Ser Val His Lys Lys Arg Ile Ala Glu Tyr Lys Thr Leu Leu Lys Val
275 280 285

<210> SEQ ID NO 69

<211> LENGTH: 867

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 69

```
atgatcggac caatatcaca aataaatatc tccggtggct tatcagaaaa agagaccagt      60
tctttaatca gtaatgaaga gcttaaaaat atcataacac agttggaaac tgatatatcg      120
gatggatcct ggttcataa aaattattca cgtatggatg tagaagtcac gcccgcatg      180
gtaatccagg cgaacaataa atatccggaa atgaatctta atcttgttac atctccattg      240
gacctttcaa tagaaataaa aaacgtcata gaaaatggag ttagatcttc ccgcttcata      300
attaacatgg gggaaggtgg aatacatctc agtgtaattg attacaaaca tataaatggg      360
aaaacatctc tgatatgtgt tgaaccagca aactttaaca gtatggggcc agcgatgctg      420
gcaataagga caaaaacggc tattgaacgt tatcaattac ctgattgccca tttctccatg      480
gtggaaatgg atattcagcg aagctcatct gaatgtggta ttttagttt tgcactggca      540
aaaaaacttt acatcgagag agatagcctg ttgaaaatac atgaagataa tataaaagg      600
atattaagtg atggtgaaaa tcttttacc cagcataagt tggaccgcta tctccggta      660
actttttaca aacatactca aggtaaaaaa cgtcttaatg aatatttaaa tactaaccg      720
cagggagttg gtactgttgt taacaaaaaa aatgaaacca tcgttaatag atttgataac      780
aataaatcca ttgtagatgg aaaggaatta tcagtttcgg tacataaaaa gagaatagct      840
gaatataaaa cacttctcaa agtataa                                     867
```

<210> SEQ ID NO 70

<211> LENGTH: 219

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 70

Met Lys Ile Ser Ser Phe Ile Ser Thr Ser Leu Pro Leu Pro Ala Ser
1 5 10 15

Val Ser Gly Ser Ser Ser Val Gly Glu Met Ser Gly Arg Ser Val Ser
20 25 30

Gln Gln Lys Ser Asp Gln Tyr Ala Asn Asn Leu Ala Gly Arg Thr Glu
35 40 45

Ser Pro Gln Gly Ser Ser Leu Ala Ser Arg Ile Ile Glu Arg Leu Ser
50 55 60

Ser Met Ala His Ser Val Ile Gly Phe Ile Gln Arg Met Phe Ser Glu
65 70 75 80

-continued

Gly Ser His Lys Pro Val Val Thr Pro Ala Leu Thr Pro Ala Gln Met
85 90 95

Pro Ser Pro Thr Ser Phe Ser Asp Ser Ile Lys Gln Leu Ala Ala Glu
100 105 110

Thr Leu Pro Lys Tyr Met Gln Gln Leu Ser Ser Leu Asp Ala Glu Thr
115 120 125

Leu Gln Lys Asn His Asp Gln Phe Ala Thr Gly Ser Gly Pro Leu Arg
130 135 140

Gly Ser Ile Thr Gln Cys Gln Gly Leu Met Gln Phe Cys Gly Gly Glu
145 150 155 160

Leu Gln Ala Glu Ala Ser Ala Ile Leu Asn Thr Pro Val Cys Gly Ile
165 170 175

Pro Phe Ser Gln Trp Gly Thr Val Gly Gly Ala Ala Ser Ala Tyr Val
180 185 190

Ala Ser Gly Val Asp Leu Thr Gln Ala Ala Asn Glu Ile Lys Gly Leu
195 200 205

Gly Gln Gln Met Gln Gln Leu Leu Ser Leu Met
210 215

<210> SEQ ID NO 71
 <211> LENGTH: 1152
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 71

gaattcccca actttgacac cgataaccgg ttcaatagta tctggaatag acagcgaaag 60
 ttgttgaaat aattgagtga tagcttggtc aaatgaatac atttgatctc ctaatagtta 120
 gataaaatat caacttaacc aaagcactct cggcagacca tcaatttttag cctataattt 180
 ttagttttta ttttgtctaa tataacaaca aaaacagcag cggtttttta tataaccacc 240
 ggctattttc ccactaagat aaccttggtt taatagccaa gggaataaat agtcatgaaa 300
 atatcatcat ttatttctac atcactgccc ctgccggcat cagtgtcagg atctagcagc 360
 gtaggagaaa tgtctgggag ctcagttcca cagcaaaaaa gtgatcaata tgcaaacaa 420
 ctggccgggc gcaactgaaag ccctcagggt tccagcttag ccagccgtat cattgagagg 480
 ttatcatcaa tggcccactc tgtgattgga tttatccaac gcatgttctc ggaggggagc 540
 cataaacagg tggtagacac agcactcacg cctgcacaaa tgccaagccc tacgtctttc 600
 agtgatagta tcaagcaact tgtgctgag acgctgccaa aatacatgca gcagttgagt 660
 agcttgatg cagagacgct gcagaaaaat catgaccagt tcgccacggg cagcgccct 720
 cttcgtggca gtatcactca atgccaaggg ctgatgcagt tttgtggtgg ggaattgcaa 780
 gctgaggcca gtgccatttt aaacacgcct gtttgtggta ttcccttctc gcagtgggga 840
 actgttggtg gggcgccag cgcgtacgtc gccagtggcg ttgatctaac gcaggcagca 900
 aatgagatca aagggtggg gcaacagatg cagcaattac tgtcattgat gtgatatgga 960
 taaaaacaag ggggtagtgt tcccccttt ttctatcaat attgogaata tcttcgtccc 1020
 tgatctttca gggcggaatc gttttttagc atgctcattg ttagaatttc tgacttatct 1080
 ctctctgtga ttactactca tactctggaa aatcctgagc atttatatct atggattgat 1140
 gcagcactcg ag 1152

-continued

```

<210> SEQ ID NO 72
<211> LENGTH: 545
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 72

Met Leu Pro Ile Asn Asn Asn Phe Ser Leu Pro Gln Asn Ser Phe Tyr
 1             5             10            15

Asn Thr Ile Ser Gly Thr Tyr Ala Asp Tyr Phe Ser Ala Trp Asp Lys
 20            25            30

Trp Glu Lys Gln Ala Leu Pro Gly Glu Glu Arg Asp Glu Ala Val Ser
 35            40            45

Arg Leu Lys Glu Cys Leu Ile Asn Asn Ser Asp Glu Leu Arg Leu Asp
 50            55            60

Arg Leu Asn Leu Ser Ser Leu Pro Asp Asn Leu Pro Ala Gln Ile Thr
 65            70            75            80

Leu Leu Asn Val Ser Tyr Asn Gln Leu Thr Asn Leu Pro Glu Leu Pro
 85            90            95

Val Thr Leu Lys Lys Leu Tyr Ser Ala Ser Asn Lys Leu Ser Glu Leu
100           105           110

Pro Val Leu Pro Pro Ala Leu Glu Ser Leu Gln Val Gln His Asn Glu
115           120           125

Leu Glu Asn Leu Pro Ala Leu Pro Asp Ser Leu Leu Thr Met Asn Ile
130           135           140

Ser Tyr Asn Glu Ile Val Ser Leu Pro Ser Leu Pro Gln Ala Leu Lys
145           150           155           160

Asn Leu Arg Ala Thr Arg Asn Phe Leu Thr Glu Leu Pro Ala Phe Ser
165           170           175

Glu Gly Asn Asn Pro Val Val Arg Glu Tyr Phe Phe Asp Arg Asn Gln
180           185           190

Ile Ser His Ile Pro Glu Ser Ile Leu Asn Leu Arg Asn Glu Cys Ser
195           200           205

Ile His Ile Ser Asp Asn Pro Leu Ser Ser His Ala Leu Pro Ala Leu
210           215           220

Gln Arg Leu Thr Ser Ser Pro Asp Tyr His Gly Pro Arg Ile Tyr Phe
225           230           235           240

Ser Met Ser Asp Gly Gln Gln Asn Thr Leu His Arg Pro Leu Ala Asp
245           250           255

Ala Val Thr Ala Trp Phe Pro Glu Asn Lys Gln Ser Asp Val Ser Gln
260           265           270

Ile Trp His Ala Phe Glu His Glu Glu His Ala Asn Thr Phe Ser Ala
275           280           285

Phe Leu Asp Arg Leu Ser Asp Thr Val Ser Ala Arg Asn Thr Ser Gly
290           295           300

Phe Arg Glu Gln Val Ala Ala Trp Leu Glu Lys Leu Ser Ala Ser Ala
305           310           315           320

Glu Leu Arg Gln Gln Ser Phe Ala Val Ala Ala Asp Ala Thr Glu Ser
325           330           335

Cys Glu Asp Arg Val Ala Leu Thr Trp Asn Asn Leu Arg Lys Thr Leu
340           345           350

Leu Val His Gln Ala Ser Glu Gly Leu Phe Asp Asn Asp Thr Gly Ala
355           360           365

```

-continued

Leu Leu Ser Leu Gly Arg Glu Met Phe Arg Leu Glu Ile Leu Glu Asp
 370 375 380
 Ile Ala Arg Asp Lys Val Arg Thr Leu His Phe Val Asp Glu Ile Glu
 385 390 395 400
 Val Tyr Leu Ala Phe Gln Thr Met Leu Ala Glu Lys Leu Gln Leu Ser
 405 410 415
 Thr Ala Val Lys Glu Met Arg Phe Tyr Gly Val Ser Gly Val Thr Ala
 420 425 430
 Asn Asp Leu Arg Thr Ala Glu Ala Met Val Arg Ser Arg Glu Glu Asn
 435 440 445
 Glu Phe Thr Asp Trp Phe Ser Leu Trp Gly Pro Trp His Ala Val Leu
 450 455 460
 Lys Arg Thr Glu Ala Asp Arg Trp Ala Gln Ala Glu Glu Gln Lys Tyr
 465 470 475 480
 Glu Met Leu Glu Asn Glu Tyr Pro Gln Arg Val Ala Asp Arg Leu Lys
 485 490 495
 Ala Ser Gly Leu Ser Gly Asp Ala Asp Ala Glu Arg Glu Ala Gly Ala
 500 505 510
 Gln Val Met Arg Glu Thr Glu Gln Gln Ile Tyr Arg Gln Leu Thr Asp
 515 520 525
 Glu Val Leu Ala Leu Arg Leu Pro Glu Asn Gly Ser Gln Leu His His
 530 535 540
 Ser
 545

<210> SEQ ID NO 73
 <211> LENGTH: 1635
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 73

atgctgccga ttaacaacaa ctttagcctg ccgcagaaca gcttttataa caccattagc	60
ggcacctatg cggattatgt tagcgcgtgg gataaatggg aaaaacaggc gctgcccggc	120
gaagaacgcg atgaagcggg gagccgcctg aaagaatgcc tgattaacaa cagcgatgaa	180
ctgcccctgg atgcctgaa cctgagcagc ctgcccggata acctgcccgc gcagattacc	240
ctgctgaacg tgagctataa ccagctgacc aacctgccgg aactgccggg gacctgaaa	300
aaactgtata gcgcgagcaa caaactgagc gaactgccgg tgctgccgcc ggcgctggaa	360
agcctgcagg tgcagcataa cgaactggaa aacctgccgg cgctgccgga tagcctgctg	420
accatgaaca ttagctataa cgaaattgtg agcctgccga gcctgccgca ggcgctgaaa	480
aacctgcgcg cgacccgcaa ctttctgacc gaactgccgg cgttttagcga aggcaacaac	540
ccggtgggtg cgcaatatgt ttttgatgc aaccagatta gccatattcc ggaaagcatt	600
ctgaacctgc gcaacgaatg cagcattcat attagcgata acccgctgag cagccatgcg	660
ctgcccgcgc tgcagcgctt gaccagcagc ccggattatc atggcccgcg catttatgtt	720
agcatgagcg atggccagca gaacacctg catcgcccgc tggcggatgc ggtgaccgcg	780
tggtttccgg aaaacaaaca gagcgatgtg agccagattt ggcattgcgt tgaacatgaa	840
gaacatgcga acacctttag cgcgtttctg gatcgctga gcgataccgt gagcgcgcg	900
aacaccagcg gcttttcgca acaggtggcg gcgtggctgg aaaaactgag cgcgagcgcg	960

-continued

```

gaactgcgcc agcagagctt tgcggtggcg gcggatgcga ccgaaagctg cgaagatcgc 1020
gtggcgctga cctggaacaa cctgcgcaaa accctgctgg tgcacaggc gagcgaaggc 1080
ctgtttgata acgataccgg cgcgctgctg agcctgggcc gcgaaatgtt tcgctggaa 1140
attctggaag atattgcgcg cgataaagtg cgcaccctgc atttgtgga tgaaattgaa 1200
gtgtatctgg cgtttcagac catgctggcg gaaaaactgc agctgagcac cgcggtgaaa 1260
gaaatgcgct tttatggcgt gagcggcgctg accgcgaacg atctgcgcac cgcggaagcg 1320
atggtgcgca gccgcgaaga aaacgaattt accgattggt ttagcctgtg gggcccgtgg 1380
catgcggtgc tgaacgcac cgaagcggat cgctgggcgc aggcggaaga acagaaatat 1440
gaaatgctgg aaaacgaata tccgcagcgc gtggcggatc gcctgaaagc gagcggcctg 1500
agcggcgatg cggatgcgga acgcgaagcg ggcgcgcagg tgatgcgcga aaccgaacag 1560
cagatttato gccagctgac cgatgaagtg ctggcgctgc gcctgccgga aaacggcagc 1620
cagctgcac atagc 1635

```

```

<210> SEQ ID NO 74
<211> LENGTH: 196
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 74

```

```

Met Lys Ile Thr Ser Thr Ile Ile Gln Thr Pro Phe Pro Phe Glu Asn
1      5      10      15
Asn Asn Ser His Ala Gly Ile Val Thr Glu Pro Ile Leu Gly Lys Leu
20     25     30
Ile Gly Gln Gly Ser Thr Ala Glu Ile Phe Glu Asp Val Asn Asp Ser
35     40     45
Ser Ala Leu Tyr Lys Lys Tyr Asp Leu Ile Gly Asn Gln Tyr Asn Glu
50     55     60
Ile Leu Glu Met Ala Trp Gln Glu Ser Glu Leu Phe Asn Ala Phe Tyr
65     70     75     80
Gly Asp Glu Ala Ser Val Val Ile Gln Tyr Gly Gly Asp Val Tyr Leu
85     90     95
Arg Met Leu Arg Val Pro Gly Thr Pro Leu Ser Asp Ile Asp Thr Ala
100    105    110
Asp Ile Pro Asp Asn Ile Glu Ser Leu Tyr Leu Gln Leu Ile Cys Lys
115    120    125
Leu Asn Glu Leu Ser Ile Ile His Tyr Asp Leu Asn Thr Gly Asn Met
130    135    140
Leu Tyr Asp Lys Glu Ser Glu Ser Leu Phe Pro Ile Asp Phe Arg Asn
145    150    155    160
Ile Tyr Ala Glu Tyr Tyr Ala Ala Thr Lys Lys Asp Lys Glu Ile Ile
165    170    175
Asp Arg Arg Leu Gln Met Arg Thr Asn Asp Phe Tyr Ser Leu Leu Asn
180    185    190
Arg Lys Tyr Leu
195

```

```

<210> SEQ ID NO 75
<211> LENGTH: 588
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

```

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 75

```

atgaaaatta ccagcaccat tattcagacc ccgtttccgt ttgaaaacaa caacagccat    60
gctgggcattg tgaccgaacc gattctgggc aaactgattg gccagggcag caccgcggaa    120
atTTTTgaag atgtgaacga tagcagcgcg ctgtataaaa aatatgatct gattggcaac    180
cagtataacg aaattctgga aatggcgctgg caggaaagcg aactgtttaa cgcgttttat    240
ggcgatgaag cgagcgctggt gattcagtat ggcggcgatg tgtatctgcg catgctgcgc    300
gtgccgggca ccccgctgag cgatattgat accgcggata ttccggataa cattgaaagc    360
ctgtatctgc agctgatttg caaactgaac gaactgagca ttattcatta tgatctgaac    420
accggcaaca tgctgtatga taaagaaagc gaaagcctgt ttccgattga ttttcgcaac    480
atTTtatgcg aatattatgc ggcgacccaaa aaagataaag aaattattga tcgccgctcg    540
cagatgcgca ccaacgattt ttatagcctg ctgaaccgca aatatctg                    588

```

<210> SEQ ID NO 76

<211> LENGTH: 462

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 76

```

Met Ser Pro Ile Leu Gly Tyr Trp Lys Ile Lys Gly Leu Val Gln Pro
1           5           10          15
Thr Arg Leu Leu Leu Glu Tyr Leu Glu Glu Lys Tyr Glu Glu His Leu
20          25          30
Tyr Glu Arg Asp Glu Gly Asp Lys Trp Arg Asn Lys Lys Phe Glu Leu
35          40          45
Gly Leu Glu Phe Pro Asn Leu Pro Tyr Tyr Ile Asp Gly Asp Val Lys
50          55          60
Leu Thr Gln Ser Met Ala Ile Ile Arg Tyr Ile Ala Asp Lys His Asn
65          70          75          80
Met Leu Gly Gly Cys Pro Lys Glu Arg Ala Glu Ile Ser Met Leu Glu
85          90          95
Gly Ala Val Leu Asp Ile Arg Tyr Gly Val Ser Arg Ile Ala Tyr Ser
100         105         110
Lys Asp Phe Glu Thr Leu Lys Val Asp Phe Leu Ser Lys Leu Pro Glu
115         120         125
Met Leu Lys Met Phe Glu Asp Arg Leu Cys His Lys Thr Tyr Leu Asn
130         135         140
Gly Asp His Val Thr His Pro Asp Phe Met Leu Tyr Asp Ala Leu Asp
145         150         155         160
Val Val Leu Tyr Met Asp Pro Met Cys Leu Asp Ala Phe Pro Lys Leu
165         170         175
Val Cys Phe Lys Lys Arg Ile Glu Ala Ile Pro Gln Ile Asp Lys Tyr
180         185         190
Leu Lys Ser Ser Lys Tyr Ile Ala Trp Pro Leu Gln Gly Trp Gln Ala
195         200         205
Thr Phe Gly Gly Gly Asp His Pro Pro Lys Ser Asp Leu Val Pro Arg
210         215         220

```

-continued

Gly Ser Met Ser Ile Glu Ile Lys Met Ile Ser Pro Ile Lys Asn Ile
 225 230 235 240
 Lys Asn Val Phe Pro Ile Asn Thr Ala Asn Thr Glu Tyr Ile Val Arg
 245 250 255
 Asn Ile Tyr Pro Arg Val Glu His Gly Tyr Phe Asn Glu Ser Pro Asn
 260 265 270
 Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Ser Met Ala Gln
 275 280 285
 Leu Lys Ile Glu Glu Phe Ile Asn Glu Lys Ser Arg Arg Leu Asn Tyr
 290 295 300
 Met Lys Thr Met Tyr Ser Pro Cys Pro Glu Asp Phe Gln Pro Ile Ser
 305 310 315 320
 Arg Asp Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
 325 330 335
 Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr His Pro
 340 345 350
 Asp Leu His Ala Leu Cys Glu Leu Pro Glu Ile Ser Cys Lys Ile Phe
 355 360 365
 Pro Lys Glu Asn Ser Asp Phe Leu Tyr Ile Ile Val Val Phe Arg Asn
 370 375 380
 Asp Ser Pro Gln Gly Glu Leu Arg Ala Asn Arg Phe Ile Glu Leu Tyr
 385 390 395 400
 Asp Ile Lys Arg Glu Ile Met Gln Val Leu Arg Asp Glu Ser Pro Glu
 405 410 415
 Leu Lys Ser Ile Lys Ser Glu Ile Ile Ile Ala Arg Glu Met Gly Glu
 420 425 430
 Leu Phe Ser Tyr Ala Ser Glu Glu Ile Asp Ser Tyr Ile Lys Gln Met
 435 440 445
 Asn Asp Arg Leu Ser Gln Ile Lys Ala Arg Met Pro Val Thr
 450 455 460

<210> SEQ ID NO 77
 <211> LENGTH: 1386
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 77

atgagcccgga ttctgggcta ttggaaaatt aaaggcctgg tgcagccgac ccgcctgctg	60
ctggaatatc tggaagaaaa atatgaagaa catctgtatg aacgcgatga aggcgataaa	120
tggcgcaaca aaaaatttga actgggcctg gaatttccga acctgccgta ttatattgat	180
ggcgatgtga aactgaccca gagcatggcg attattcgct atattgcgga taaacataac	240
atgctggggcg gctgcccga aagaacgcgc gaaattagca tgctggaagg cgcggtgctg	300
gatattcgct atggcgtgag ccgcattgcy tatagcaaag attttgaac cctgaaagtg	360
gattttctga gcaaaactgcc ggaaatgctg aaaatgtttg aagatcgcc gtgccataaa	420
acctatctga acggcgatca tgtgacccat ccggatttta tgctgtatga tgcgctggat	480
gtggtgctgt atatggatcc gatgtgcctg gatgcgtttc cgaaactggt gtgctttaa	540
aaacgcattg aagcgattcc gcagattgat aaatatctga aaagcagcaa atatattgcg	600
tggccgctgc agggctggca ggcgacctt ggccggcgcg atcatccgcc gaaaagcgat	660
ctggtgccgc gcggcagcat gacgattgaa attaaaatga ttagcccgat taaaaacatt	720

-continued

```

aaaaacgtgt ttccgattaa caccgcgaac accgaatata ttgtgcgcaa ctttatccg 780
cgcggtggaac atggctatatt taacgaaagc ccgaacattt atgataaaaa atatattagc 840
ggcattaccc gcagcatggc gcagctgaaa attgaagaat ttattaacga aaaaagccgc 900
cgctgaact atatgaaaac catgtatagc ccgtgcccg aagattttca gccgattagc 960
cgcgatgaag cgagcaccac ggaaggcagc tggctgaccg tgattagcgg caaacgcccg 1020
atgggccagt ttagcgtgga tagcctgtat catccggatc tgcatgcgct gtgcgaactg 1080
ccggaatta gctgcaaaat tttccgaaa gaaaacagcg attttctgta tattattgtg 1140
gtgtttcgca acgatagccc gcaggcgcaa ctgcgcgcga accgctttat tgaactgtat 1200
gatattaaac gcgaattat gcagggtgctg ccgatgaaa gcccggaact gaaaagcatt 1260
aaaagcgaaa ttattattgc gcgcgaaatg gccgaactgt ttagctatgc gagcgaagaa 1320
attgatagct atattaaaca gatgaacgat cgctgagcc agattaaagc gcgcgatgcc 1380
gtgacc 1386

```

```

<210> SEQ ID NO 78
<211> LENGTH: 212
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 78

```

```

Met Ile Asn Gly Val Ser Leu Gln Gly Thr Ala Gly Tyr Glu Ala His
1             5             10            15

Thr Glu Glu Gly Asn Val Asn Val Lys Lys Leu Leu Glu Ser Leu Asn
20            25            30

Ser Lys Ser Leu Gly Asp Met Asp Lys Asp Ser Glu Leu Ala Ala Thr
35            40            45

Leu Gln Lys Met Ile Asn Pro Ser Gly Gly Asp Gly Asn Cys Ser Gly
50            55            60

Cys Ala Leu His Ala Cys Met Ala Met Leu Gly Tyr Gly Val Arg Glu
65            70            75            80

Ala Pro Val Pro Asn Glu Ile Ser Glu Tyr Met Thr Gly Phe Phe His
85            90            95

Arg His Leu Glu Gln Ile Asp Ser Glu Gly Ile Val Ser His Pro Asn
100           105           110

Glu Thr Tyr Ser Lys Phe Arg Glu Arg Ile Ala Glu Asn Ile Leu Gln
115           120           125

Asn Thr Ser Lys Gly Ser Val Val Met Ile Ser Ile Glu Gln Ala Thr
130           135           140

His Trp Ile Ala Gly Phe Asn Asp Gly Glu Lys Ile Met Phe Leu Asp
145           150           155           160

Val Gln Thr Gly Lys Gly Phe Asn Leu Tyr Asp Pro Val Glu Lys Ser
165           170           175

Pro Asp Ala Phe Val Asp Glu Asn Ser Ser Val Gln Val Ile His Val
180           185           190

Ser Asp Gln Glu Phe Asp His Tyr Ala Asn Ser Ser Ser Trp Lys Ser
195           200           205

Lys Arg Leu Cys
210

```

-continued

```

<210> SEQ ID NO 79
<211> LENGTH: 636
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

<400> SEQUENCE: 79

atgattaacg gcgtgagcct gcagggcacc gcgggctatg aagcgcatac cgaagaaggc      60
aacgtgaacg tgaaaaaact gctggaaagc ctgaacagca aaagcctggg cgatatggat      120
aaagatagcg aactggcggc gaccctgcag aaaatgatta acccgagcgg cgcgcatggc      180
aactgcagcg gctgcgcgct gcatgcgtgc atggcgatgc tgggctatgg cgtgcgcgaa      240
gcgcgggtgc cgaacgaaat tagcgaatat atgaccggct tttttcatcg ccatctggaa      300
cagattgata gcgaaggcat tgtgagccat ccgaacgaaa cctatagcaa atttcgcgaa      360
cgcatctgcg aaaaacattct gcagaacacc agcaaaggca gcgtggtgat gattagcatt      420
gaacaggcga cccattggat tgcgggcttt aacgatggcg aaaaaattat gtttctggat      480
gtgcagaccg gcaaaggctt taacctgtat gatccggtag aaaaaagccc ggatgcgttt      540
gtggatgaaa acagcagcgt gcaggtgatt catgtgagcg atcaggaatt tgatcattat      600
gcgaacagca gcagctggaa aagcaaacgc ctgtgc                                636

```

```

<210> SEQ ID NO 80
<211> LENGTH: 298
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 80

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser
1          5          10          15

Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val
20          25          30

Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu
35          40          45

Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln
50          55          60

Asn Ser Ser Phe Leu Lys Ile Ile Asp Ile Asn Pro Val Thr Asn Thr
65          70          75          80

Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys
85          90          95

Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn
100         105         110

Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu
115         120         125

Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr
130         135         140

Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser
145         150         155         160

Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
165         170         175

Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro
180         185         190

```

-continued

Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe
 195 200 205
 Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn
 210 215 220
 Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr
 225 230 235 240
 Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu
 245 250 255
 Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu
 260 265 270
 Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile
 275 280 285
 Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 290 295

<210> SEQ ID NO 81
 <211> LENGTH: 315
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 81

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser
 1 5 10 15
 Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val
 20 25 30
 Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu
 35 40 45
 Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln
 50 55 60
 Asn Ser Ser Phe Leu Lys Ile Ile Asp Pro Ala Pro Ala Pro
 65 70 75 80
 Ala Pro Ala Pro Ala Pro Ala Pro Pro Ala Ile Asn Pro Val Thr Asn
 85 90 95
 Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val
 100 105 110
 Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro
 115 120 125
 Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala
 130 135 140
 Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser
 145 150 155 160
 Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile
 165 170 175
 Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile
 180 185 190
 Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn
 195 200 205
 Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile
 210 215 220
 Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg
 225 230 235 240
 Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu
 245 250 255

-continued

Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro
 260 265 270

Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly
 275 280 285

Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr
 290 295 300

Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 305 310 315

<210> SEQ ID NO 82

<211> LENGTH: 314

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 82

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser
 1 5 10 15

Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val
 20 25 30

Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu
 35 40 45

Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln
 50 55 60

Asn Ser Ser Phe Leu Lys Ile Ile Asp Gly Ser Gly Ser Gly Ser Gly
 65 70 75 80

Ser Gly Ser Gly Ser Gly Ser Gly Ser Ile Asn Pro Val Thr Asn Thr
 85 90 95

Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys
 100 105 110

Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn
 115 120 125

Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu
 130 135 140

Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr
 145 150 155 160

Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser
 165 170 175

Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
 180 185 190

Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro
 195 200 205

Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe
 210 215 220

Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn
 225 230 235 240

Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr
 245 250 255

Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu
 260 265 270

Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu
 275 280 285

Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile

-continued

290	295	300
-----	-----	-----

Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 305 310

<210> SEQ ID NO 83
 <211> LENGTH: 298
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 83

Pro	Glu	Leu	Pro	Pro	His	Leu	Glu	Ser	Leu	Val	Ala	Ser	Cys	Asn	Ser
1				5					10					15	
Leu	Thr	Glu	Leu	Pro	Glu	Leu	Pro	Gln	Ser	Leu	Lys	Ser	Leu	Gln	Val
		20						25					30		
Asp	Asn	Asn	Asn	Leu	Lys	Ala	Leu	Ser	Asp	Leu	Pro	Pro	Ser	Leu	Glu
	35					40					45				
Phe	Leu	Ala	Ala	Gly	Asn	Asn	Gln	Leu	Glu	Glu	Leu	Pro	Glu	Leu	Gln
	50				55					60					
Asn	Ser	Ser	Phe	Leu	Lys	Ile	Ile	Asp	Ile	Asn	Pro	Val	Thr	Asn	Thr
65				70					75					80	
Gln	Gly	Val	Ser	Pro	Ile	Asn	Thr	Lys	Tyr	Ala	Glu	His	Val	Val	Lys
			85					90						95	
Asn	Ile	Tyr	Pro	Lys	Ile	Lys	His	Asp	Tyr	Phe	Asn	Glu	Ser	Pro	Asn
			100					105					110		
Ile	Tyr	Asp	Lys	Lys	Tyr	Ile	Ser	Gly	Ile	Thr	Arg	Gly	Val	Ala	Glu
	115					120					125				
Leu	Lys	Gln	Glu	Glu	Phe	Val	Asn	Glu	Lys	Ala	Arg	Arg	Phe	Ser	Tyr
	130				135					140					
Met	Lys	Thr	Met	Tyr	Ser	Val	Cys	Pro	Glu	Ala	Phe	Glu	Pro	Ile	Ser
145				150					155					160	
Arg	Asn	Glu	Ala	Ser	Thr	Pro	Glu	Gly	Ser	Trp	Leu	Thr	Val	Ile	Ser
			165					170						175	
Gly	Lys	Ala	Pro	Met	Gly	Gln	Phe	Ser	Val	Asp	Ser	Leu	Tyr	Asn	Pro
		180					185						190		
Asp	Leu	His	Ala	Leu	Cys	Glu	Leu	Pro	Asp	Ile	Cys	Cys	Lys	Ile	Phe
	195				200						205				
Pro	Lys	Glu	Asn	Asn	Asp	Phe	Leu	Tyr	Ile	Val	Val	Val	Tyr	Arg	Asn
	210				215					220					
Asp	Ser	Pro	Leu	Gly	Glu	Gln	Arg	Ala	Asn	Arg	Phe	Ile	Glu	Leu	Tyr
225				230				235					240		
Asn	Ile	Lys	Arg	Asp	Ile	Met	Gln	Glu	Leu	Asn	Tyr	Glu	Leu	Pro	Glu
		245						250					255		
Leu	Lys	Ala	Val	Lys	Ser	Glu	Met	Ile	Ile	Ala	Arg	Glu	Met	Gly	Glu
		260					265						270		
Ile	Phe	Ser	Tyr	Met	Pro	Gly	Glu	Ile	Asp	Ser	Tyr	Met	Lys	Tyr	Ile
	275					280					285				
Asn	Asn	Lys	Leu	Ser	Lys	Ile	Glu	Gly	Ser						
290					295										

<210> SEQ ID NO 84
 <211> LENGTH: 315
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 84

```

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser
1           5           10           15
Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val
20           25           30
Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu
35           40           45
Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln
50           55           60
Asn Ser Ser Phe Leu Lys Ile Ile Asp Pro Ala Pro Ala Pro Ala Pro
65           70           75           80
Ala Pro Ala Pro Ala Pro Ala Pro Pro Ala Ile Asn Pro Val Thr Asn
85           90           95
Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val
100          105          110
Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro
115          120          125
Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala
130          135          140
Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser
145          150          155          160
Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile
165          170          175
Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile
180          185          190
Ser Gly Lys Ala Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn
195          200          205
Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile
210          215          220
Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg
225          230          235          240
Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu
245          250          255
Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro
260          265          270
Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly
275          280          285
Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr
290          295          300
Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
305          310          315

```

<210> SEQ ID NO 85

<211> LENGTH: 314

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 85

```

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser
1           5           10           15

```

-continued

```

Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val
      20      25      30
Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu
      35      40      45
Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln
      50      55      60
Asn Ser Ser Phe Leu Lys Ile Ile Asp Gly Ser Gly Ser Gly Ser Gly
      65      70      75      80
Ser Gly Ser Gly Ser Gly Ser Gly Ser Ile Asn Pro Val Thr Asn Thr
      85      90      95
Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys
      100      105      110
Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn
      115      120      125
Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu
      130      135      140
Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr
      145      150      155      160
Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser
      165      170      175
Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
      180      185      190
Gly Lys Ala Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro
      195      200      205
Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe
      210      215      220
Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn
      225      230      235      240
Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr
      245      250      255
Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu
      260      265      270
Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu
      275      280      285
Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile
      290      295      300
Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
305      310

```

<210> SEQ ID NO 86

<211> LENGTH: 292

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 86

```

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser
1      5      10      15
Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val
      20      25      30
Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu
      35      40      45
Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln
      50      55      60

```

-continued

Asn Ser Ser Phe Leu Lys Ile Ile Asp Ile Asn Pro Val Thr Asn Thr
 65 70 75 80
 Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys
 85 90 95
 Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn
 100 105 110
 Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu
 115 120 125
 Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr
 130 135 140
 Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser
 145 150 155 160
 Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
 165 170 175
 Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro
 180 185 190
 Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe
 195 200 205
 Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn
 210 215 220
 Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr
 225 230 235 240
 Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu
 245 250 255
 Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu
 260 265 270
 Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile
 275 280 285
 Asn Asn Lys Leu
 290

<210> SEQ ID NO 87

<211> LENGTH: 309

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 87

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser
 1 5 10 15
 Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val
 20 25 30
 Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu
 35 40 45
 Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln
 50 55 60
 Asn Ser Ser Phe Leu Lys Ile Ile Asp Pro Ala Pro Ala Pro Ala Pro
 65 70 75 80
 Ala Pro Ala Pro Ala Pro Ala Pro Pro Ala Ile Asn Pro Val Thr Asn
 85 90 95
 Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val
 100 105 110
 Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro

-continued

115	120	125
Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala		
130	135	140
Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser		
145	150	155
Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile		
165	170	175
Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile		
180	185	190
Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn		
195	200	205
Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile		
210	215	220
Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg		
225	230	235
Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu		
245	250	255
Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro		
260	265	270
Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly		
275	280	285
Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr		
290	295	300
Ile Asn Asn Lys Leu		
305		

<210> SEQ ID NO 88

<211> LENGTH: 308

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 88

Pro Glu Leu Pro Pro His Leu Glu Ser Leu Val Ala Ser Cys Asn Ser		
1	5	10
Leu Thr Glu Leu Pro Glu Leu Pro Gln Ser Leu Lys Ser Leu Gln Val		
20	25	30
Asp Asn Asn Asn Leu Lys Ala Leu Ser Asp Leu Pro Pro Ser Leu Glu		
35	40	45
Phe Leu Ala Ala Gly Asn Asn Gln Leu Glu Glu Leu Pro Glu Leu Gln		
50	55	60
Asn Ser Ser Phe Leu Lys Ile Ile Asp Gly Ser Gly Ser Gly Ser Gly		
65	70	75
Ser Gly Ser Gly Ser Gly Ser Gly Ser Ile Asn Pro Val Thr Asn Thr		
85	90	95
Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr Ala Glu His Val Val Lys		
100	105	110
Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr Phe Asn Glu Ser Pro Asn		
115	120	125
Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile Thr Arg Gly Val Ala Glu		
130	135	140
Leu Lys Gln Glu Glu Phe Val Asn Glu Lys Ala Arg Arg Phe Ser Tyr		
145	150	155
		160

-continued

Met Lys Thr Met Tyr Ser Val Cys Pro Glu Ala Phe Glu Pro Ile Ser
165 170 175

Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser Trp Leu Thr Val Ile Ser
180 185 190

Gly Lys Arg Pro Met Gly Gln Phe Ser Val Asp Ser Leu Tyr Asn Pro
195 200 205

Asp Leu His Ala Leu Cys Glu Leu Pro Asp Ile Cys Cys Lys Ile Phe
210 215 220

Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile Val Val Val Tyr Arg Asn
225 230 235 240

Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn Arg Phe Ile Glu Leu Tyr
245 250 255

Asn Ile Lys Arg Asp Ile Met Gln Glu Leu Asn Tyr Glu Leu Pro Glu
260 265 270

Leu Lys Ala Val Lys Ser Glu Met Ile Ile Ala Arg Glu Met Gly Glu
275 280 285

Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp Ser Tyr Met Lys Tyr Ile
290 295 300

Asn Asn Lys Leu
305

<210> SEQ ID NO 89
<211> LENGTH: 224
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 89

Asn Pro Val Thr Asn Thr Gln Gly Val Ser Pro Ile Asn Thr Lys Tyr
1 5 10 15

Ala Glu His Val Val Lys Asn Ile Tyr Pro Lys Ile Lys His Asp Tyr
20 25 30

Phe Asn Glu Ser Pro Asn Ile Tyr Asp Lys Lys Tyr Ile Ser Gly Ile
35 40 45

Thr Arg Gly Val Ala Glu Leu Lys Gln Glu Glu Phe Val Asn Glu Lys
50 55 60

Ala Arg Arg Phe Ser Tyr Met Lys Thr Met Tyr Ser Val Cys Pro Glu
65 70 75 80

Ala Phe Glu Pro Ile Ser Arg Asn Glu Ala Ser Thr Pro Glu Gly Ser
85 90 95

Trp Leu Thr Val Ile Ser Gly Lys Arg Pro Met Gly Gln Phe Ser Val
100 105 110

Asp Ser Leu Tyr Asn Pro Asp Leu His Ala Leu Cys Glu Leu Pro Asp
115 120 125

Ile Cys Cys Lys Ile Phe Pro Lys Glu Asn Asn Asp Phe Leu Tyr Ile
130 135 140

Val Val Val Tyr Arg Asn Asp Ser Pro Leu Gly Glu Gln Arg Ala Asn
145 150 155 160

Arg Phe Ile Glu Leu Tyr Asn Ile Lys Arg Asp Ile Met Gln Glu Leu
165 170 175

Asn Tyr Glu Leu Pro Glu Leu Lys Ala Val Lys Ser Glu Met Ile Ile
180 185 190

Ala Arg Glu Met Gly Glu Ile Phe Ser Tyr Met Pro Gly Glu Ile Asp
195 200 205

-continued

Ser Tyr Met Lys Tyr Ile Asn Asn Lys Leu Ser Lys Ile Glu Gly Ser
 210 215 220

<210> SEQ ID NO 90
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 6xHis tag

<400> SEQUENCE: 90

His His His His His His
 1 5

<210> SEQ ID NO 91
 <211> LENGTH: 4
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 91

Pro Ala Pro Ala
 1

<210> SEQ ID NO 92
 <211> LENGTH: 4
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 92

Gly Ser Gly Ser
 1

<210> SEQ ID NO 93
 <211> LENGTH: 672
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 93

atgattaatc ctgttactaa tactcagggc gtgtccccta taaatactaa atatgctgaa	60
catgtggtga aaaatattta cccgaaaatt aaacatgatt actttaatga atcacccaat	120
atatatgata agaagtatat atccgggtata accagaggag tagctgaact aaacaggaa	180
gaatttgta acgagaaagc cagacgggtt tcttatatga agactatgta ttctgtatgt	240
ccagaagcgt ttgaacctat ttccagaaat gaagccagta caccggaagg aagctggcta	300
acagttatat cggaaaaacg cccaatgggg cagttttctg tagatagttt atacaatcct	360

-continued

gatttacatg cattatgtga gcttccggac atttgttgta agatcttccc taaagaaaat	420
aatgattttt tatacatagt tgttgtgtac agaaatgaca gccctctagg agaacaacgg	480
gcaaatagat ttatagaatt atataatata aaaagagata tcatgcagga attaaattat	540
gagttaccag agttaagggc agtaaaatct gaaatgatta tcgcacgtga aatgggagaa	600
atctttagct acatgcctgg ggaaatagac agttatatga aatacataaa taataaaactt	660
tctaaaattg ag	672

The invention claimed is:

1. A composition comprising a set of paired peptides, wherein the set of paired peptides is linked to a protein transduction domain, and wherein the set of paired peptides comprises a first bacterial effector polypeptide linked to a second bacterial effector polypeptide, wherein the protein transduction domain is a YopM protein transduction domain, an SspH1 protein transduction domain, or an 1paH protein transduction domain wherein the protein transduction domain and the set of paired peptides comprise a fusion protein and the amino acid sequence of the fusion protein is at least 85% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24.

2. The composition of claim 1, wherein the first bacterial effector polypeptide and second bacterial effector polypeptide are different.

3. The composition of claim 1, wherein a first bacterial effector polypeptide and the second bacterial effector polypeptide are immunomodulatory.

4. The composition of claim 1, wherein the first bacterial effector polypeptide and the second bacterial effector polypeptide recognize a different molecular target or modulate a different inflammatory pathway.

5. The composition of claim 4, wherein the inflammatory pathway is the NFκB pathway, the JNK pathway, the p38 pathway or the STING pathway.

6. The composition of claim 1, wherein a linker is positioned between the first bacterial effector polypeptide and the second bacterial effector polypeptide.

7. The composition of claim 1, wherein the protein transduction domain is a YopM protein transduction domain.

8. The composition of claim 7, wherein the protein transduction domain comprises SEQ ID NO. 5.

9. The composition of claim 1, wherein

(a) the first bacterial effector polypeptide is a YopM polypeptide or an NLeE polypeptide; or

(b) the second bacterial effector polypeptide is a YopM polypeptide or an NLeE polypeptide.

10. The composition of claim 1, wherein the first bacterial effector polypeptide is a YopM polypeptide and the second bacterial effector polypeptide is an NLeE polypeptide.

11. A fusion protein comprising a set of paired peptides, wherein the set of paired peptides comprises a first bacterial effector polypeptide linked to a second bacterial effector polypeptide wherein the amino acid sequence of the fusion protein is at least 85% identical to the sequence set forth in SEQ ID NO. 10, 13, 16, 19, 22, or 24.

12. A composition comprising a protein transduction domain polypeptide linked to a first bacterial effector polypeptide and at least one additional bacterial effector polypeptides wherein the first bacterial effector polypeptide is a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79 and the at least one additional effector polypeptide is a polypeptide having 90% sequence identity to an amino acid sequence set forth in the group consisting of SEQ ID NOs. 3, 89, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, and 79.

* * * * *